

# Shareholder Voice and Executive Compensation\*

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## Abstract

I estimate a model of CEO compensation with advisory shareholder votes (Say-on-Pay). Because the Board sets pay anticipating dissent, the vote disciplines pay ex ante even though it is nonbinding. Removing the vote raises pay by 11.9% and lowers shareholder welfare by 0.51%, despite a 6% failure rate. I study two binding designs. Reversion restores the last approved policy after failure; revision adjusts the proposal using the vote's information. Relative to advisory voting, reversion lowers shareholder welfare because the Board controls both the proposal and the status quo; revision raises it by making vote-based information consequential for the current policy.

**Keywords:** shareholder voting, nonbinding voting, binding voting, corporate governance, executive compensation, CEO compensation, say-on-pay, structural estimation

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# 1. Introduction

Shareholders elect the Board of Directors but the Board need not represent their interests (Shleifer and Vishny, 1997). A primary manifestation is in executive compensation: the Board should set compensation to align the interests of management and shareholders, yet directors may have an incentive to favor the CEO (Bebchuk and Fried, 2003).

When shareholders disagree with compensation decisions and exerting control is not viable, shareholders can convey dissent through *voice* (Hirschman, 1970; Cuñat et al., 2016). Say-on-Pay (SOP), a nonbinding shareholder vote on CEO compensation policy, is an important channel through which shareholders may voice dissent. While advisory, SOP functions as a public performance evaluation of the compensation committee and a vote of confidence in the Board’s pay-setting choices.

Yet the impact of SOP is unclear. Compensation policies receive over 90% support on average and only about 6% of SOP votes in the US fail.<sup>1</sup> An important consideration is that these approval votes are endogenous outcomes, occurring after compensation has been set and firm performance realized. The pay package brought to a vote is itself an equilibrium object: anticipating how shareholders translate pay and realized performance into dissent, and internalizing the cost of that dissent, the Board may set compensation to make the vote pass. Whether low dissent ex post is consistent with effective governance ex ante is therefore an empirical question.

I estimate a structural model of compensation and shareholder voting to quantify the disciplinary effect of the vote. The Board sets compensation and may differ from shareholders over the appropriate pay level. Shareholders hold the vote; failure may punish the Board, but public dissent on a prominent policy may itself be costly to shareholders.<sup>2</sup> Estimating the model recovers the compensation the Board would have chosen without the threat of dissent, which observed votes cannot reveal.

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<sup>1</sup>SOP was formalized in the US under Dodd-Frank in 2010. Shareholders vote ex post on the named executive officers’ compensation for the previous fiscal year, of which the CEO’s is the largest component, and the vote is nonbinding. I use “failure” to refer to votes receiving less than 70% support, the threshold commonly used by academics and practitioners (see Dey et al., 2024; ISS, 2022, Section 5, “Compensation”).

<sup>2</sup>In surveys, institutional investors report reluctance to vote against SOP because they prefer to resolve concerns through engagement, and because negative votes can trigger costly follow-up with the firm (Edmans et al., 2023).

A key challenge is to separate the disciplinary effect of the vote from other forces affecting compensation and voting. For example, the Board and shareholders do not observe CEO skill directly and may hold different beliefs about ability (Taylor, 2010). Accounting for these forces and others is necessary for the counterfactual.

Identification comes from moments that summarize the equilibrium interaction between the Board and shareholders: an ex post voting equation that maps pay and performance into dissent, and an ex ante wage-setting equation that maps anticipated dissent into pay. These moments separate the disagreement over pay that creates a role for shareholder discipline from the potential costs that both sides may bear when such discipline is exercised.

I use the estimated model to answer two questions. First, how much does an advisory vote affect compensation and value? Removing the voting institution raises CEO pay by 11.9%, lowers operating value by 1.56%, and lowers shareholder welfare (operating value net of shareholders' costs of dissent), by 0.51%. As such, shareholder voice, via nonbinding shareholder voting, provides value to shareholders.

For SOP to provide this disciplinary value, the Board and shareholders must disagree over the appropriate level of CEO pay.<sup>3</sup> I call this the governance wedge: a large wedge means the Board places more weight on CEO pay and prefers higher compensation than shareholders. The estimated wedge is positive and larger at more co-opted boards (Coles et al., 2014), providing scope for the vote to discipline the Board's pay-setting decision.

Moreover, whether discipline is effective depends on the consequences of dissent. The estimation implies a cost of failure to the Board, so the threat of dissent enters the compensation decision even when failure itself is uncommon. Voting against the Board is also costly to shareholders, limiting dissent observed in equilibrium. Importantly, the model allows these costs to be zero, but the data select positive values. Together, these costs make SOP a costly punishment mechanism: failure disciplines the Board, while shareholders use it selectively.

The compensation policy brought to a vote is an equilibrium object: the Board sets proposals to manage failure risk, so infrequent dissent is consistent with effective discipline operating through the threat of dissent. This logic extends beyond SOP. In many shareholder votes on

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<sup>3</sup>This statement concerns SOP's disciplinary role. If the Board and shareholders have aligned pay-setting objectives, the vote may still convey information, but there is no pay-setting distortion for the threat to correct.

board- or management-sponsored policies, such as director elections (Aggarwal et al., 2019), major transactions (Becht et al., 2016), or equity issuance (Holderness, 2018), the proposal itself, or whether it reaches a vote at all, may be shaped by anticipated shareholder dissent.

These results are directly relevant to policy. In May 2026, the SEC proposed exempting about 80% of public companies from recurring Say-on-Pay votes.<sup>4</sup> Because the vote rarely fails and is nonbinding, its benefit is difficult to infer from observed outcomes. The estimates instead measure that benefit through the proposals the vote shapes.

Having established that advisory voting can be an effective governance mechanism, I turn to the paper's second question: should shareholder dissent remain advisory, or should rejection carry consequence, and if so, what form should it take? I consider two counterfactual binding designs, which I label "reversion" and "revision."

Under binding reversion, the design of the UK's binding policy vote and the default under the EU directive, a rejected policy leaves the last approved policy in force; the model stylizes this as reversion to the last approved pay level.<sup>5</sup> Binding revision requires the Board to revise the current proposal using the information revealed by shareholder dissent under vote failure.

The two designs have sharply different welfare effects. Relative to the advisory vote, binding reversion raises operating value by +0.69%. It also produces more dissent, and this cost for shareholders offsets the gain, reducing shareholder welfare by -1.21% below the advisory benchmark. Under reversion, passage today sets tomorrow's status quo, giving the Board dynamic agenda-setting power (Romer and Rosenthal, 1979). A higher wage weakens shareholders' future threat to reject, inducing the Board to propose higher pay and shareholders to reject more often to preserve a lower status quo.

Because today's decision becomes tomorrow's status quo, dynamic agenda setting makes current choices strategically valuable (Baron, 1996; Diermeier and Fong, 2011). This creates an intertemporal commitment problem for shareholders. At each meeting, they choose the voting threshold that is optimal given the inherited wage, but cannot commit to how they

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<sup>4</sup>SEC Release No. 33-11419 (May 19, 2026). All public companies with float below \$2 billion would be exempt, which covers about 80% of public companies.

<sup>5</sup>UK companies hold a binding policy vote at least every three years (Companies Act 2006, s. 439A); the EU directive at least every four. In both, a rejected policy restores the last approved policy until a new vote. The directive lets member states make the policy vote advisory (several have). Swiss binding votes are on pay levels (Wagner and Wenk, 2026).

will vote once today's proposal governs tomorrow. The resulting sequence is optimal at each meeting but leaves shareholders worse off.

Revision, however, raises shareholder welfare by 0.24% relative to the advisory vote. A failed vote changes the current proposal using the information revealed by dissent, making shareholder information consequential for the compensation implemented today. Because the failure outcome depends on the Board's current proposal and the current vote rather than on an inherited fallback, revision avoids reversion's intertemporal agenda-setting distortion.

The counterfactuals point to a broader distinction between vote designs. In the one-shot agenda-setting model of [Matsusaka and Ozbas \(2017\)](#), the right to approve is weak, while the right to propose can help or hurt shareholders. A recurring vote adds a third margin: control of the status quo. Under reversion, the Board controls it, so making the vote binding lowers shareholder welfare. Under revision, dissent instead allows shareholders' information to amend the proposal, so a binding vote raises welfare. The value of approval rights therefore depends on what happens if shareholders do not approve the policy.<sup>6</sup>

While the model captures key features of SOP, it abstracts from two aspects of the setting. First, I do not model shareholder engagement before or after the annual meeting (e.g., [Dey et al., 2024](#); [Heath et al., 2026](#)). [Barry et al. \(2026a\)](#) jointly model private engagement and public voting and show that engagement can affect both subsequent compensation and observed dissent. The estimates and counterfactuals therefore hold the engagement environment fixed.

Second, I summarize the shareholder base with a representative voting rule, abstracting from strategic voting and the horizontal margin of how dispersed information aggregates ([Maug and Rydqvist, 2009](#); [Levit and Malenko, 2011](#); [Malenko and Malenko, 2019](#); [Barry et al., 2026b](#)). My focus is vertical: how the Board responds to the probability of failure, and what a failure reveals about the CEO. The estimation disciplines both objects using the observed failure rule and the failure-pay response. The counterfactuals then abstract from how strategic voting might change under a new design. This isolates the paper's main mechanism: how boards internalize the threat of dissent when setting compensation.

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<sup>6</sup>This result requires a vote for which rejection preserves a previously approved policy or authorization. It therefore applies naturally to recurring pay-policy votes, equity-compensation plans, increases in authorized shares, and similar votes over amendments to existing corporate policies.

**Literature review.** A large literature studies SOP and related pay-vote regulations (e.g., Cai and Walkling, 2011; Ferri and Maber, 2013; Cuñat et al., 2016; Correa and Lel, 2016; Ferri and Göx, 2018), and finds that expanding shareholder voice can affect compensation and firm value, even though failed votes are rare and need not directly change pay (Armstrong et al., 2013). This relates to work showing that rarely exercised governance threats can affect firm behavior (Fos, 2017; Johnson and Swem, 2021). My estimates speak directly to the adoption evidence in Cuñat et al. (2016) and Correa and Lel (2016).<sup>7</sup> I analyze the mechanism underlying these reduced-form effects. Because Boards choose compensation anticipating shareholder dissent, observed voting outcomes may not capture the vote’s disciplinary effect.

Wagner and Wenk (2026) study policy changes that introduced binding SOP in Switzerland and document shareholder-value effects across designs. My counterfactuals connect binding designs to agenda-setting models. A setter’s power depends on the reversion level (Romer and Rosenthal, 1979), and Ingberman (1985) makes today’s outcome tomorrow’s reversion level. In the modern version of Barseghyan and Coate (2014), neither side can commit and equilibrium paths are inefficient. Such an endogenous status quo makes players disagree more often in Dziuda and Loeper (2016) and can raise efficiency in Bowen et al. (2014). I estimate this tradeoff for a shareholder vote: at the estimated wedge, the Board’s control of the fallback lowers shareholder welfare, because shareholders cannot commit across meetings.

The analysis also complements Levit and Malenko (2011), who show that binding and non-binding voting can differ in their informational and disciplinary roles. I hold the vote binding and vary the consequence attached to rejection, which governs information use, dynamic incentives, and the value of the approval right.

Blonien et al. (2025) study how proxy-advisor information affects shareholder voting, while Barry et al. (2026b) estimate a model that separates blockholder private information and strategic voting from underlying preferences toward management. Barry et al. (2026a) jointly model private engagement and public voting as dynamic governance channels, emphasizing how each affects information transmission and the use of the other. My contribution is to es-

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<sup>7</sup>Using the introduction of SOP, these papers find economically meaningful effects on firm outcomes: Cuñat et al. (2016) estimate, from close votes on US shareholder-sponsored SOP proposals, a roughly 5% increase in market value with limited effects on the level of pay, and Correa and Lel (2016) find in cross-country data that CEO pay growth is roughly 7% lower after mandatory SOP laws.

timate a dynamic model in which the Board’s policy choice and shareholder voting are jointly determined, and then use the model to quantify how the voting institution itself disciplines policy and how alternative consequences of dissent affect welfare.

## **2. Institutional background and data**

### **2.1. Institutional details of Say-on-Pay**

SOP became a standard feature of US corporate governance after the 2010 Dodd-Frank Act mandated that public firms provide shareholders with an advisory vote on executive compensation beginning in 2011, with some smaller firms phased in later. In practice, nearly all large US firms hold SOP votes annually. The vote is ex post and advisory: shareholders vote on compensation outcomes disclosed in the proxy statement for the prior fiscal year, rather than on compensation policy going forward. The vote covers executive compensation disclosed under Item 402 of Regulation S-K, including the Compensation Discussion and Analysis (CD&A), which contextualizes the Board’s rationale for its compensation policy. The model’s action and information-revelation sequence is chosen to replicate these institutional features.

SOP is advisory, so no threshold forces the compensation committee to revise pay. Instead, what matters is the level of dissent at which key governance intermediaries and boards treat the outcome as negative. A central benchmark is 70% support, or 30% votes against. Institutional Shareholder Services (ISS) treats SOP support below 70% as a low-support outcome that triggers heightened scrutiny of the firm’s response and may lead ISS to recommend votes against the subsequent SOP proposal and/or directors (ISS, 2022, Section 5, “Compensation”). Consistent with this, firms experience sharp changes in institutional investor engagement around the 70% cutoff, while placebo threshold analyses do not show comparable discontinuities at alternative cutoffs such as 50% support (Dey et al., 2024). For these reasons, throughout the paper I define “SOP failure” as at least 30% of votes cast against the SOP proposal.<sup>8</sup>

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<sup>8</sup>Table A.2 reports that compensation-committee directors are more likely to leave the board within three and five years after a SOP vote receiving less than 70% support. This evidence is consistent with the 70% margin being salient for the Board. As a robustness check, I re-estimate the model using the statutory majority threshold of 50% support to define failure. The parameter estimates and conclusions are very similar.

## 2.2. Data and summary statistics

For the estimation described in Section 4, I merge vote outcomes from Institutional Shareholder Services (ISS) with CEO compensation from Execucomp and firm-level accounting data from Compustat. The sample covers S&P 1500 firms over 2011–2024. Table 1 reports summary statistics for the key variables used in estimation. Mean opposition to SOP is 8.2%, and 6.1% of votes exceed 30% against, the failure rate targeted in estimation. CEO total expected pay (TDC1) averages \$7.47 million, with an interquartile range for annual pay changes of  $-10\%$  to  $+18\%$ . Median CEO tenure is seven years, informing the model’s learning process.

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| Table 1 about here |
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## 3. Model

### 3.1. Technology and environment

Time is annual. The firm is infinitely-lived and consists of three actors: the Board of Directors, which sets the CEO’s pay each period; the CEO, who provides a productive input to the firm in return for pay; and a shareholder base  $S$ , which holds an approval vote over the Board’s pay policy. The wage in the model represents the level of total (expected) pay, with no distinction between base and incentive pay; one interpretation is that the Board and CEO negotiate a pay level that determines the CEO input used in production that year (Taylor, 2013).<sup>9</sup>

To link CEO compensation to firm performance in the simplest possible way, I assume a reduced-form relation between pay and observed output:

$$y_t = \eta A_t w_t^\alpha, \tag{1}$$

where  $w_t$  is total CEO compensation,  $A_t > 0$  is firm productivity,  $\alpha \in (0, 1)$  governs the curvature of output in pay, and  $\eta$  maps model output into observed firm revenues. Appendix IA1.1 shows that (1) can replicate the reduced-form of a linear contract in which the CEO chooses

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<sup>9</sup>Empirically,  $w_t$  is measured by Execucomp’s total compensation (TDC1), which bundles salary, bonus, and equity-based pay.

costly input, so the curvature  $\alpha$  summarizes both the CEO's input response to pay and diminishing returns in production.

Firm productivity is centered around CEO skill  $a$  but influenced by a mean-zero shock  $\varepsilon_{yt}$ ,

$$\ln A_t = a + \varepsilon_{yt}, \quad \varepsilon_{yt} \sim N(0, \sigma_y^2). \quad (2)$$

Type  $a$  is not observed by the Board and shareholders, they make predictions about  $a$  based on information they observe (as detailed in Section 3.2). Eq. (2) defines the notion of CEO skill: higher types achieve higher average productivity.

Operating income in year  $t$  is

$$\Pi_t = \eta A_t w_t^\alpha - \eta \kappa w_t.$$

The parameters  $\eta$  and  $\kappa > 0$  set the units of output and pay. They do not affect the model's strategic incentives, but are needed to translate equilibrium pay choices into observed compensation and operating-value effects (Appendix IA1.1). Normalizing operating income by  $\eta$  (so  $\pi_t \equiv \Pi_t/\eta$ ), we have

$$\pi_t = A_t w_t^\alpha - \kappa w_t. \quad (3)$$

The Board of Directors (B) sets the wage  $w_t$ . A key friction is that the Board's pay-setting objective may differ from the shareholders' objective. In particular, the Board may place additional weight on CEO compensation relative to shareholder operating income. Abstracting from dynamic considerations and Say-on-Pay, and expressing pay-setting objectives in normalized units, the Board chooses  $w_t$  to maximize

$$\pi_t^B = A_t w_t^\alpha - (1 - \lambda_\tau) \kappa w_t,$$

where  $\lambda_\tau \in [0, 1)$  is the governance wedge between the Board and shareholders over the optimal level of CEO compensation. When  $\lambda_\tau = 0$ , the Board internalizes the full marginal cost of pay; when  $\lambda_\tau > 0$ , it acts as if this cost is lower and therefore prefers higher compensation

than shareholders do. The wedge may vary with CEO tenure and, in the empirical implementation, across firms; I suppress the firm index for expositional simplicity. Section 4 specifies these tenure and cross-sectional components.<sup>10</sup>

Shareholders hold an annual Say-on-Pay (SOP) vote on the Board’s CEO pay policy. The vote is nonbinding: failure does not mechanically alter compensation. It occurs at the end of the period, after compensation and output are realized. A failed vote imposes a cost  $\chi_s \geq 0$  borne by shareholders and internalized by the Board, as well as an additional private cost  $\delta \geq 0$  on directors. The Board’s per-period utility is

$$A_t w_t^\alpha - \kappa w_t + \lambda_\tau \kappa w_t - (\chi_s + \delta) \mathbf{1}[\text{SOP fail}_t]. \quad (4)$$

Shareholders value operating income (3) but also incur a cost when they dissent from the Board. When proposed compensation is high relative to their beliefs about CEO ability, they may vote against the pay policy, trading off the gain from disciplining compensation against the cost of a failed vote. The parameter  $\chi_s$  captures this dissent cost, which may reflect, for example, the reputational cost of publicly opposing the Board or the engagement burden that follows a negative vote.<sup>11</sup> Shareholders’ per-period utility is

$$A_t w_t^\alpha - \kappa w_t - \chi_s \mathbf{1}[\text{SOP fail}_t]. \quad (5)$$

### 3.2. Beliefs, signals and model timeline

Each period the firm separates from the CEO with exogenous probability  $f_\tau$  and matches with a new CEO of tenure  $\tau = 0$ .<sup>12</sup> At the start of each match, the Board and shareholders begin with the prior about CEO skill,

$$a \sim N(\mu_0, \sigma_0^2), \quad (6)$$

<sup>10</sup>Allowing the wedge to vary with CEO tenure is motivated by the idea that CEOs may accumulate influence over Board composition as their tenure increases (Hermalin and Weisbach, 1998; Coles et al., 2014).

<sup>11</sup>Surveyed institutional investors report reluctance to vote against SOP because they prefer to resolve concerns through engagement rather than public dissent, and because a negative vote can trigger repeated follow-up requests from the firm, “imposing a significant time cost” (Edmans et al., 2023, Online Appendix A).

<sup>12</sup>The Board’s perceived tenure-specific separation hazard is exogenous. Section 4 explains the failure-dependent turnover adjustment used in the simulated panels.

which matches the distribution of ability in the CEO talent pool. At the annual compensation committee meeting, the Board receives its private signal about the CEO,

$$z_{bt} = a + \varepsilon_{bt}, \quad \varepsilon_{bt} \sim N(0, \sigma_{z_b}^2). \quad (7)$$

The signal  $z_{bt}$  will inform the Board's wage decision, and may represent operational interaction with the CEO. Shareholders do not observe  $z_{bt}$ , which means that the Board is asymmetrically informed (more precise beliefs) about CEO skill when setting pay. After pay is set, both output and productivity realize (that is, B does not observe  $A_t$  before setting compensation).

At the annual shareholder meeting, the shareholder base, denoted S, observes public information and a private signal about CEO type and evaluates the Board's proposed wage. I model S as a representative shareholder.<sup>13</sup> At the end of each period  $t$ , S observes an aggregate private signal about CEO type,

$$z_{st} = a + \varepsilon_{st}, \quad \varepsilon_{st} \sim N(0, \sigma_{z_s}^2). \quad (8)$$

S also observes output  $y_t$  and the CEO's pay  $w_t$ . These observables imply a public signal about productivity  $A_t$ , which is informative of the CEO's type. I label this signal

$$z_{yt} = \ln A_t = a + \varepsilon_{yt}, \quad \varepsilon_{yt} \sim N(0, \sigma_y^2). \quad (9)$$

Both the private signal  $z_{st}$  and the public signal  $z_{yt}$  will affect the SOP result.

**Model timeline.** Figure 1 displays the per-period sequence of events. The Board and Shareholder start the period with the shared belief  $a \sim N(\mu_{at}, \sigma_{at}^2)$ . At the compensation committee, B receives its signal  $z_{bt}$ , which informs their wage decision. Then, operations take place: the CEO supplies their input, productivity realizes and output occurs. Finally, S receives the private signal  $z_{st}$ ; output, productivity, and wages are revealed; and the SOP vote occurs. Within

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<sup>13</sup>Appendix IA1.2 microfound this representation from an atomistic shareholder base casting individual votes. With many small holders, the aggregate vote-against share is a sufficient statistic for the common component of shareholder information, and the threshold at which the aggregate vote fails coincides with the representative Shareholder's decision rule.

a period, the Board and Shareholder therefore hold different information sets: the Board sees its private signal  $z_{bt}$  before setting pay, while the Shareholder sees  $z_{st}$ , the public productivity signal  $z_{yt}$ , and the wage before voting. Both signals are disclosed by the end of the period, so beliefs coincide again at the start of the next (Appendix [IA1.4](#)).

Figure 1 about here

**Board and Shareholder beliefs.** Both B and S use Bayes' rule to update beliefs about CEO ability after their signals. I use subscript  $a$  to refer to beliefs shared by B and S:  $(\mu_{at}, \sigma_{at}^2)$  refers to the shared prior at the beginning  $t$ . Subscripts  $b$  and  $s$  refer to when B and S can have different beliefs. CEO tenure fully determines the variance of beliefs,  $\sigma_{at}^2 = \sigma_a^2(\tau_t)$ , which declines with tenure, so  $\mu_{at}$  tends toward the CEO's true ability.

### 3.3. The vote decision

This section formalizes the shareholder voting strategy, taking the Board's wage choice as given. At each state, S chooses a screen before the period's signals are drawn; this screen maps the proposed wage and the information available at the annual meeting into a probability of SOP failure. The screen is a scalar chosen at the start of the period, conditional on the current belief state, and applied at the meeting as chosen. Conditional on that state, the voting rule depends only on the current proposed wage and current signals.<sup>14</sup>

Informally, S fails the SOP when the CEO's wage is "too high" given S' beliefs about CEO type and the cost of vote failure. Formally, conditional on the wage choice, S' strategy can be represented as the threshold:

$$k_{st} = s_t \times w_t, \tag{10}$$

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<sup>14</sup>Two restrictions. First, the Shareholder commits within the period to a scalar screen, chosen before the Board sets pay and applied at the meeting as chosen. This is the institutional structure of SOP voting: proxy advisors and the large institutions publish pay-voting policies before each proxy season, those policies screen a proposal on its pay relative to performance, and compensation committees set pay knowing the screen their proposal will face (Ertimur et al., 2013; Larcker et al., 2015; Malenko and Shen, 2016). The screen is chosen anew at each state, so the commitment horizon is one policy cycle. Absent within-period adherence, an advisory vote would never fail; 6.1% of votes do. Second, conditional on the state, the screen depends only on the current proposal and the meeting signals: the benchmark is formed from those signals, and the proposed wage enters the vote only through its gap from the benchmark, not as a signal from which the Board's information is inverted. Appendix [IA1.3](#) discusses both restrictions and shows that, under the advisory vote, the state-by-state screen coincides with the screen the Shareholder would choose under commitment to a state-contingent path.

where  $s_t$  is the Shareholder's choice variable. The multiplicative form is the simplest specification that delivers a wage-elastic failure probability, which is the equilibrium object of focus in the estimation. After observing signals at the shareholder meeting, they compute a benchmark wage implied by the meeting information, which I label  $w_t^S$  and define below. The vote fails whenever the benchmark falls below the threshold:

$$\text{SOP fail}_t \iff w_t^S \leq k_{st} = s_t \times w_t.$$

Thus,  $s_t$  controls the sensitivity of failure probability to changes in the Board's wage choice  $w_t$ : a lower  $s_t$  makes shareholders more tolerant of deviations from their benchmark.

At the annual shareholder meeting, the Shareholder combines its private signal  $z_{st}$  and the public productivity signal  $z_{yt} = \ln A_t$  into a signal about CEO type,

$$Z_{synt} = pz_{st} + (1 - p)z_{yt} = a + \varepsilon_{synt}, \quad (11)$$

where  $p$  is the precision weight on the private signal and  $\varepsilon_{synt}$  has variance  $\sigma_{sy}^2$ .

The combined signal implies the benchmark wage against which the proposal is screened, the wage that maximizes operating income when the meeting signal is taken at face value:

$$w_t^S = \arg \max_{w_t} \exp(Z_{synt}) w_t^\alpha - \kappa w_t = \left( \frac{\alpha}{\kappa} \exp(Z_{synt}) \right)^{\frac{1}{1-\alpha}}.$$

Thus, S evaluates the proposed wage against the wage justified by the information available at the meeting. Ex ante, at the time S chooses its strategy,  $w_t^S$  is lognormal given the shared belief  $(\mu_{at}, \sigma_{at}^2)$ ; let  $F_t^S$  denote its CDF. Conditional on the Board's wage choice  $w_t$ , the ex ante probability of failure is

$$\Pr(\text{SOP fail}_t) = \Pr(w_t^S \leq s_t \times w_t) = F_t^S(s_t \times w_t). \quad (12)$$

This is the probability from the Shareholder's information set at the start of the period. The Board sets pay after observing  $z_{bt}$ , so the failure probability it faces at a proposed wage,  $F_{t|z_{bt}}^S[s_t \times w_t]$ , is the same expression at its posterior after  $z_{bt}$ . Because the vote is nonbinding,

shareholders' only influence on the current wage is ex ante, through the Board's anticipation of the voting rule. Written as a dynamic problem, S solves

$$V^S(\mu_{at}, \tau_t) = \max_{s_t} \int_{z_b} f(z_b) \left[ \mathbb{E}_{b|z_b}[A_t] w_t(z_{bt}, s_t)^\alpha - \kappa w_t(z_{bt}, s_t) - \chi_s \times F_{t|z_{bt}}^S[s_t \times w_t(z_{bt}, s_t)] \right] dz_b + \beta \left[ (1 - f_{\tau_t}) \mathbb{E}_{at} [V^S(\mu_{at+1}, \tau_t + 1)] + f_{\tau_t} V^{S,R} \right], \quad (13)$$

with  $V^{S,R} \equiv V^S(\mu_0, \tau = 0)$ . Under the advisory vote, the continuation value is independent of  $s_t$ : neither the screen nor the wage it induces becomes a state variable. It therefore drops out of the maximization, leaving a state-by-state choice of the screen.<sup>15</sup>

### 3.4. The compensation decision

Each period, the Board observes its private signal  $z_{bt}$  and chooses the CEO's wage  $w_t$ , taking the Shareholder's SOP policy  $s_t$  as given. The signal updates the belief to  $(\mu_{bt|z_b}, \sigma_{bt|z_b}^2)$ , which determines expected productivity and the failure probability  $F_{t|z_{bt}}^S$  (Appendix IA1.4).

The Board's wage policy solves

$$V(\mu_{at}, \tau_t, z_{bt}) = \max_{w_t} \mathbb{E}_{b|z_b}[A_t] w_t^\alpha - (1 - \lambda_{\tau_t}) \kappa w_t - (\chi_s + \delta) F_{t|z_{bt}}^S[s_t w_t] + \beta \left[ (1 - f_{\tau_t}) \mathbb{E}_{b|z_b}[V(\mu_{at+1}, \tau_t + 1)] + f_{\tau_t} V^R \right],$$

$$V(\mu_{at}, \tau_t) = \mathbb{E}_{at}[V(\mu_{at}, \tau_t, z_{bt})]. \quad (14)$$

The Board trades off expected operating income, its governance wedge, and the expected cost of a failed vote. The state is the shared belief mean  $\mu_{at}$  and tenure  $\tau_t$ , which pins down belief dispersion. Continuation values integrate over next period's belief using the Board's posterior; with exogenous separation hazard  $f_{\tau_t}$ , beliefs reset to the entry state, so  $V^R = V(\mu_0, 0)$ .<sup>16</sup>

<sup>15</sup>This simplification is specific to the advisory design. In counterfactuals, when I study a binding vote design with reversion to a previously chosen wage, the implemented wage becomes a state variable, so the continuation value can depend on the current screen. In that case, the Markov screen need not coincide with the screen S would choose under commitment; Appendix B.3 characterizes this distinction.

<sup>16</sup>I follow Taylor (2010) in computing the tenure-specific separation hazards. Details on the empirical construction of  $f_\tau$  and the treatment of turnover following failed votes are given in Section 4.

### 3.5. Model solution, operating value and shareholder welfare

I compute a Markov-perfect equilibrium in stationary policies. At each state, the Shareholder chooses a screen  $s(\mu_{at}, \tau_t)$  maximizing (13); after observing  $z_{bt}$ , the Board chooses a wage  $w(\mu_{at}, \tau_t, z_{bt})$  maximizing (14); and the screen is applied at the meeting. Neither side can commit to future policies. Because neither  $s_t$  nor  $w_t$  becomes a state variable, continuation values do not depend on current actions, so the equilibrium reduces to a sequence of state-by-state Stackelberg problems. Appendices IA1.5 and IA2.1 give the derivation and numerical solution.

To evaluate the real effects of the voting institution, I report *operating value*,  $V^\Pi(\mu_{at}, \tau_t)$ , the discounted present value of operating income (3) under the equilibrium policy. Operating value is the model's counterpart of firm market value.<sup>17</sup> I measure *shareholder welfare* using the Shareholder value function  $V^S$  in (13), which is operating value net of the Shareholder's expected costs of dissent. The Board's private cost  $\delta$  affects equilibrium pay through (14) but is not a firm cash flow and therefore enters neither operating value nor Shareholder welfare.

## 4. Estimation

I estimate the model's structural parameters using indirect inference (Gouriéroux et al., 1993), implemented as a simulated method of moments (McFadden, 1989): I choose the vector of parameters which minimizes the difference between the moments of an auxiliary model estimated on observed and simulated data, computing the same auxiliary statistics in both. As is standard, I weight moments by the inverse of the estimated variance-covariance matrix of the data moments, clustered by firm-CEO spell. Appendix IA2 gives the simulation design, weighting matrix, covariance estimation, and numerical implementation.

To implement the model empirically, I make several functional form assumptions and calibrate several parameters externally (see Table A.1). The governance wedge is parameterized as

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<sup>17</sup>Percentage effects on operating value and welfare are percentage changes in the discounted operating income of (3), in which the operating cost base scales with the CEO's input; they are not percentages of CEO pay. The pay effects of the counterfactuals do not depend on this mapping, and the value effects depend on it through the curvature  $\alpha$  alone, since  $\eta$  and  $\kappa$  are units of account (Appendix IA1.1).

$$\lambda_{\tau k} = \lambda_0 + \lambda_1 \tau + \gamma(\bar{Z}_k - \bar{Z}), \quad (15)$$

where  $\tau$  is CEO tenure and  $k$  indexes governance cells based on the board co-option measure of Coles et al. (2014), residualized with respect to tenure. Firms are split at the median of co-option, and the model is solved separately by cell. The three parameters,  $\lambda_0$ ,  $\lambda_1$  and  $\gamma$ , are to be estimated.

Separation hazards  $f_\tau$  equal the empirical CEO separation rates by tenure. Because failed SOP votes are followed by higher CEO turnover in the data, simulated separation hazards rise after a failed vote to reproduce this selection in CEO spells, but neither the Board nor the Shareholder prices this adjustment.<sup>18</sup> I set  $\kappa$ , a pay-scale normalization, at 0.366, at which the model reproduces average log pay to within one standard error given the estimated parameters, and set the discount factor to  $\beta = 0.90$ .

## 4.1. Identification strategy

I show there is a tight relation between the reduced-form moments and structural parameters. Panel A of Table IA.2 lists the targeted moments, Panel B reports pairwise correlations.

### 4.1.1. Output and CEO skill parameters

Because the structural model has neither firm nor time heterogeneity, I purge these components from observed revenues using a longer panel (1992–2024). I regress log revenue on firm, CEO and year effects, identifying firm effects from CEOs observed at more than one firm, and define  $\log y_{it}^{\text{data}}$  as log revenue net of the estimated firm and year effects, retaining the CEO component. In the structural model,  $\log y_{it} = \log \eta + a_i + \alpha \log w_{it} + \varepsilon_{yit}$ . Since  $\mathbb{E}[a_i]$  and  $\log \eta$  are not separately identified, I normalize  $\mu_0 = \mathbb{E}[a_i] = 0$  and target  $\mathbb{E}[\log y_{it}^{\text{data}}]$  to pin down

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<sup>18</sup>The post-failure increase in separation is imposed only when simulating CEO spells and does not enter either player's continuation value or decision rule. It therefore captures the observed change in the composition of surviving CEOs without introducing a causal channel through which either player can use the vote to force out a low-type CEO.

$\log \eta$ . To discipline  $(\alpha, \sigma_y, \sigma_0)$ , I estimate on the 2011–2024 sample

$$\log y_{it}^{\text{data}} = y_1 \log w_{it} + \mu_i^y + \epsilon_{it}^y. \quad (16)$$

I target  $\hat{y}_1$ ,  $\text{Var}(\hat{\epsilon}_{it}^y)$ , and  $\text{Var}(\hat{\mu}_i^y)$  using the same procedures in observed and simulated data. These moments discipline  $\alpha$ ,  $\sigma_y$ , and  $\sigma_0$ , respectively.<sup>19</sup>

#### 4.1.2. The Board-Shareholder game

The identification argument largely centers on matching equations that describe (in reduced-form) the model's equilibrium interaction between the Board and Shareholder.

**The probability of SOP failure.** In the model, the vote fails when the benchmark wage  $w_t^S$  falls below  $s_{it} w_{it}$ . Conditional on the Board's information after its signal,  $\mathcal{I}_{it}$ , and its wage choice, the probability of failure is

$$\Pr(\text{SOP fail}_{it} \mid \mathcal{I}_{it}, w_{it}) = \Phi\left(\frac{\log(s_{it} w_{it}) - m_{it}^S}{\sqrt{v_{it}^S}}\right), \quad m_{it}^S = \frac{\mu_{bit|z_b} + \log \frac{\alpha}{\kappa}}{1 - \alpha}, \quad v_{it}^S = \frac{\sigma_{bit|z_b}^2 + \sigma_{sy}^2}{(1 - \alpha)^2}, \quad (17)$$

with  $\sigma_{sy}^2$  the noise in the combined meeting signal and  $(\mu_{bit|z_b}, \sigma_{bit|z_b}^2)$  the Board's posterior. The wage enters only through  $\log(s_{it} w_{it})$ , so  $s_{it}$  shifts the location of the failure rule and  $v_{it}^S$  governs how steeply failure responds to pay.

The level of the failure rate is informative about the Shareholder's aversion to failure  $\chi_s$ : a higher  $\chi_s$  lowers the average screen  $s_{it}$  and the failure rate. The tenure profile carries further information. From (17), the failure rule steepens as  $\sigma_{b|z_b}^2(\tau)$  falls with tenure, so the rising failure rate reflects both this steepening and any tenure drift in the Board's proposal relative to the benchmark, captured by  $\lambda_1$  in (15). The model-implied determination of SOP failure maps to the probit regression

$$\Pr(\text{SOP fail}_{it} \mid w_{it}, \epsilon_{it}^y, \mu_i^y) = \Phi(s_0 + s_1 \log w_{it} + s_2 \epsilon_{it}^y + s_3 \mu_i^y) \quad (18)$$

<sup>19</sup>Pay is total compensation (TDC1), which values equity grants at the grant-date price. Appendix Table IA.1 shows that this price component does not drive  $\hat{y}_1$ : a 10% change in market value moves total pay by about 1% within-spell, and controlling for it changes  $\hat{y}_1$  by 1.6% of its value.

where  $\epsilon_{it}^y$  and  $\mu_i^y$  are the residual and the CEO effect from the output regression (16); the CEO effect is a control and  $s_3$  is not targeted.<sup>20</sup>

I target the average marginal effects of log compensation and the output residual. The sensitivity of failure to pay is informative mainly about private-signal precision and prior ability dispersion: from (17), the wage slope is  $\phi(\cdot)/\sqrt{v_{it}^S}$ , and  $v_{it}^S$  rises with  $\sigma_{z_s}$ , so a noisier Shareholder signal flattens the response of failure to pay.  $\chi_s$  also dampens this sensitivity but is disciplined mainly by the level and distribution of dissent. Including the output residual isolates variation attributable to private signal realizations as shareholders learn about CEO ability from output innovations.

**Co-option.** Governance varies across firms in the model only through the wedge, via  $\gamma$  in (15). I target three contrasts for firm-years above the median co-option index minus those at or below it, denoted  $\Delta^Z$ : the gap in the failure rate, the gap in the within-spell slope of the against-share on log pay, and the gap in the size-adjusted level of pay.<sup>21</sup> Because governance varies across cells only through the wedge, the failure gap is informative about  $\gamma$ , while the pay gap, which reflects how far the Board's response to the threat is tempered by its cost of failure, helps separate the wedge from the dissent-cost parameters.

**The Board's compensation decision.** The Board's pay choice depends on its beliefs about CEO ability, the governance wedge, and the anticipated cost of SOP failure. Abstracting from dynamics, its static program is

$$\max_{w_{it}} \mathbb{E}_{bt}[A_{it}]w_{it}^\alpha - \kappa(1 - \lambda_{\tau k})w_{it} - (\chi_s + \delta)\Pr(\text{SOP fail}_{it}),$$

which gives the approximation

$$(1 - \alpha) \log w_{it} \approx \log \frac{\alpha}{\kappa} + \log \mathbb{E}_{bt}[A_{it}] - \log(1 - \lambda_{\tau k}) - \frac{\chi_s + \delta}{\kappa(1 - \lambda_{\tau k})} \frac{\partial \Pr(\text{SOP fail}_{it} \mid \mathbf{I}_{it})}{\partial w_{it}}. \quad (19)$$

<sup>20</sup>Under Gaussian signals, posterior beliefs are affine in a persistent ability component, output innovations, and the Shareholder's private signal, and  $\log(s_{it} w_{it})$  is an equilibrium object measurable with respect to  $\mathbf{I}_{it}$ . The index is therefore linear in compensation and output innovations, with  $z_{sit}$  absorbed into the latent error.

<sup>21</sup>The pay level is the cell mean residual from pooled OLS of log pay on log sales within three-digit SIC.

Thus, compensation reflects beliefs about CEO ability, the governance wedge, and the marginal discipline from the vote. Given the skill parameters, the level of the wedge  $\lambda_0$  is informed by the output-wage elasticity and the level of failure, and mean log pay then sets the scale  $\kappa$ .

I also estimate

$$\log w_{it} = b_1 \mathbb{1}[\text{SOP fail}_{it-1}] + b_2 \tau_{it} + \mu_i^b + \epsilon_{it}^b. \quad (20)$$

The coefficient  $\hat{b}_1$  measures how the Board revises pay after learning from a failed vote. Because failure selects low realizations of the Shareholder's signal, this response is informative about the precision of  $z_{st}$  relative to  $z_{bt}$ , conditional on the equilibrium screen and wage proposal. I also match the variance of the CEO fixed effect, which is informative about signal precision through the dispersion of persistent Board beliefs.

**Dissent costs.** The Shareholder chooses the screen taking into account  $\chi_s$ , while the Board responds to failure through the total cost  $\chi_s + \delta$ . Hence the level and distribution of dissent discipline  $\chi_s$ , while compensation responds to the equilibrium threat of failure discipline the total cost and thereby  $\delta$ , particularly through the pay moments, co-option pay gap, and responses of pay to lagged failure, dissent and pay.

**Private information.** The Board and Shareholder signal variances are identified jointly from voting and wage moments. On the Shareholder side, residual co-movement between failure and the against-share, conditional on (18), is informative about  $\sigma_{z_s}$  and  $\sigma_\varepsilon$ ; Appendix IA2.3 gives the construction. Failure sensitivities to pay and output provide additional information through the steepness of the voting rule. On the Board side, more precise private information increases the dispersion of persistent beliefs and CEO wage effects. The post-failure wage response  $\hat{b}_1$ , together with the failure-output slope, helps separate  $\sigma_{z_b}$  from  $\sigma_{z_s}$  by measuring how strongly the Board updates when Shareholder information is revealed.

## 4.2. Estimated model fit

Table 2 shows the model fit. The model matches most moments closely, with meaningful misses concentrated in three moments: the variance of the CEO effect in output (moment 4),

the share of votes with 20% to 30% dissent (moment 14), and the sensitivity of SOP failure to output shocks (moment 16). I discuss these misses with the corresponding moment blocks.

|                    |
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| Table 2 about here |
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**Output and CEO skill moments.** The output moments are broadly matched. Average log output and the wage elasticity are close in the data and model; I interpret the latter as a reduced-form elasticity because wages respond to beliefs formed from past output. The main miss is the variance of the CEO fixed effect, 0.641 in the model versus 0.450 in the data. This likely reflects the empirical decomposition of firm and CEO effects, whereas the model has no firm fixed effect and assigns persistent match heterogeneity to the CEO spell.

**Shareholder moments.** The Shareholder moments are generally well matched. The model reproduces the tenure profile of SOP failure, the sensitivity of failure to compensation, the co-option gaps, and the failure rate near the 30% dissent threshold. The main miss is the sensitivity of failure to output innovations,  $-0.045$  in the data versus  $-0.091$  in the model. This is consistent with the model’s parsimonious information structure: current output is the only public performance signal, whereas shareholders in the data can condition on a broader information set. This miss is unlikely to drive the counterfactuals, which depend on the failure response to pay, the costs of failure, and the information revealed by failure, all of which are closely matched. Excess-dissent dispersion (moment 10) is also slightly lower in the model.<sup>22</sup>

**Board moments.** The Board moments fit closely. The model matches the level and dispersion of compensation, the post-failure wage response, the response of pay to lagged dissent, the autocovariance of log pay, and the change in pay after failed relative to passed votes (moments 18, 19 and 21–24). The main miss is the size-adjusted pay gap across co-option cells (moment 20), 0.098 in the data versus 0.033 in the model, although the corresponding differences in failure and the dissent–pay slope are matched.<sup>23</sup> The remaining gap plausibly reflects determinants of pay at co-opted boards the model omits.

<sup>22</sup>Appendix Figure A.2 shows that the model matches the distribution of observed dissent well, beyond the moments targeted in the estimation.

<sup>23</sup>This does not impact full-sample counterfactuals studied below as the cell shift  $\gamma(\bar{Z}_k - \bar{Z})$  is mean-zero across cells by construction (Appendix IA1.1); only the co-option splits are affected.

#### 4.2.1. Parameter–moment sensitivity

The moment sensitivities of Andrews et al. (2017), the change in each estimate in standard-error units from a one-standard-error perturbation of each targeted moment (Figure IA.2, Internet Appendix), support the identification argument.  $\chi_s$  responds most to the level and distribution of dissent and  $\delta$  to the wage moments;  $\gamma$  to the failure gap across governance cells;  $\lambda_1$  to the late-tenure failure rate; and  $\sigma_{z_s}$  and  $\sigma_{z_b}$  load with opposite signs on the failure slopes and on the wage moments, which separates Board from Shareholder information. Internet Appendix IA2.3 gives the full analysis.

#### 4.2.2. Model fit: Changes in CEO pay after shareholder disapproval

Figure 2 compares changes in CEO compensation from  $t$  to  $t+1$  across SOP disapproval rates in year  $t$ . The relation is similar in the observed and simulated data. In the model, greater dissent reveals more negative shareholder information about CEO type, leading the Board to revise beliefs and reduce subsequent compensation. The figure shows that this revision channel is also present in the data. The slope in the figure is not targeted, but it is not independent of the targeted moments, since the estimation targets the related responses of pay to lagged failure and lagged dissent and the change in pay after a failed vote (moments 21, 22 and 24).

Figure 2 about here

## 5. Results

This section presents the estimated structural parameters and then quantifies how the voting institution affects compensation, operating value, and shareholder welfare.

### 5.1. Estimated structural parameters and economic implications

Table 3 reports the estimated parameters. I estimate positive costs of SOP failure for both shareholders and the Board. The Shareholder cost,  $\chi_s = 0.472$ , governs the willingness to dissent. The Board internalizes this cost and an additional private cost,  $\delta = 0.274$ . Because these are utility costs rather than mechanical reductions in firm value, their magnitude depends on

the normalization. Panel B translates them into annual flow units. Per failed vote, the Shareholder cost is 19.5% of annual operating income and the Board’s incremental cost is 48.2% of annual captured rent; weighting by the equilibrium failure rate gives annual expected costs of 1.33% and 3.27%. Consistent with a director-side cost, compensation-committee directors are 4.4 percentage points more likely to leave the board within three years of a failed vote (Table A.2).<sup>24</sup>

Table 3 about here

The estimates also imply a positive Board–shareholder wedge in pay-setting, but little variation over CEO tenure:  $\lambda_0 = 0.191$  and  $\lambda_1 = 0.0024$ . The more important variation is cross-sectional. The wedge is larger at more co-opted boards in (15), where co-option is measured using Coles et al. (2014) and residualized with respect to CEO tenure. The gap between the more and less co-opted halves of the sample is 0.033. The three co-option contrasts are targeted, so this pattern is a fit rather than a test. What the estimate establishes is the sign and size of the wedge needed to reproduce them: the wedge is larger where the Coles et al. (2014) measure of CEO influence over the board is higher, consistent with its interpretation as Board capture. The untargeted evidence on the Board’s side is Table A.2, which documents the director career consequences of failure that the estimated  $\delta$  requires.

The estimated signal variances imply economically meaningful private information on both sides. Board and Shareholder signal noise ( $\sigma_{z_b} = 0.876$ ,  $\sigma_{z_s} = 0.583$ ) is comparable to the prior dispersion of CEO ability, so each signal substantially reduces uncertainty. Learning is concentrated early in CEO tenure: the standard deviation of beliefs falls from 0.763 at hire to 0.15 by the median tenure of seven years. The estimates imply substantial persistent heterogeneity in CEO productivity ( $\sigma_0 = 0.763$ ) together with sizable transitory output volatility ( $\sigma_y = 0.795$ ). The curvature parameter  $\alpha = 0.500$  governs how compensation maps into output in the production technology.

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<sup>24</sup>Direct evidence on the Shareholder cost is more limited. Edmans et al. (2023) survey institutional investors and find that they may be reluctant to vote against compensation even when they disagree with aspects of the proposal, consistent with public dissent itself being costly.

## 5.2. How much does Say-on-Pay affect compensation and value?

I re-solve the model removing the voting institution, holding the remaining structural parameters fixed. The Board no longer faces the threat of failure. The comparison therefore isolates SOP's disciplinary channel from its informational role.<sup>25</sup> Table 4 reports changes in compensation and value. Removing SOP would raise mean total compensation by 11.9%. This magnitude is broadly consistent with [Correa and Lel \(2016\)](#), who find that CEO pay falls by roughly 7% following SOP adoption in a cross-country setting.

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| Table 4 about here |
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The effect is heterogeneous. SOP has a larger effect on compensation at more co-opted boards (12.28% versus 9.01%) and for CEOs late in tenure (14.07% versus 7.67%). The co-option difference reflects the larger estimated governance wedge at more co-opted boards. Across tenure, learning makes shareholder beliefs more precise and the failure rule steeper, increasing the disciplinary effect of the vote even though the estimated tenure drift in the wedge is small. The model implies that SOP raises operating value by 1.56% on average and Shareholder welfare, which nets out the costs of dissent, by 0.51%.<sup>26</sup> The welfare effect is 0.62% for observations above the median baseline probability of failure and 0.39% for those below it.<sup>27</sup>

More broadly, the results speak to shareholder voting when proposals are endogenous to anticipated dissent. In votes on director elections ([Aggarwal et al., 2019](#)), major transactions ([Becht et al., 2016](#)), and equity issuance ([Holderness, 2018](#)), the policy put to shareholders, or whether it reaches a vote, may already reflect that threat. In my setting, the Board similarly adjusts compensation to manage failure risk. High pass rates can therefore coexist with effective discipline because voting operates partly through the proposals selected ex ante, not only through realized rejection.

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<sup>25</sup>In untabulated results, further restricting information makes the effects on operating value and Shareholder welfare substantially larger.

<sup>26</sup>The operating value effect is larger than the savings in CEO compensation alone, because pay also changes output and the operating costs that scale with the CEO's input through the technology in (3).

<sup>27</sup>Table A.3 shows 95% parameter-uncertainty bands, drawing parameter vectors from the joint parameter distribution and re-solving the model at each draw. Every cell retains its sign in every draw: the vote lowers pay and raises operating value and Shareholder welfare throughout the sampling distribution of the estimator.

### 5.3. Robustness

**Majority-defeat threshold.** Table [IA.3](#) re-estimates the model defining failure at 50% dissent and rebuilding all failure-based moments at that threshold. The estimates are very similar to the baseline: 0.464 versus 0.472 for the Shareholder cost, 0.292 versus 0.274 for the Board cost, and similarly for  $\lambda_0$  and  $\gamma$ . Stability arises because moving the threshold changes which votes are failures, but not the underlying incentives governing dissent and pay.

**No Board capture.** Table [IA.4](#) re-estimates the model imposing  $\lambda_0 = \lambda_1 = \gamma = 0$ . This model fits much worse, with  $J = 1576.1$  against 441.4 in the main estimate, and does so only at extreme parameter values: the Shareholder’s dissent cost falls to zero and the Board’s signal becomes 20 times noisier than the Shareholder’s. Thus, the no-capture model is strongly rejected by the data and requires implausible parameters to preserve observed failure rates.

**Passive ownership.** Table [IA.5](#) re-estimates the model above and below median ownership of Vanguard, BlackRock, and State Street. The Shareholder’s estimated cost of dissent is substantially higher in the high-passive-ownership subsample, while the Board’s total cost is similar (0.83 versus 0.76). This may reflect that passive institutions have weaker incentives to incur costly monitoring and engagement because they capture only part of the benefits (e.g., [Bebchuk and Hirst, 2019](#); [Heath et al., 2022](#)). More broadly, the split suggests that the estimated cost reflects meaningful frictions in exercising voice.

## 6. Counterfactual analysis of binding voting

Section [5](#) shows that advisory SOP disciplines compensation through the anticipation of shareholder dissent. I now ask whether attaching a binding consequence to rejection creates additional value, and whether that value depends on the consequence. I consider two designs. Under binding *reversion*, the design of the UK’s binding policy vote and the EU directive’s default, a failed vote restores the previously approved wage; under binding *revision*, it requires the Board to revise the proposal using the shareholder information revealed by the vote.

The key distinction between the two designs is what rejection implements. Under *reversion* it is an inherited wage, so today’s proposal also shapes tomorrow’s fallback. A setter’s

power depends on the reversion level (Romer and Rosenthal, 1978, 1979), and here the setter chooses it: today's approved wage is tomorrow's reversion level (Ingberman, 1985; Baron, 1996; Diermeier and Fong, 2011; Barseghyan and Coate, 2014). This creates a commitment problem for shareholders: the screen that is optimal given an inherited fallback does not generate the sequence of policies shareholders prefer ex ante.

## 6.1. The two binding designs

Figure A.1 compares the timing of the two designs with the advisory baseline. Under the advisory vote, compensation has already been implemented when shareholders vote, so failure affects future compensation only through the information revealed by the vote. Under either binding design, the vote instead determines the wage implemented in the current period.

As in the baseline model, the Shareholder chooses the screen  $s_t$  and the Board then observes  $z_{bt}$  and proposes  $w_t^{\text{pass}}$ . At the meeting, shareholders observe their information and vote. The implemented wage is

$$w_t = \begin{cases} w_t^{\text{pass}} & \text{if SOP pass,} \\ w_{t-1} & \text{if SOP fail, binding reversion,} \\ w_t^{\text{rev}} & \text{if SOP fail, binding revision,} \end{cases} \quad (21)$$

where

$$w_t^{\text{rev}} = w_t^{\text{pass}} e^{\Delta_t},$$

and  $\Delta_t$  is the log revision in compensation implied by updating beliefs about CEO ability conditional on vote failure (Appendix B).<sup>28</sup> Binding revision therefore incorporates the information revealed by shareholder dissent into the wage implemented in the current period, while retaining the Board's pay-setting wedge.

In either design, failure continues to impose costs  $\chi_s$  on shareholders and  $\chi_s + \delta$  on the Board. The model is re-solved under each rule. Under binding revision, the inherited wage is not a state variable and the equilibrium retains the state-by-state structure of Section 3.5.

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<sup>28</sup>To be clear,  $\Delta_t$  varies by state: it is not a fixed constant in the counterfactual.

Under binding reversion, the inherited wage becomes a state because today’s proposal determines tomorrow’s fallback. Appendix B gives the solution method for each counterfactual.

## 6.2. Results

Table 5 reports the two designs relative to advisory SOP. Binding reversion raises operating value by 0.69%: the veto constrains the Board’s preferred compensation, and the gain comes from the more co-opted boards, where the wedge is larger. But it produces over-dissent, dissent beyond what the Shareholder would choose if she could commit across meetings. The failure rate more than doubles, from 6.0% to 15.0%, and the cost of that dissent more than offsets the gain: Shareholder welfare falls by 1.21%, below the no-vote benchmark.<sup>29</sup> The veto does not tame the Board’s proposals, which rise by 2.94%; it contests them. Binding revision instead raises Shareholder welfare by 0.24% and operating value by 0.58%, with a failure rate of 7.3%. Average compensation is 0.67% higher than under the advisory vote. Thus, the gain from revision does not come from mechanically reducing pay, but from using shareholder information to improve the wage implemented after dissent.

Table 5 about here

The difference between the two designs comes from the consequence attached to failure. Table A.5 decomposes the welfare effects by first imposing the new failure consequence while holding advisory policies fixed, then allowing the Board to revise its proposal, and finally allowing the Shareholder to re-optimize its screen. Under binding reversion, the mechanical effect of reversion is essentially zero and the Board’s response adds 0.03%, while the final step lowers Shareholder welfare by 1.31%. Thus, the welfare loss arises from the equilibrium response of the voting screen and the Board’s proposal to the inherited fallback, rather than from reversion itself: the ratchet and the gain from rejection of Proposition 1.

<sup>29</sup>Binding votes on pay amounts rarely fail in practice: [Wagner and Wenk \(2026\)](#) report three rejections in the first years of the Swiss regime. Such restraint is consistent with the model’s commitment logic. Markov reversion leaves the Shareholder below the no-vote benchmark, while commitment to the advisory screen lowers failure from 15% to 6% and raises welfare (Appendix B). A literal binding vote would also replace the advisory 30% dissent threshold with majority rejection. Finally, the Swiss designs do not make the Board’s last approved proposal the fallback: a rejected retrospective amount is not paid, while the prospective vote approves a budget for the coming year, so both are state-free in the sense of Proposition 1(c).

The welfare loss under reversion comes from the inability to commit across meetings. If shareholders can commit to applying in future meetings the same voting rule they would use under the advisory regime, the over-dissent disappears and reversion becomes welfare-improving. At a new match, committed reversion is worth 0.7 to 1.0% more than the Markov regime and exceeds both the advisory and no-vote benchmarks; on the stock convention of Table 5 it lies 0.04% above the advisory vote (Appendix B). Thus, it is not binding approval itself that lowers welfare under reversion, but the interaction between binding approval and the endogenous future status quo.

Under binding revision, the failure consequence initially raises Shareholder welfare by 0.76%. The Board's proposal response gives back 0.69%, while the Shareholder's subsequent voting response restores 0.16%. The remaining gain is informational: failure causes the current wage to incorporate information that otherwise affects compensation only in future periods. Revision therefore preserves a binding consequence of dissent without giving the Board intertemporal control of the status quo.

Board control of the fallback drives the reversion loss. When failure instead implements a lagged wage that the Board cannot manipulate through its current proposal, shareholder welfare rises by 0.71%, even more than under revision (Appendix Table A.5, Panel C). Under revision, a higher proposal also raises the wage implemented after failure, so the Board can undo much of the mechanical gain. With a fixed fallback, a higher proposal only raises the probability of rejection.

Dynamic bargaining models show that an endogenous status quo can either exacerbate conflict or improve welfare (Dziuda and Loeper, 2016; Bowen et al., 2014). Figure 3 locates the estimated governance wedge within this tradeoff. I vary the governance wedge  $\lambda_0$  while holding the other primitives fixed. Near the estimated wedge, binding reversion lowers Shareholder welfare, but it becomes beneficial as Board-shareholder disagreement grows large: a larger wedge raises the value of constraining the Board's preferred compensation enough to offset the dynamic cost created by its control of the reversion point. Within the range spanned by the two co-option groups in the data, binding revision continues to dominate the advisory vote, while reversion becomes attractive only at larger wedges.

Figure 3 about here

The main conclusions are also present when I account for parameter uncertainty. Binding revision raises Shareholder welfare in every parameter draw, and binding reversion lowers it in at least 99 of 100 draws, for the full sample and for each subsample. The 95 percent band on the reversion loss runs from  $-2.05\%$  to  $-0.27\%$ . The ranking of the three regimes also holds at parameter vectors moved toward the two fit misses most directly related to the counterfactual mechanisms: the failure–output slope, which governs the information a vote reveals, and the co-option pay gap, which governs the wedge the vote disciplines (Appendix Table A.6). Appendix Tables A.4 and A.5 report these parameter-uncertainty draws and the decomposition of the design effects.

As with any regime counterfactual, I hold the structural primitives fixed when changing the voting rule. A binding regime could itself alter the costs of dissent, engagement practices, or the Board–shareholder relationship. The results should therefore be interpreted as the effect of changing the consequence of failure under the preferences and information structure estimated from the advisory regime.

## 7. Conclusion

Say-on-Pay is an advisory shareholder vote on a Board-sponsored policy. Because the proposal itself is chosen in anticipation of shareholder dissent, observed vote outcomes do not reveal how much discipline the vote creates ex ante. I estimate a dynamic model of the Board’s pay decision and the Shareholder’s vote to recover that discipline. The estimates imply that the Board and shareholders disagree over the appropriate level of pay, that failed votes are costly to both sides, and that the threat of dissent materially changes the compensation policy brought to the vote. Importantly, the voting institution itself is valuable to shareholders.

The structural model also lets me change what happens after shareholders reject the Board’s proposal, holding preferences and information fixed. Under binding revision, information revealed by dissent affects the policy implemented today, and shareholder welfare rises. Binding reversion raises operating value, but today’s approved policy becomes tomorrow’s fallback, so the Board shapes the state against which shareholders vote. The result is

over-dissent: failures more than double, their cost more than offsets the gain, and shareholders end up below the no-vote benchmark.

The broader implication is that the consequence of shareholder dissent is itself a governance design choice. It changes both the proposal the Board chooses ex ante and the Shareholder's willingness to reject it. The ex ante discipline of a vote applies to any Board-sponsored proposal. The dynamic agenda-setting distortion requires more: a recurring vote whose fallback is the last approved decision, as in pay-policy votes and equity-plan share authorizations, and not a one-shot vote with an exogenous fallback, such as a merger.

One future direction is to introduce heterogeneous shareholders with private information and strategic voting, as in [Barry et al. \(2026b\)](#), while allowing the Board's proposal to be endogenous to those strategic voting incentives, as in [Bond and Eraslan \(2010\)](#). This would allow the model to study how information aggregation among shareholders interacts with the ex ante discipline imposed on the policy setter.

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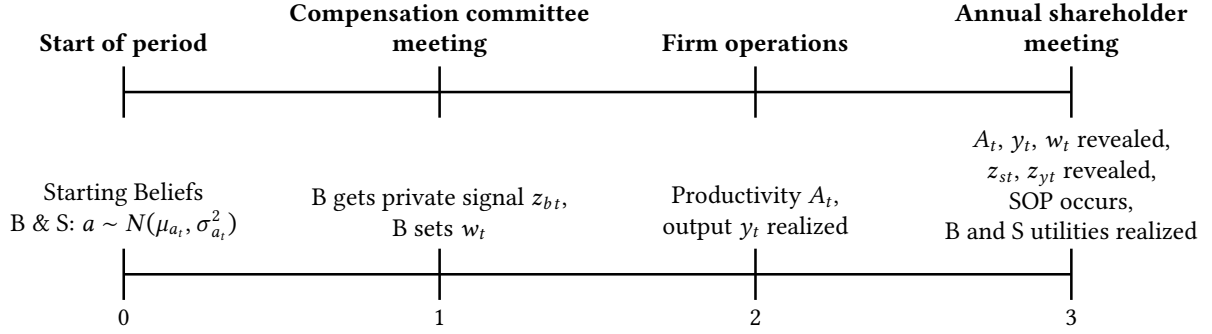
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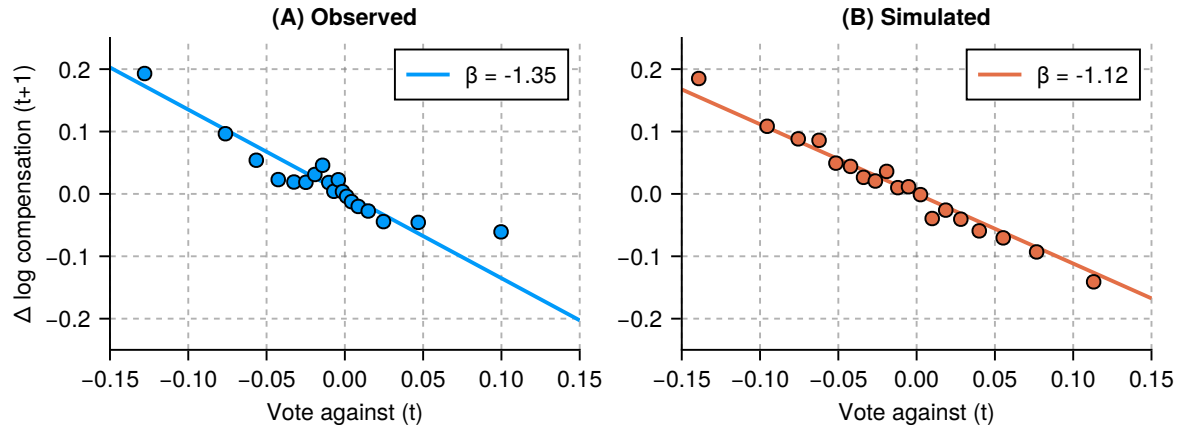
**Figure 1. Model timeline**

This figure displays the within-period model timeline. The top timeline maps the model to practice. The bottom timeline reports the sequence of events within each period  $t$ . Beliefs about CEO ability update with the year's output and the signals revealed at the meeting. Under the advisory vote, the wage is not carried into  $t + 1$  as a state. Appendix IA1.4 provides a detailed timeline of the Board's and Shareholder's strategies.

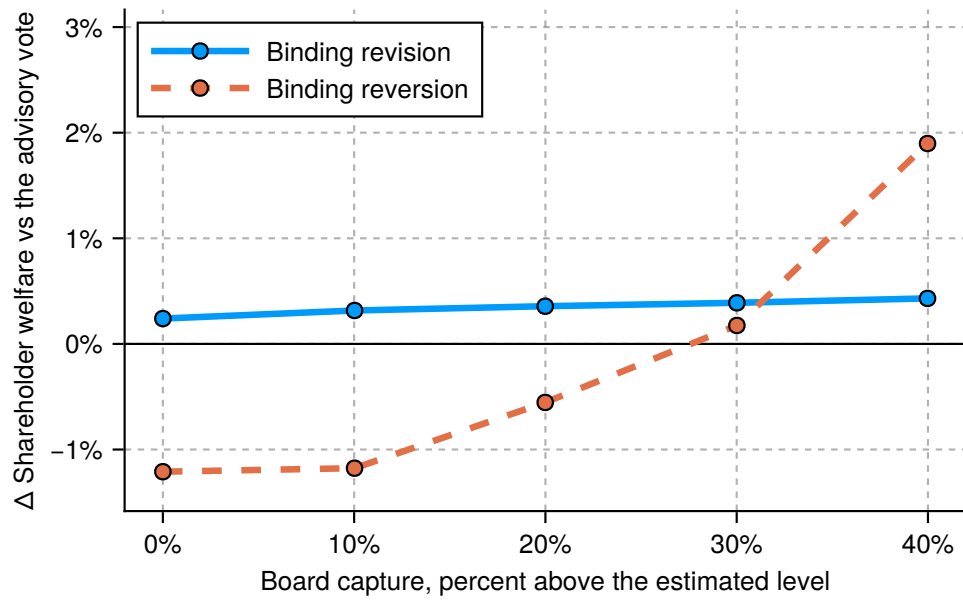


**Figure 2.** Model fit: The relation between changes in CEO pay and SOP vote outcomes

This figure displays the relation between changes in CEO pay and the SOP vote outcome in the observed and simulated data. Each panel displays a binned scatterplot from a regression of the change in log CEO pay from  $t$  to  $t + 1$  on the against-share of the SOP vote at  $t$ . Both variables are residualized on log revenue and CEO-spell fixed effects and grouped into twenty quantile bins. Panel A reports the observed panel. Panel B reports the dynamic model's simulated panel at the estimated parameter vector. The slope reported in each panel is the coefficient on the against-share, with standard errors clustered by CEO spell. The slope is not targeted.



**Figure 3.** Counterfactual shareholder welfare effects of binding votes across board capture  
This figure varies  $\lambda_0$  from its estimate, 0.191, holding all other estimated primitives fixed. The vertical axis reports Shareholder welfare, operating value net of the Shareholder's costs of dissent, in percentage points relative to an advisory SOP solved at the same altered  $\lambda_0$ .



**Table 1.** Summary statistics

This table reports descriptive statistics for the variables used in the model estimation. The sample merges Compustat, Execucomp, and ISS for fiscal years 2011–2024. All dollar variables are deflated to 2018 dollars. “Vote against SOP” is the fraction of votes cast against the Say-on-Pay proposal. “SOP failure” and the other indented rows are indicators equal to one when the vote-against share exceeds 30%, 20%, and 50%, respectively. “Total compensation” is total expected pay in millions, measured by TDC1. “Current compensation share” is current compensation divided by total compensation. “Change in total pay” is the change in log total compensation from the prior fiscal year within a CEO spell. “CEO tenure” is the number of years since the CEO assumed office. “Length of CEO tenure” is the total duration of the CEO’s spell at the firm, with one observation per CEO–firm spell. “Assets”, “Revenues”, and “Market capitalization” are in billions. Market capitalization is measured at the end of the fiscal year. “Return on assets” is operating income divided by assets. “Profit margin” is operating income divided by sales.

| Variable                              | Unit  | N      | Mean   | Std dev | 25%    | 50%   | 75%    |
|---------------------------------------|-------|--------|--------|---------|--------|-------|--------|
| <b>Say-on-Pay</b>                     |       |        |        |         |        |       |        |
| Vote against SOP (%)                  | %     | 13,109 | 0.082  | 0.099   | 0.025  | 0.046 | 0.085  |
| SOP failure: 1[More than 30% against] | {0,1} | 13,109 | 0.061  |         |        |       |        |
| 1[More than 20% against]              | {0,1} | 13,109 | 0.105  |         |        |       |        |
| 1[More than 50% against]              | {0,1} | 13,109 | 0.018  |         |        |       |        |
| <b>CEO</b>                            |       |        |        |         |        |       |        |
| Total compensation (TDC1)             | \$mil | 13,109 | 7.469  | 7.815   | 3.041  | 5.643 | 9.800  |
| Current compensation share            | %     | 13,109 | 0.228  | 0.188   | 0.111  | 0.166 | 0.269  |
| Change in total pay                   | %     | 11,659 | 0.019  | 0.420   | -0.101 | 0.016 | 0.177  |
| CEO tenure                            | years | 13,109 | 6.278  | 5.733   | 2      | 5     | 9      |
| Length of CEO tenure                  | years | 2,324  | 8.725  | 6.159   | 4      | 7     | 12     |
| <b>Firm</b>                           |       |        |        |         |        |       |        |
| Assets                                | \$bil | 13,109 | 31.392 | 162.531 | 1.281  | 4.084 | 13.643 |
| Revenues                              | \$bil | 13,109 | 10.819 | 33.231  | 0.824  | 2.323 | 7.628  |
| Market capitalization                 | \$bil | 13,109 | 18.896 | 74.511  | 1.251  | 3.454 | 12.044 |
| Return on assets                      | %     | 13,109 | 0.124  | 0.090   | 0.068  | 0.117 | 0.170  |
| Profit margin                         | %     | 13,109 | 0.208  | 0.161   | 0.103  | 0.175 | 0.289  |

**Table 2.** Model fit

This table reports the estimated model's fit. I estimate the model in Section 3 by indirect inference, following the identification strategy in Section 4.1 and the procedure in Appendix IA2. The table reports the twenty-four targeted moments in the data and in the simulated model evaluated at the estimated parameter vector. Standard errors are in parentheses beneath the observed values. "Difference" is the observed moment minus the simulated moment. The final column ( $t$ -stat) is the simulated moment minus the observed moment divided by the observed moment's standard error. The SMM objective value is 441.4.

**Panel A.** Output and CEO skill

| Description                  | Observed         | Simulated | Difference | $t$ -stat |
|------------------------------|------------------|-----------|------------|-----------|
| (1) Mean log output          | 7.859<br>(0.038) | 7.910     | -0.052     | 1.353     |
| (2) Output-wage elasticity   | 0.284<br>(0.017) | 0.274     | 0.010      | -0.591    |
| (3) Output residual variance | 0.471<br>(0.010) | 0.518     | -0.047     | 4.792     |
| (4) Output FE variance       | 0.450<br>(0.019) | 0.641     | -0.191     | 9.930     |

**Panel B. Shareholder and Board**

| <b>Description</b>                      | <b>Observed</b>   | <b>Simulated</b> | <b>Difference</b> | <b><i>t</i>-stat</b> |
|-----------------------------------------|-------------------|------------------|-------------------|----------------------|
| <b>Shareholder</b>                      |                   |                  |                   |                      |
| (5) Failure rate, tenure $\leq 7$       | 0.054<br>(0.003)  | 0.058            | -0.003            | 1.100                |
| (6) Failure rate, tenure 8–15           | 0.066<br>(0.006)  | 0.081            | -0.015            | 2.559                |
| (7) Failure rate, tenure $\geq 16$      | 0.099<br>(0.013)  | 0.104            | -0.006            | 0.446                |
| (8) Failure rate, co-option gap         | 0.033<br>(0.006)  | 0.028            | 0.005             | -0.875               |
| (9) Failure rate, 25–35% dissent        | 0.452<br>(0.023)  | 0.424            | 0.028             | -1.178               |
| (10) Failure–dissent covariance         | 0.119<br>(0.006)  | 0.106            | 0.013             | -2.014               |
| (11) Mean dissent, pass                 | 0.061<br>(0.001)  | 0.062            | -0.001            | 1.741                |
| (12) Mean dissent, fail                 | 0.455<br>(0.005)  | 0.434            | 0.021             | -4.241               |
| (13) Votes below 5% dissent             | 0.534<br>(0.007)  | 0.551            | -0.017            | 2.286                |
| (14) Votes with 20–30% dissent          | 0.044<br>(0.002)  | 0.066            | -0.023            | 11.068               |
| (15) Failure–wage coef                  | 0.058<br>(0.006)  | 0.058            | 0.000             | -0.029               |
| (16) Failure–output shock coef          | -0.045<br>(0.006) | -0.091           | 0.047             | -7.585               |
| (17) Dissent–pay slope, co-option       | 0.018<br>(0.007)  | 0.023            | -0.005            | 0.651                |
| <b>Board</b>                            |                   |                  |                   |                      |
| (18) Mean log wage                      | 1.659<br>(0.019)  | 1.677            | -0.019            | 0.993                |
| (19) Wage FE variance                   | 0.571<br>(0.016)  | 0.538            | 0.032             | -2.024               |
| (20) Wage residual, co-option           | 0.098<br>(0.018)  | 0.033            | 0.065             | -3.501               |
| (21) Log wage–lagged failure coef       | -0.104<br>(0.028) | -0.085           | -0.019            | 0.673                |
| (22) Log wage–lagged dissent coef       | -0.321<br>(0.065) | -0.313           | -0.008            | 0.126                |
| (23) Log wage autocovariance            | 0.594<br>(0.021)  | 0.621            | -0.027            | 1.317                |
| (24) $\Delta$ Log wage, fail minus pass | -0.351<br>(0.034) | -0.336           | -0.014            | 0.423                |

**Table 3. Estimated parameters**

This table reports estimates of the parameters in Section 3 using the estimation strategy in Section 4. Panel A reports each parameter estimate with its standard error in parentheses. Panel B expresses each dissent cost in flow units as a share of the annual flow that the party bearing the cost has at stake. The Shareholder's cost is divided by the firm's expected annual operating income,  $\Pi = \mathbb{E}[A_t w_t^\alpha - \kappa w_t]$ . The Board's incremental cost is divided by the annual rent the CEO captures through the pay-setting wedge,  $R = \mathbb{E}[\lambda_\tau \kappa w_t]$ . Both normalizers are evaluated in the estimated equilibrium on the estimation design. The first pair of columns reports the cost of one failed vote. The second pair reports the expected annual cost, calculated as the cost per failed vote multiplied by the equilibrium probability of failure.

| Panel A. Parameter estimates                |  |                      |  |                 |          |                   |         |
|---------------------------------------------|--|----------------------|--|-----------------|----------|-------------------|---------|
| Description                                 |  | Notation             |  | Value           |          |                   |         |
| Output & CEO skill                          |  |                      |  |                 |          |                   |         |
| Prior std dev of CEO ability                |  | $\sigma_0$           |  | 0.7635          | (0.0189) |                   |         |
| Std dev of productivity shock               |  | $\sigma_y$           |  | 0.7953          | (0.0075) |                   |         |
| Output–CEO wage elasticity                  |  | $\alpha$             |  | 0.5000          | (0.0115) |                   |         |
| Scaling factor (output)                     |  | $\log \eta$          |  | 7.0665          | (0.0406) |                   |         |
| Shareholder                                 |  |                      |  |                 |          |                   |         |
| Std dev of Shareholder signal               |  | $\sigma_{z_s}$       |  | 0.5826          | (0.0324) |                   |         |
| Shareholder SOP failure cost                |  | $\chi_s$             |  | 0.4720          | (0.2425) |                   |         |
| Std dev of dispersed shareholder thresholds |  | $\sigma_\varepsilon$ |  | 1.3715          | (0.0416) |                   |         |
| Board                                       |  |                      |  |                 |          |                   |         |
| Std dev of Board signal                     |  | $\sigma_{z_b}$       |  | 0.8759          | (0.1123) |                   |         |
| Board incremental private cost              |  | $\delta$             |  | 0.2743          | (0.1335) |                   |         |
| CEO board capture (constant)                |  | $\lambda_0$          |  | 0.1906          | (0.0403) |                   |         |
| CEO board capture (growth)                  |  | $\lambda_1$          |  | 0.0024          | (0.0009) |                   |         |
| CEO board capture (co-option)               |  | $\gamma$             |  | 0.2318          | (0.0611) |                   |         |
|                                             |  |                      |  |                 |          |                   |         |
| Panel B. Dissent costs in flow units        |  |                      |  |                 |          |                   |         |
| Description                                 |  | Notation             |  | Per failed vote |          | Expected per year |         |
| Shareholder cost                            |  | $\frac{\chi_s}{\Pi}$ |  | 19.50%          | (10.02%) | 1.33%             | (0.68%) |
| Annual op. inc.                             |  | $\Pi$                |  |                 |          |                   |         |
| Board incremental cost                      |  | $\frac{\delta}{R}$   |  | 48.20%          | (23.45%) | 3.27%             | (1.59%) |
| Annual captured rent                        |  | $R$                  |  |                 |          |                   |         |

**Table 4.** The impact of Say-on-Pay on compensation and shareholder welfare

This table reports the impact of Say-on-Pay on compensation policy, operating value, and shareholder welfare. I re-solve the model with the vote removed, holding every estimated primitive fixed, and report the effect of the advisory vote as the percentage difference between the estimated institution and the No-SOP economy. Effects are computed observation by observation on the advisory panel and random numbers are held constant across the advisory and no-SOP models. The first row reports the percentage change in the level of pay. The second row reports the change in operating value, defined as the operating value of the firm (discounted value of Eq. 3 at the Board's optimal strategy). The third row reports the change in Shareholder welfare, defined as operating value net of the Shareholder's aggregated costs of dissent (the value implied by Eq. 13). The fourth row reports the probability that the vote fails in the advisory baselines. The first column reports the full sample. The remaining columns split the sample at its median. Columns two and three split on the co-option index, the paper's measure of Board capture. Columns four and five split on CEO tenure. Columns six and seven split on the baseline probability of SOP failure.

|                     | (1)                | (2)              | (3)     | (4)           | (5)     | (6)             | (7)     |
|---------------------|--------------------|------------------|---------|---------------|---------|-----------------|---------|
|                     |                    | <b>Co-option</b> |         | <b>Tenure</b> |         | <b>Pr(Fail)</b> |         |
|                     | <b>Full sample</b> | Below            | Above   | Early         | Late    | Low             | High    |
| Pay level           | -10.64%            | -9.01%           | -12.28% | -7.67%        | -14.07% | -10.36%         | -10.93% |
| Operating value     | +1.56%             | +1.12%           | +2.00%  | +1.38%        | +1.76%  | +1.26%          | +1.85%  |
| Shareholder welfare | +0.51%             | +0.33%           | +0.69%  | +0.43%        | +0.60%  | +0.39%          | +0.62%  |
| Pr(Fail), baseline  | 6.02%              | 4.56%            | 7.48%   | 4.49%         | 7.78%   | 3.32%           | 8.72%   |

**Table 5.** Design counterfactuals: Binding votes relative to the advisory vote

This table reports counterfactual outcomes under the two binding designs of Section 6, each relative to the advisory vote in the United States. Under binding reversion, the UK design and the EU directive's default, a failed vote implements last year's approved pay. Under binding revision, a failed vote implements the Board's proposal revised by the information that the vote reveals. I re-solve the model under each design, holding every estimated primitive fixed. Each design is a Markov-perfect equilibrium. The sample, splits, and convention are those of Table 4. Entries are percentage differences from the advisory vote, computed observation by observation. The first three rows report changes in the level of pay, operating value, and Shareholder welfare. The last two rows report the probability of failure under the advisory vote and under the design. Panel A reports binding reversion. Panel B reports binding revision.

**Panel A. Binding reversion**

|                     | (1)                | (2)              | (3)    | (4)           | (5)    | (6)             | (7)    |
|---------------------|--------------------|------------------|--------|---------------|--------|-----------------|--------|
|                     |                    | <b>Co-option</b> |        | <b>Tenure</b> |        | <b>Pr(Fail)</b> |        |
|                     | <b>Full sample</b> | Below            | Above  | Early         | Late   | Low             | High   |
| Pay level           | +2.94%             | +3.81%           | +2.06% | +7.48%        | -2.29% | +4.52%          | +1.36% |
| Operating value     | +0.69%             | +0.12%           | +1.25% | +0.64%        | +0.74% | +0.64%          | +0.73% |
| Shareholder welfare | -1.21%             | -1.19%           | -1.23% | -1.34%        | -1.05% | -1.01%          | -1.40% |
| Pr(Fail), baseline  | 6.02%              | 4.56%            | 7.48%  | 4.49%         | 7.78%  | 3.32%           | 8.72%  |
| Pr(Fail)            | 14.98%             | 9.10%            | 20.85% | 14.93%        | 15.02% | 5.86%           | 24.10% |

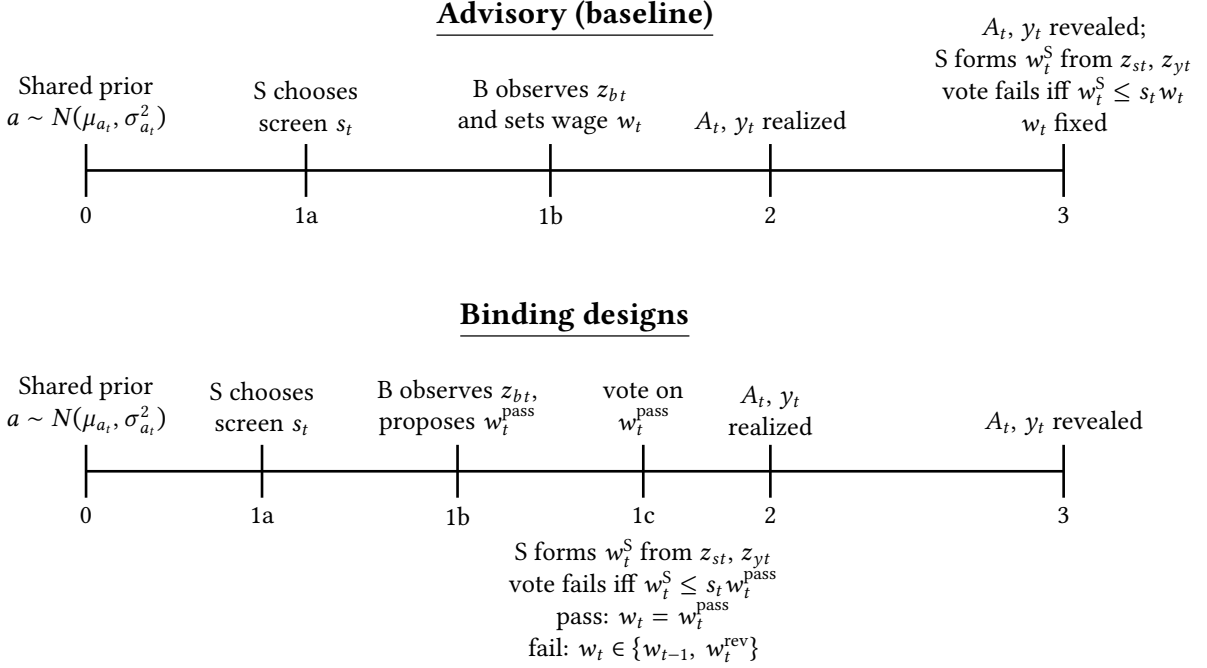
**Panel B. Binding revision**

|                     | (1)                | (2)              | (3)    | (4)           | (5)    | (6)             | (7)    |
|---------------------|--------------------|------------------|--------|---------------|--------|-----------------|--------|
|                     |                    | <b>Co-option</b> |        | <b>Tenure</b> |        | <b>Pr(Fail)</b> |        |
|                     | <b>Full sample</b> | Below            | Above  | Early         | Late   | Low             | High   |
| Pay level           | +0.67%             | +0.49%           | +0.85% | +0.96%        | +0.34% | +0.37%          | +0.97% |
| Operating value     | +0.58%             | +0.54%           | +0.62% | +0.62%        | +0.54% | +0.61%          | +0.55% |
| Shareholder welfare | +0.24%             | +0.22%           | +0.26% | +0.25%        | +0.23% | +0.25%          | +0.23% |
| Pr(Fail), baseline  | 6.02%              | 4.56%            | 7.48%  | 4.49%         | 7.78%  | 3.32%           | 8.72%  |
| Pr(Fail)            | 7.30%              | 5.71%            | 8.88%  | 6.59%         | 8.11%  | 4.38%           | 10.21% |

## A. Appendix figures and tables

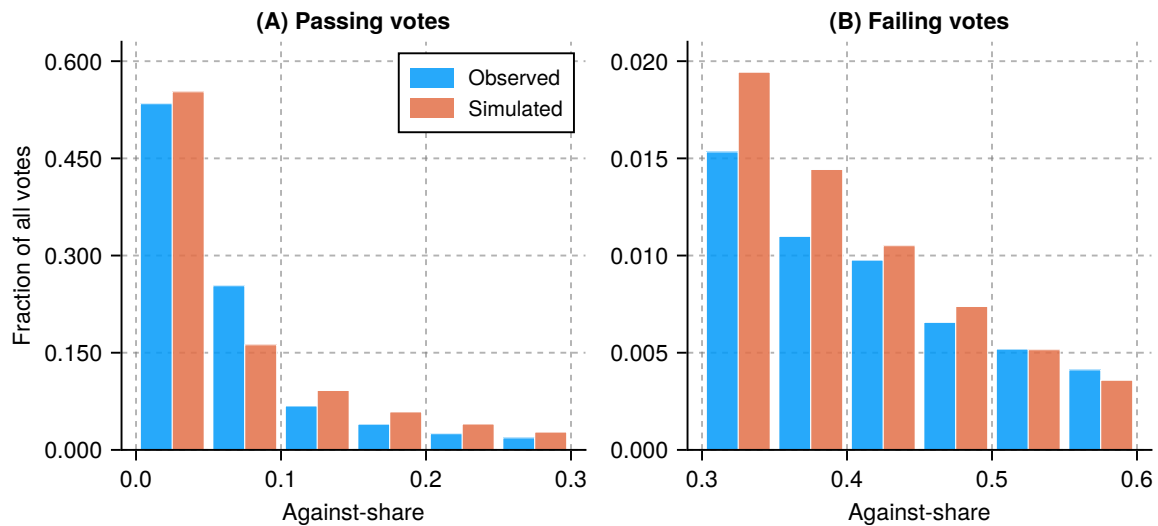
**Figure A.1.** Timing of the two binding designs

This figure shows the within-period timing of the two binding designs in Section 6, alongside the advisory baseline of Figure 1. Under the advisory vote, the Board sets pay before the annual meeting, so failure only generates the cost of dissent. Under either binding design, the Shareholder votes on the Board's proposal  $w_t^{\text{pass}}$  before pay is implemented. If the vote passes, the proposal is implemented. If it fails, binding reversion implements last year's approved pay,  $w_{t-1}$ , while binding revision implements  $w_t^{\text{rev}} = w_t^{\text{pass}} e^{\Delta_t}$ , incorporating the belief update revealed by failure. The Shareholder chooses her screen  $s_t$  state by state, and each design is a Markov-perfect equilibrium of the same model.



**Figure A.2. Model fit: The distribution of shareholder dissent**

This figure displays the distribution of the against-share in SOP votes in the observed and simulated data. Panel A reports passing votes, with an against-share below the 30% failure threshold. Panel B reports failing votes, with an against-share between 30% and 60%. Each bar is the fraction of all votes in a 5-percentage-point bin. The two panels use the same units and separate vertical scales. The observed sample has 13,109 votes, and the simulated data are five panels of 1,500 firms at the estimated parameter vector. Four features of the distribution are targeted: mean dissent among passing and failing votes, the share of votes below 5%, and the share between 20% and 30% dissent, and the proportion of votes above 30% dissent. The remaining bins are not targeted. Panel B omits votes with dissent above 60% for visual clarity, though the bins are well-matched.



**Table A.1.** Calibrated parameters

This table lists the parameters of the model that are calibrated rather than estimated, their values, and source for calibration.

| Description                      | Notation | Value     | Source and identifying object                                                                                                                                                                                                                         |
|----------------------------------|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pay scale                        | $\kappa$ | 0.366     | Unit of account for pay (Internet Appendix <a href="#">IA1.1</a> ): set at a value at which the model reproduces average log pay to within one standard error given the estimated parameters; no other moment depends on it.                          |
| Board discount factor            | $\beta$  | 0.9       | Set at the annual value used in the compensation literature; not identified by the targeted moments.                                                                                                                                                  |
| Prior mean of CEO ability        | $\mu_0$  | 0         | Normalization; the level of output is carried by $\log \eta$ .                                                                                                                                                                                        |
| CEO separation hazards           | $f_\tau$ | by tenure | Separation rates by tenure in the estimation sample, the share of CEO spells observed at tenure $\tau$ that end at $\tau$ , constructed as in <a href="#">Taylor (2010)</a> ; $f_0 = 0$ and $f_{25} = 1$ at the tenure cap; one-year spells excluded. |
| Hazard shift after a failed vote | –        | 0.15      | Chosen to reproduce the coefficient of a failed vote in a regression of CEO separation on failure, tenure, firm and year effects (0.033, s.e. 0.015); applied in the simulated panels, not priced by the Board.                                       |
| Failure threshold                | –        | 0.30      | The support level below which proxy advisors flag a vote and boards respond ( <a href="#">ISS, 2022</a> ; <a href="#">Dey et al., 2024</a> ); the failure indicator in the data and the screen event in the model.                                    |

**Table A.2.** Director career consequences of SOP failure

This table reports differences in director career outcomes following an SOP vote below 70% support, using the matched design of Fos and Tsoutsoura (2014) applied to SOP failures in BoardEx seat-level histories linked to failed-vote meetings from 2011–2024. Each failed-vote firm is matched to the same-industry (Fama–French 12) firm of closest market capitalization that year whose vote received at least 80% support, with no failure in the surrounding five-year window. The match conditions on industry, size, director characteristics, and assets but not on pre-event performance, and the control window includes post-event years. Treated and control directors are the compensation-committee members of the two firms at their meeting dates. The outcomes, by column, are departure from the event board, departure from its compensation committee, holding no listed board seat, holding no S&P 1500 seat, and holding no S&P 500 seat. Panel A measures the outcomes three years after the vote. Panel B measures them five years after the vote. Panel C reports specifications at the three-year horizon restricted to each firm’s first failure, restricted to directors under 72 at the event, and defining failure as a majority defeat. Coefficients are in percentage points, with control means and sample sizes below. Regressions control for the director’s age, board tenure, number of board seats, number of listed seats, and the firm’s log assets, with indicators for missing age and tenure and match-group fixed effects for each failed-vote firm and its matched control. Standard errors are two-way clustered by event board and director and reported in parentheses. The last rows report the within-group  $R^2$  and the specification. Directors who die within the horizon are excluded. \*\*\*, \*\*, \* denote significance at 1%, 5%, 10%.

|                                                                 | (1)<br>Off<br>board | (2)<br>Off comp.<br>committee | (3)<br>Holds no<br>listed seat | (4)<br>Holds no<br>S&P 1500 seat | (5)<br>Holds no<br>S&P 500 seat |
|-----------------------------------------------------------------|---------------------|-------------------------------|--------------------------------|----------------------------------|---------------------------------|
| <b>Panel A.</b> Three years after the vote ( $k = 3$ )          |                     |                               |                                |                                  |                                 |
| Failed vote (pp)                                                | 4.41***<br>(1.24)   | 4.41***<br>(1.29)             | 1.76*<br>(0.99)                | 7.82***<br>(1.70)                | 4.76***<br>(1.37)               |
| Control mean (%)                                                | 22.13               | 30.53                         | 16.65                          | 31.55                            | 66.95                           |
| $N$                                                             | 13,157              | 13,157                        | 13,157                         | 13,157                           | 13,157                          |
| $R^2$ (within)                                                  | 0.008               | 0.004                         | 0.019                          | 0.026                            | 0.075                           |
| <b>Panel B.</b> Five years after the vote ( $k = 5$ )           |                     |                               |                                |                                  |                                 |
| Failed vote (pp)                                                | 7.10***<br>(1.69)   | 6.64***<br>(1.61)             | 3.75***<br>(1.44)              | 7.54***<br>(1.72)                | 4.20***<br>(1.33)               |
| Control mean (%)                                                | 37.08               | 46.93                         | 26.95                          | 41.20                            | 72.10                           |
| $N$                                                             | 10,298              | 10,298                        | 10,298                         | 10,298                           | 10,298                          |
| $R^2$ (within)                                                  | 0.011               | 0.006                         | 0.027                          | 0.028                            | 0.065                           |
| <b>Panel C.</b> Robustness, failed-vote coefficient ( $k = 3$ ) |                     |                               |                                |                                  |                                 |
| First failure only                                              | 4.21***<br>(1.37)   | 4.38***<br>(1.49)             | 2.15**<br>(1.09)               | 6.30***<br>(1.61)                | 4.02***<br>(1.34)               |
| Age < 72 at event                                               | 5.05***<br>(1.27)   | 5.54***<br>(1.32)             | 2.17**<br>(0.93)               | 7.95***<br>(1.68)                | 5.10***<br>(1.47)               |
| Majority fail                                                   | 8.05***<br>(2.37)   | 10.35***<br>(2.43)            | 3.20*<br>(1.82)                | 9.38***<br>(2.98)                | 6.92***<br>(2.25)               |
| Director and firm controls                                      | Yes                 | Yes                           | Yes                            | Yes                              | Yes                             |
| Match-group fixed effects                                       | Yes                 | Yes                           | Yes                            | Yes                              | Yes                             |
| SE clustered by board and director                              | Yes                 | Yes                           | Yes                            | Yes                              | Yes                             |

**Table A.3.** Parameter uncertainty in the impact of Say-on-Pay

This table repeats Table 4 with 95% parameter-uncertainty intervals beneath each estimate. I draw 100 parameter vectors from the estimated joint asymptotic distribution using the seed-averaged covariance matrix and re-solve both the estimated and No-SOP economies at each draw. The reported intervals are the 2.5th and 97.5th percentiles of the resulting counterfactual outcomes. Draw 0 is the estimated parameter vector and reproduces the reported point estimates.

|                     | (1)                        | (2)                       | (3)                        | (4)                       | (5)                        | (6)                        | (7)                        |
|---------------------|----------------------------|---------------------------|----------------------------|---------------------------|----------------------------|----------------------------|----------------------------|
|                     |                            | <b>Co-option</b>          |                            | <b>Tenure</b>             |                            | <b>Pr(Fail)</b>            |                            |
|                     | <b>Full sample</b>         | Below                     | Above                      | Early                     | Late                       | Low                        | High                       |
| Pay level           | −10.64%<br>[−15.75, −3.81] | −9.01%<br>[−13.72, −3.13] | −12.28%<br>[−18.41, −4.48] | −7.67%<br>[−12.68, −2.36] | −14.07%<br>[−21.30, −5.23] | −10.36%<br>[−15.98, −2.80] | −10.92%<br>[−16.81, −4.23] |
| Operating value     | +1.56%<br>[+0.37, +4.55]   | +1.12%<br>[+0.25, +3.42]  | +2.00%<br>[+0.42, +5.67]   | +1.38%<br>[+0.34, +4.08]  | +1.76%<br>[+0.41, +5.09]   | +1.26%<br>[+0.27, +3.85]   | +1.85%<br>[+0.40, +5.25]   |
| Shareholder welfare | +0.51%<br>[+0.12, +1.53]   | +0.33%<br>[+0.08, +1.06]  | +0.69%<br>[+0.17, +2.00]   | +0.43%<br>[+0.10, +1.32]  | +0.60%<br>[+0.13, +1.77]   | +0.39%<br>[+0.09, +1.26]   | +0.62%<br>[+0.15, +1.80]   |
| Pr(Fail), baseline  | 6.02%<br>[4.32, 18.26]     | 4.56%<br>[2.93, 17.18]    | 7.48%<br>[5.40, 20.57]     | 4.49%<br>[2.82, 16.99]    | 7.78%<br>[5.82, 20.45]     | 3.32%<br>[1.85, 13.18]     | 8.72%<br>[5.94, 23.77]     |

**Table A.4.** Parameter uncertainty in the design counterfactuals

This table repeats Table 5 with 95 percent parameter-uncertainty bands beneath each entry, constructed as in Table A.3. Panel A reports binding reversion relative to the advisory vote. Panel B reports binding revision relative to the advisory vote.

| <b>Panel A.</b> Binding reversion relative to advisory |                           |                          |                           |                           |                          |                          |                           |
|--------------------------------------------------------|---------------------------|--------------------------|---------------------------|---------------------------|--------------------------|--------------------------|---------------------------|
|                                                        | (1)                       | (2)                      | (3)                       | (4)                       | (5)                      | (6)                      | (7)                       |
|                                                        |                           | <b>Co-option</b>         |                           | <b>Tenure</b>             |                          | <b>Pr(Fail)</b>          |                           |
|                                                        | <b>Full sample</b>        | Below                    | Above                     | Early                     | Late                     | Low                      | High                      |
| Pay level                                              | +2.94%<br>[+0.38, +10.36] | +3.81%<br>[+0.51, +9.90] | +2.06%<br>[−0.72, +11.56] | +7.48%<br>[+3.79, +14.90] | −2.29%<br>[−4.85, +5.71] | +4.52%<br>[+1.68, +8.70] | +1.36%<br>[−3.00, +14.25] |
| Operating value                                        | +0.69%<br>[−0.32, +3.72]  | +0.12%<br>[−0.36, +2.39] | +1.25%<br>[−0.34, +5.05]  | +0.64%<br>[−0.35, +3.68]  | +0.74%<br>[−0.23, +3.76] | +0.64%<br>[−0.32, +3.66] | +0.73%<br>[−0.32, +3.78]  |
| Shareholder welfare                                    | −1.21%<br>[−2.05, −0.27]  | −1.19%<br>[−2.01, −0.28] | −1.23%<br>[−2.32, −0.12]  | −1.34%<br>[−2.35, −0.34]  | −1.05%<br>[−1.70, −0.12] | −1.01%<br>[−1.86, −0.21] | −1.40%<br>[−2.23, −0.30]  |
| Pr(Fail), baseline                                     | 6.02%<br>[4.32, 18.26]    | 4.56%<br>[2.93, 17.18]   | 7.48%<br>[5.40, 20.57]    | 4.49%<br>[2.82, 16.99]    | 7.78%<br>[5.82, 20.45]   | 3.32%<br>[1.85, 13.18]   | 8.72%<br>[5.94, 23.77]    |
| Pr(Fail)                                               | 14.98%<br>[10.27, 50.19]  | 9.10%<br>[5.70, 46.26]   | 20.85%<br>[13.86, 54.12]  | 14.93%<br>[8.81, 41.14]   | 15.02%<br>[9.45, 56.90]  | 5.86%<br>[3.01, 26.78]   | 24.09%<br>[16.00, 73.34]  |

**Panel B.** Binding revision relative to advisory

|                     | (1)                      | (2)                      | (3)                      | (4)                      | (5)                      | (6)                      | (7)                      |
|---------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|                     |                          | <b>Co-option</b>         |                          | <b>Tenure</b>            |                          | <b>Pr(Fail)</b>          |                          |
|                     | <b>Full sample</b>       | Below                    | Above                    | Early                    | Late                     | Low                      | High                     |
| Pay level           | +0.67%<br>[+0.32, +3.18] | +0.49%<br>[+0.12, +2.96] | +0.85%<br>[+0.54, +3.41] | +0.96%<br>[+0.39, +4.61] | +0.34%<br>[+0.20, +1.53] | +0.37%<br>[+0.02, +2.54] | +0.97%<br>[+0.61, +4.35] |
| Operating value     | +0.58%<br>[+0.42, +0.68] | +0.54%<br>[+0.37, +0.63] | +0.62%<br>[+0.45, +0.73] | +0.62%<br>[+0.44, +0.72] | +0.54%<br>[+0.39, +0.63] | +0.61%<br>[+0.43, +0.72] | +0.55%<br>[+0.40, +0.64] |
| Shareholder welfare | +0.24%<br>[+0.10, +0.54] | +0.22%<br>[+0.07, +0.53] | +0.26%<br>[+0.12, +0.55] | +0.25%<br>[+0.10, +0.57] | +0.23%<br>[+0.10, +0.51] | +0.25%<br>[+0.09, +0.57] | +0.23%<br>[+0.10, +0.50] |
| Pr(Fail), baseline  | 6.02%<br>[4.32, 18.26]   | 4.56%<br>[2.93, 17.18]   | 7.48%<br>[5.40, 20.57]   | 4.49%<br>[2.82, 16.99]   | 7.78%<br>[5.82, 20.45]   | 3.32%<br>[1.85, 13.18]   | 8.72%<br>[5.94, 23.77]   |
| Pr(Fail)            | 7.30%<br>[5.77, 22.12]   | 5.71%<br>[4.06, 20.84]   | 8.88%<br>[7.35, 24.14]   | 6.59%<br>[4.51, 23.33]   | 8.11%<br>[6.29, 21.23]   | 4.38%<br>[2.87, 15.64]   | 10.21%<br>[7.87, 28.55]  |

**Table A.5.** Decomposing the binding designs' effects on shareholder welfare

This table decomposes each binding design's effect on Shareholder welfare into three steps. Entries are percentage points of the advisory value and are evaluated at the advisory panel's distribution of states, following the convention of Table 5. The first row imposes the design's consequence of failure while holding the Board's proposal policy and the Shareholder's voting rule at their advisory values. The second row adds the Board's proposals re-optimized against the new consequence while holding the Shareholder's advisory voting rule fixed. The third row adds the Shareholder's re-optimized screen and the Board's re-optimization against that screen, restoring the equilibrium of Table 5. The three steps add to the total by construction. Columns one and two split the sample at the median of the co-option index. Column three reports all firms. Panel A reports binding reversion. Panel B reports binding revision. Panel C compares binding reversion with an auxiliary fallback, a lagged wage outside the Board's control (last year's advisory-equilibrium wage), solved as a Markov-perfect equilibrium and reported for all firms relative to the advisory vote. Columns one and three of Panel C are ratios of panel-weighted mean values for existing matches, column two is the value at a new match, and column four is the failure probability on the same weights, which is 6.0% under the advisory vote.

**Panel A. Binding reversion**

|                                      | (1)              | (2)    | (3)              |
|--------------------------------------|------------------|--------|------------------|
|                                      | <b>Co-option</b> |        |                  |
|                                      | Below            | Above  | <b>All firms</b> |
| Mandated reversion alone             | −0.01%           | +0.02% | +0.00%           |
| Board's proposal response            | +0.03%           | +0.03% | +0.03%           |
| Shareholders' screen response        | −1.24%           | −1.37% | −1.31%           |
| Total: binding reversion vs advisory | −1.22%           | −1.32% | −1.27%           |

**Panel B. Binding revision**

|                                     | (1)              | (2)    | (3)              |
|-------------------------------------|------------------|--------|------------------|
|                                     | <b>Co-option</b> |        |                  |
|                                     | Below            | Above  | <b>All firms</b> |
| Mandated revision alone             | +0.57%           | +0.97% | +0.76%           |
| Board's proposal response           | −0.50%           | −0.88% | −0.69%           |
| Shareholders' screen response       | +0.15%           | +0.16% | +0.16%           |
| Total: binding revision vs advisory | +0.21%           | +0.26% | +0.23%           |

**Panel C. Lagged fallback with and without Board control**

|                                         | (1)                        | (2)       | (3)             | (4)      |
|-----------------------------------------|----------------------------|-----------|-----------------|----------|
|                                         | <b>Shareholder welfare</b> |           |                 |          |
|                                         | Existing matches           | New match | Operating value | Pr(Fail) |
| Wage implemented after a failed vote    |                            |           |                 |          |
| Lagged wage outside the Board's control | +0.71%                     | +0.91%    | +1.00%          | 6.2%     |
| Last approved wage (binding reversion)  | −1.27%                     | −0.81%    | +0.64%          | 15.0%    |

**Table A.6.** The counterfactuals at parameter vectors moved toward the fit misses most related to the counterfactual mechanisms

This table repeats the full-sample rows of Tables 4 and 5 at two alternative parameter vectors, each the first-order estimate of where the estimate would move if one targeted moment alone sat at its data value:  $\hat{\theta}_k = \hat{\theta} + f_k \Lambda_k (m_k^{\text{data}} - m_k^{\text{model}})$ , with  $\Lambda = (G'WG)^{-1}G'W$  the sensitivity matrix of Andrews et al. (2017) behind Figure IA.2 and  $f_k \leq 1$  the largest fraction of the step that keeps the vector inside the estimation box. The two moments are the fit misses of Table 2 most directly related to the counterfactual mechanisms: the failure–output slope (moment 16) governs how much information a vote reveals, which the revision design uses, and the co-option pay gap (moment 20) governs the cross-sectional wedge the vote disciplines. The other misses, the output fixed-effect variance (moment 4) and the middle of the dissent distribution (moment 14), enter no counterfactual directly. Column (1) is the estimate of Table 3. Column (2) uses the failure–output slope, moment 16 ( $f = 0.903$ , at which the Board’s total cost of failure reaches its upper bound). Columns (3) to (5) use the co-option pay gap, moment 20 ( $f = 1.000$ ), for the full sample and the two co-option halves. All twelve estimated parameters move;  $\kappa$  and the calibrated inputs do not. Each column re-solves No-SOP, the advisory vote and both binding designs in both co-option cells and evaluates them on the advisory panel with the convention of Table 4. The memo rows report the parameters that move most and the two moments re-simulated at the vector, against their data values; the linear step closes each gap only in part.

|                                                           | (1)<br>Estimate | (2)<br>Moment 16 vector | (3)         | (4)             | (5)             |
|-----------------------------------------------------------|-----------------|-------------------------|-------------|-----------------|-----------------|
|                                                           | Full sample     | Full sample             | Full sample | Co-option below | Co-option above |
| <i>Advisory vote relative to No-SOP</i>                   |                 |                         |             |                 |                 |
| Pay level                                                 | -10.64%         | -12.86%                 | -7.17%      | -5.77%          | -8.57%          |
| Operating value                                           | +1.56%          | +1.45%                  | +1.23%      | +0.86%          | +1.61%          |
| Shareholder welfare                                       | +0.51%          | +0.53%                  | +0.33%      | +0.21%          | +0.46%          |
| <i>Binding reversion relative to advisory</i>             |                 |                         |             |                 |                 |
| Pay level                                                 | +2.94%          | +0.88%                  | +5.82%      | +5.17%          | +6.46%          |
| Operating value                                           | +0.69%          | +0.39%                  | +0.93%      | +0.51%          | +1.35%          |
| Shareholder welfare                                       | -1.21%          | -0.72%                  | -1.18%      | -1.08%          | -1.29%          |
| Pr(Fail)                                                  | 14.98%          | 9.56%                   | 12.74%      | 8.09%           | 17.39%          |
| <i>Binding revision relative to advisory</i>              |                 |                         |             |                 |                 |
| Pay level                                                 | +0.67%          | +0.42%                  | +0.69%      | +0.53%          | +0.86%          |
| Operating value                                           | +0.58%          | +0.66%                  | +0.46%      | +0.40%          | +0.51%          |
| Shareholder welfare                                       | +0.24%          | +0.34%                  | +0.13%      | +0.12%          | +0.15%          |
| Pr(Fail)                                                  | 7.30%           | 6.25%                   | 5.93%       | 4.44%           | 7.42%           |
| <i>Memo: parameters and the two moments at the vector</i> |                 |                         |             |                 |                 |
| $\sigma_{z_s}$                                            | 0.583           | 0.466                   | 0.622       | 0.622           | 0.622           |
| $\sigma_{z_b}$                                            | 0.876           | 1.116                   | 0.742       | 0.742           | 0.742           |
| $\gamma$                                                  | 0.232           | 0.229                   | 0.233       | 0.233           | 0.233           |
| $\chi_s$                                                  | 0.472           | 0.542                   | 0.471       | 0.471           | 0.471           |
| $\delta$                                                  | 0.274           | 0.458                   | 0.081       | 0.081           | 0.081           |
| $\lambda_0$                                               | 0.191           | 0.166                   | 0.220       | 0.220           | 0.220           |
| $\alpha$                                                  | 0.500           | 0.511                   | 0.489       | 0.489           | 0.489           |
| Moment 16 (data -0.045)                                   | -0.091          | -0.075                  | -0.084      | -0.084          | -0.084          |
| Moment 20 (data 0.098)                                    | 0.033           | 0.025                   | 0.044       | 0.044           | 0.044           |

## B. Counterfactual appendix

This appendix states the two binding designs of Section 6.1 formally. I first describe the counterfactual model: its timing and information structure, the information revealed by failure, the Shareholder's and Board's problems, and the numerical solution. All structural primitives remain fixed at the estimates in Table 3; only the timing of the vote and the consequence attached to vote failure change.

### B.1. Model

#### B.1.1. Timing and information

The state at the start of period  $t$  is the belief pair  $(\mu_{at}, \sigma_a^2(\tau_t))$  from Section 3.2. Under binding reversion, it additionally includes the inherited wage  $w_{t-1}$ . Within the period,

- (i) the Shareholder chooses the screen  $s_t$  at the current state, before the period's signals are drawn;
- (ii) the Board observes  $z_{bt}$ , updates according to (IA.10)–(IA.11), and proposes  $w_t^{\text{pass}}(z_{bt}, s_t)$ ;
- (iii) at the meeting, the Shareholder observes its meeting information, forms the benchmark  $w_t^S$  from (11), and the vote fails exactly when  $w_t^S \leq s_t w_t^{\text{pass}}$ ;
- (iv) if the vote passes,  $w_t^{\text{pass}}$  is implemented. If it fails, the implemented wage is  $w_{t-1}$  under binding reversion and  $w_t^{\text{rev}} = w_t^{\text{pass}} e^{\Delta_t}$  under binding revision. Failure costs the Shareholder  $\chi_s$  and the Board  $\chi_s + \delta$ ;
- (v) production occurs at the implemented wage and beliefs update into period  $t + 1$ .

To isolate the consequence of the voting rule from a change in information, I hold fixed the amount of information arriving each period. In the advisory model, the public productivity signal is observed before the vote because the vote follows the year's operations. A binding vote must instead occur before the wage it selects is implemented. I therefore let the Shareholder observe at the meeting a public signal

$$z_{yt} = a + \varepsilon_{yt}, \quad \varepsilon_{yt} \sim N(0, \sigma_y^2),$$

with the same precision as the advisory model's public productivity signal. In the counterfactual, this is a meeting signal rather than the realized productivity of the year to which the wage applies. Actual productivity is drawn separately after the vote and determines output at the implemented wage. Thus, the meeting provides the same amount of information about CEO ability as under the advisory vote while allowing the vote to determine current compensation. Realized output generates no second belief update, so learning proceeds at the same rate under every design.

Conditional on the Board's signal,  $\log w_t^S$  is normal with mean

$$\frac{\mu_{bt|z_b} + \log(\alpha/\kappa)}{1 - \alpha}$$

and variance

$$\frac{\sigma_{bt|z_b}^2 + \sigma_{sy}^2}{(1 - \alpha)^2},$$

as in Section 3.4. I denote its distribution function by  $\tilde{F}_t^S$ . At proposal  $w_t^{\text{pass}}$ , the failure probability from the Board's information set is

$$p_t = \tilde{F}_t^S[s_t w_t^{\text{pass}}],$$

which the Shareholder integrates over  $z_{bt}$  when choosing the screen.

At  $\tau_t = 0$ , binding reversion has no inherited wage. I therefore use a common entry convention for both counterfactual designs: the first vote of a new match is advisory, so failure leaves the proposal in place and imposes the failure costs, and the binding rule begins with the second meeting. Entry values are the corresponding values at  $(\mu_0, 0)$ .

### B.1.2. The belief a failed vote reveals

Let

$$u_t = \frac{Z_{syt} - \mu_{bt|z_b}}{\sqrt{\sigma_{bt|z_b}^2 + \sigma_{sy}^2}}$$

be the combined meeting signal of (11), standardized at the Board's posterior, and let  $q_t \equiv \Phi^{-1}(p_t)$  denote the failure quantile. The vote fails exactly when  $u_t \leq q_t$ , so the posterior mean

of CEO ability conditional on the Board's signal and failure is

$$\mathbb{E}[a \mid \text{fail}] = \mu_{bt|z_b} - \frac{\sigma_{bt|z_b}^2}{\sqrt{\sigma_{bt|z_b}^2 + \sigma_{sy}^2}} \frac{\phi(q_t)}{p_t}. \quad (\text{B.1})$$

The revision in (21) is the corresponding change in the log benchmark wage,

$$\Delta_t = \frac{\mathbb{E}[a \mid \text{fail}] - \mu_{bt|z_b}}{1 - \alpha} < 0. \quad (\text{B.2})$$

Because the meeting signal is informative about CEO type but not the subsequent transitory productivity shock, failure changes expected productivity only through its information about  $a$ . Let

$$k_t \equiv \frac{\sigma_{bt|z_b}^2}{\sqrt{\sigma_{bt|z_b}^2 + \sigma_{sy}^2}}.$$

Then expected productivity in the failure branch is

$$\mathbb{E}_{bt|z_b}[A_t \mid \text{fail}] = \mathbb{E}_{bt|z_b}[A_t] \frac{\Phi(q_t - k_t)}{p_t},$$

while the pass branch follows from the law of total expectation,

$$(1 - p_t)\mathbb{E}[A_t \mid \text{pass}] = \mathbb{E}[A_t] - p_t\mathbb{E}[A_t \mid \text{fail}].$$

### B.1.3. The Shareholder's problem

Write  $x_t$  for the state,  $w^{\text{pass}}$  for the Board's proposal at  $(z_{bt}, s_t)$ ,  $w^{\text{fail}}$  for the wage implemented after failure, and  $p_t$  for the failure probability. The Shareholder chooses  $s_t$  to maximize expected operating income across the two vote outcomes, net of the expected cost of dissent and including continuation value:

$$\begin{aligned} V^S(x_t) = \max_{s_t} \int_{z_b} f(z_b) & \left\{ (1 - p_t) \left[ \mathbb{E}[A_t \mid \text{pass}](w^{\text{pass}})^\alpha - \kappa w^{\text{pass}} \right] \right. \\ & \left. + p_t \left[ \mathbb{E}[A_t \mid \text{fail}](w^{\text{fail}})^\alpha - \kappa w^{\text{fail}} - \chi_s \right] \right\} \end{aligned}$$

$$+ \beta \left[ (1 - f_{\tau_t}) \left( (1 - p_t) \mathbb{E} [V^S(x_{t+1}^{\text{pass}})] + p_t \mathbb{E} [V^S(x_{t+1}^{\text{fail}})] \right) + f_{\tau_t} V^{S,R} \right] dz_b. \quad (\text{B.3})$$

Vote outcomes also affect next period's beliefs because failure selects low realizations of the meeting signal. Conditional on  $z_{bt}$  and before the vote,  $\mu_{at+1}$  is normal with mean  $\mu_{bt|z_b}$  and variance  $\sigma_{bt|z_b}^2 - \sigma_a^2(\tau_t + 1)$ . Failure selects its lower tail and passage its complement.

The two designs differ through the state transition. Under binding revision, the implemented wage is not carried forward as a state, so the probability-weighted continuation values aggregate to the unconditional continuation value, independent of the current screen. The Shareholder's problem therefore remains state by state as under the advisory vote. Under binding reversion, the implemented wage becomes next period's fallback: passage carries forward  $w_t^{\text{pass}}$ , while failure preserves  $w_{t-1}$ . The current screen therefore affects both today's rejection probability and next period's fallback. Never dissenting,  $s_t = 0$ , remains feasible at every state.

#### B.1.4. The Board's problem

Given  $s_t$  and its posterior after observing  $z_{bt}$ , the Board chooses the proposal knowing that passage and failure imply different implemented wages, information about CEO ability, and, under reversion, continuation states. For  $j \in \{\text{pass}, \text{fail}\}$ , define

$$\mathcal{V}^j(w) = (1 - f_{\tau_t}) \mathbb{E}_{bt|z_b} [V(x_{t+1}^j(w)) | j] + f_{\tau_t} V^R.$$

The Board's problem is

$$\begin{aligned} V(x_t, z_{bt}) = \max_{w^{\text{pass}}} (1 - p_t) & \left[ \mathbb{E}_{bt|z_b} [A_t | \text{pass}] (w^{\text{pass}})^\alpha - (1 - \lambda_{\tau_t k}) \kappa w^{\text{pass}} + \beta \mathcal{V}^{\text{pass}}(w^{\text{pass}}) \right] \\ & + p_t \left[ \mathbb{E}_{bt|z_b} [A_t | \text{fail}] (w^{\text{fail}})^\alpha - (1 - \lambda_{\tau_t k}) \kappa w^{\text{fail}} + \beta \mathcal{V}^{\text{fail}}(w^{\text{fail}}) \right] \\ & - (\chi_s + \delta) p_t. \end{aligned} \quad (\text{B.4})$$

Here

$$w^{\text{fail}} = \begin{cases} w_{t-1}, & \text{binding reversion,} \\ w^{\text{pass}} e^{\Delta_t}, & \text{binding revision.} \end{cases}$$

Relative to (14), the Board's failure cost is unchanged. What changes is the wage implemented after failure and, under reversion, the continuation state. The wedge  $\lambda_{\tau,k}$  applies in both branches, so binding revision updates the Board's proposal rather than replacing it with the Shareholder's preferred wage.

### B.1.5. Solution

Each design is solved as a Markov-perfect equilibrium in stationary policies. At every state, the screen maximizes (B.3) given the Board's proposal policy, and the proposal maximizes (B.4) at every realization of  $z_{bt}$  given the screen. Within each period, the Shareholder moves first and the Board responds, as in Section 3.5. Under binding revision there is no inherited-wage state, so the equilibrium retains the state-by-state structure of Section 3.5 with the branch-contingent payoffs above.

Under binding reversion, the inherited wage is a state on a grid and the Board and Shareholder play a fully dynamic game. I solve it by backward induction on tenure: tenure is bounded, beliefs reset to the entry state upon separation, and the terminal layer is pinned to the entry values. Each guess of the entry values therefore requires one backward sweep, followed by a fixed point in those values. The screen is searched on a grid up to a ceiling; under reversion the ceiling binds in about one tenth of the states, and tripling it moves the reversion value by at most 0.2% without changing any ranking.

Simulated panels in both counterfactuals use the same sequence of shocks as the advisory baseline.

## B.2. Quantifying effects

I use the solved counterfactuals for additional exercises that quantify the mechanisms behind the welfare results and their robustness.

### B.2.1. Commitment

The first exercise compares each regime with a commitment never to dissent, which delivers the No-SOP value. Under binding reversion, the Shareholder prefers this opt-out in both governance cells. At a new match, the gains are 0.28 and 0.08 units of value, approximately 0.8% and 0.2%. Under binding revision, the equilibrium instead dominates the opt-out in both cells by 0.15 and 0.24 units. Because  $s_t = 0$  is already available at every state, the reversion comparison is not about a one-period deviation, but the inability of the Markov Shareholder to commit across meetings never to use a veto that is attractive ex post.

I also compute a lower bound on the value of commitment using two one-parameter families of voting policies,  $a s^{\text{MPE}}(\cdot)$  and  $a s^{\text{A}}(\cdot)$ , which scale either the regime's Markov screen or the advisory screen. The Shareholder chooses  $a$  ex ante and commits to the resulting rule, while the Board's dynamic proposal problem is re-solved against it. Under reversion, the best rule in these families is the advisory screen itself,  $a = 1$ . It raises the Shareholder's value at a new match by 0.36 and 0.25 units above the Markov equilibrium in the two governance cells, or 1.0% and 0.7%. Committed reversion then lies above no vote by 0.08 and 0.17 units and above the advisory vote by 0.02 and 0.03. For the stock of existing matches, on the convention of Table 5, committed reversion changes Shareholder welfare by +0.04% relative to the advisory vote, and 6.0% of its votes fail. Under revision, the Markov screen is itself the best rule in both families, so this lower bound on the commitment premium is zero, as Proposition 1(c) implies for a state-free design.

### B.2.2. Isolating control of the fallback

Binding reversion differs from revision because the fallback is both inherited and previously chosen by the Board. To separate these channels, I consider an auxiliary design in which failure still implements a one-year-lagged wage, but one taken from the advisory equilibrium and therefore unaffected by the Board's past or current proposals. This preserves the staleness of the reversion fallback while removing the Board's control over it. Panel C of Table A.5 compares this design with binding reversion. When the lagged fallback lies outside the Board's control, Shareholder welfare for existing matches is +0.71% relative to the advisory vote and

6.2% of votes fail, compared with  $-1.27\%$  and  $15.0\%$  under reversion. Thus, the reversion loss is generated by the Board's control of the fallback rather than its age.

### B.2.3. Using the full vote signal

The revision rule conditions only on whether the vote fails, not on the realized against-share, so it uses less than the information the vote reveals; the gain it delivers is the gain from this coarse rule. Holding the proposal and screen fixed, allowing the full signal to enter the wage decision would strengthen the informational effect of the revision design.

### B.2.4. Valuation conventions

Figure 3 and Tables 4, 5, A.3, A.4, and A.5 evaluate each design's value functions at the advisory panel's distribution of beliefs, tenure, and inherited pay. These comparisons therefore evaluate the stock of existing matches and report percentage differences relative to the advisory vote. The shares of parameter draws in which the cells of Tables A.3 and A.4 retain their sign use the same convention.

## B.3. Endogenous reversion and the commitment problem

A say-on-pay design maps the Board's proposal, the vote outcome, and the information revealed by the vote into the implemented wage. Write  $r_t$  for the *fallback*, the wage implemented if the vote fails. I call a design *state-free* if  $r_t$  depends only on the current proposal  $w_t$  and the vote's information. The advisory vote,  $r_t = w_t$ , and binding revision,  $r_t = w_t e^{\Delta_t}$ , are state-free. Binding reversion is not:  $r_t = w_{t-1}$ , so a proposal that passes at  $t - 1$  becomes the fallback the Shareholder faces at  $t$ . The fallback is therefore a state of the game that the Board controls. The result below isolates the two dynamic terms this state adds to the problems above.

Under binding reversion the state is  $x = (\mu_a, \tau, r)$ . Let  $V(x)$  and  $V^S(x)$  denote the Board's and Shareholder's values in a Markov-perfect equilibrium with screen  $s^M(x)$ . At proposal  $w$  and screen  $s$ , let the vote fail with probability  $p = \tilde{F}_t^S(s \times w)$ , and write  $x'(w) = (\mu'_a, \tau + 1, w)$  for next period's state when  $w$  becomes the fallback. Let  $V^C$  denote the Shareholder's value when she can commit, before the Board proposes, to a policy mapping the fallback into a

screen, and  $V^M$  her value under the Markov policy. Since the Markov policy is feasible under commitment,  $V^C \geq V^M$ ; I call the difference the commitment premium.

**Proposition 1** (The two dynamic terms of endogenous reversion). (a) The Board's ratchet.

*Under binding reversion, the Board's objective (B.4) equals the objective it would have if its proposal could not move next year's fallback, plus*

$$D^B(w) = \beta(1 - f_\tau)(1 - p) \mathbb{E}[V(x'(w)) - V(x'(r)) \mid z_b, \text{pass}].$$

*This is the discounted value of installing the current proposal as next year's fallback, weighted by the probabilities that the proposal passes and the match continues. If  $V$  is nondecreasing in the fallback,  $D^B(w)$  is nonnegative for  $w \geq r$  and nonpositive for  $w \leq r$ : the Board gains from proposals that raise the fallback and loses from proposals that lower it.*

(b) The Shareholder's gain from rejection. *Under binding reversion, the Shareholder's objective (B.3) equals the objective she would have if the vote could not move next year's fallback, minus*

$$D^S(s) = \beta(1 - f_\tau) \mathbb{E}_{z_b} [(1 - p) \mathbb{E}[V^S(x'(r)) - V^S(x'(w)) \mid z_b, \text{pass}]].$$

*This is the expected discounted loss from approving a proposal that becomes next year's fallback. Rejection avoids this loss. If  $V^S$  is nonincreasing in the fallback and the Board's proposals lie at or above the current fallback,  $D^S(s) \geq 0$ .*

(c) State-free designs. *Under a state-free design, the Board's proposal, the Shareholder's screen, the failure probability, and both parties' values do not depend on  $w_{t-1}$ . Both dynamic terms are zero, and there is no commitment problem through the inherited fallback.*

*Proof.* For part (a), conditional on  $z_b$ , passage and failure partition the meeting signal. For any fixed fallback  $w$ ,

$$(1 - p) \mathbb{E}[V(x'(w)) \mid \text{pass}] + p \mathbb{E}[V(x'(w)) \mid \text{fail}] = \mathbb{E}[V(x'(w)) \mid z_b].$$

Applying this identity at  $w = r$  to the continuation value in (B.4) gives

$$\begin{aligned} & (1 - p)\mathbb{E}[V(x'(w)) \mid \text{pass}] + p\mathbb{E}[V(x'(r)) \mid \text{fail}] \\ &= \mathbb{E}[V(x'(r)) \mid z_b] + (1 - p)\mathbb{E}[V(x'(w)) - V(x'(r)) \mid \text{pass}]. \end{aligned}$$

The first term is the continuation value if the current proposal cannot change the fallback. The second, multiplied by  $\beta(1 - f_\tau)$ , is  $D^B(w)$ . If  $V$  is nondecreasing in the fallback,  $V(x'(w)) - V(x'(r))$  is nonnegative for  $w \geq r$  and nonpositive for  $w \leq r$  at every realization of  $\mu'_a$ , and so is  $D^B(w)$ .

For part (b), the same decomposition applied to the Shareholder's continuation value gives

$$\mathbb{E}[V^S(x'(r)) \mid z_b] - (1 - p)\mathbb{E}[V^S(x'(r)) - V^S(x'(w)) \mid \text{pass}].$$

The second term, integrated over  $z_b$  and multiplied by  $\beta(1 - f_\tau)$ , is  $D^S(s)$ . If  $V^S$  is nonincreasing in the fallback and every proposal lies at or above  $r$ , each bracket is nonnegative, so  $D^S(s) \geq 0$ .

For part (c), under a state-free design the implemented wage after either vote outcome depends only on the current proposal and current vote information. The inherited wage therefore enters neither current payoffs nor the transition. By backward induction, the Board's proposal, the Shareholder's screen, the failure probability, and both value functions are independent of  $w_{t-1}$ . Hence  $D^B = D^S = 0$ . Because nothing the Board chooses at  $t$  then enters the Shareholder's problem at  $t + 1$ , a policy mapping the fallback into a screen can do no better than the Markov screen, so there is no commitment problem through the inherited fallback. ■

The two terms reinforce one another. When approving a high proposal installs a high fallback, the Shareholder has an additional reason to reject today in order to preserve a lower fallback tomorrow. At the same time, the Board values a high proposal not only for the policy implemented today but because, if it passes, it becomes a more favorable fallback in the future. Each period's screen can therefore be optimal given the inherited fallback even though the resulting sequence of screens is not the one the Shareholder would choose ex ante.

This is the commitment problem under binding reversion. The Shareholder would like to commit not to relax the voting rule after the Board succeeds in raising the fallback. Once that higher fallback is inherited, however, rejecting a proposal provides less benefit relative to the cost of dissent, so a more lenient screen is optimal. The weakness is therefore not the veto itself, but that the party the veto is meant to discipline also controls the state against which future proposals are evaluated. Advisory voting and binding revision are state-free and give the Board no such lever.

## Internet Appendix

## IA1. Model appendix

This appendix contains the technical details of the model in Section 3. Internet Appendix IA1.1 derives the pay–output technology and the governance wedge of Section 3.1, shows that  $\kappa$  and  $\eta$  are scale parameters, and tests total compensation as the pay measure. Internet Appendix IA1.2 derives the representative Shareholder of Sections 3.2 and 3.3 from the votes of many holders. Internet Appendix IA1.3 states the timing assumption on the Shareholder’s screen in Section 3.3 and its institutional basis, shows why the advisory vote requires no commitment across meetings, and gives the sources of the two failure costs. Internet Appendix IA1.4 gives the belief recursions of Section 3.2, the distribution of the benchmark wage and the failure probability of Section 3.3, and the Board’s posterior of Section 3.4. Internet Appendix IA1.5 derives the state-by-state solution of Section 3.5 and gives the recursion for operating value. Internet Appendix IA1.6 derives the flow-unit normalizers of Table 3. The numerical solution is in Internet Appendix IA2.1.

### IA1.1. Technology and the pay-setting wedge

This appendix derives the two reduced-form objects Section 3 takes as primitives, the pay–output technology (1) and the Board–shareholder wedge  $\lambda_\tau$ , and then shows that  $\kappa$  and  $\eta$  are scale parameters and tests total compensation as the pay measure. The derivations of the technology and the wedge are static and hold the vote fixed; the scale result holds in the full dynamic model.

#### IA1.1.1. Technology

**CEO input choice and the pay–output reduced form.** The Board offers a linear contract that pays  $\omega$  per unit of CEO input  $n$ , so  $w = \omega n$ . The CEO supplies input at convex cost,

$$\max_{n \geq 0} \{ \omega n - n^{1/\theta} \}, \quad \theta \in (0, 1),$$

as in the convex-cost formulations surveyed in [Edmans and Gabaix \(2016\)](#). The first-order condition  $\omega = \theta^{-1} n^{1/\theta-1}$  gives  $w = \omega n = \theta^{-1} n^{1/\theta}$ , so input supply is a power function of *total*

pay,

$$n(w) = (\theta w)^\theta.$$

Output is produced with a concave technology in CEO input,  $y = \eta A n^\varphi$  with  $\varphi \in (0, 1)$ . Substituting  $n(w)$  and absorbing the constant  $\theta^{\varphi\theta}$  into the scale parameter  $\eta$  gives (1),

$$y = \eta A w^\alpha, \quad \alpha \equiv \varphi\theta \in (0, 1).$$

Total compensation is a sufficient statistic for the input the contract induces. The model can therefore set a pay level and read off the input level without carrying the contract itself. Only  $\alpha = \varphi\theta$  enters, so the CEO's input response to pay,  $\theta$ , and diminishing returns in production,  $\varphi$ , are not separately identified;  $\alpha$  summarizes both.

**The cost term.** The cost side of (3) is the cost of the operation the CEO's input runs, not the pay itself. If the operating inputs the CEO manages are hired in fixed proportion to the CEO's input and at the same factor price, their cost is proportional to the CEO's pay, which is the form  $\kappa w$  takes. The cost base is therefore the operating cost of the inputs the CEO's effort scales, not the pay itself, and its dollar level is set by the scale parameters  $\eta$  and  $\kappa$ . Those parameters are units of account (below): every percentage effect in the counterfactuals is invariant to them, so the size of the cost base relative to pay is not an input into the welfare results. What is an input is the curvature  $\alpha$ , which governs how strongly output and the cost base respond to pay; it is estimated to within 0.012 from the within-spell output–pay elasticity, and the price component of equity pay is shown below not to drive that elasticity. The operating-value effects of the counterfactuals run through output and this cost base rather than through compensation itself.

**The shareholder-optimal wage.** In the normalized units of (3), operating income  $\pi(A, w) = Aw^\alpha - \kappa w$  is strictly concave in  $w$ , so the unique shareholder-optimal wage is

$$w_s(A) = \arg \max_{w>0} \{Aw^\alpha - \kappa w\} = \left( \frac{\alpha A}{\kappa} \right)^{\frac{1}{1-\alpha}}. \quad (\text{IA.1})$$

Neither player observes  $A_t$  when pay is set. The Board evaluates (IA.1) at its posterior  $\mathbb{E}_{b_t|z_b}[A_t]$ , while the Shareholder evaluates it at the meeting signal  $\exp(Z_{sy_t})$  to form the post-meeting benchmark  $w_t^S$  in (IA.12).

### IA1.1.2. The pay-setting wedge

**The Board's private benefit.** The Board does not maximize (3). Its flow payoff, (4), is

$$\underbrace{A_t w_t^\alpha - \kappa w_t}_{\pi_t} + \lambda_\tau \kappa w_t - (\chi_s + \delta) \mathbb{1}[\text{SOP fail}_t],$$

so the Board receives a private benefit proportional to the wage bill  $\kappa w_t$ . This perceived payoff of the directors is not an accounting flow: it enters neither operating income (3) nor Shareholder welfare (13), and reaches shareholders only through the wage policy it induces. Because the benefit has the same slope as the cost the Board internalizes, its linearity rescales the marginal cost of pay,

$$\pi_t + \lambda_\tau \kappa w_t = A_t w_t^\alpha - (1 - \lambda_\tau) \kappa w_t.$$

Appendix IA1.3 gives the sources of the failure cost  $\chi_s + \delta$ , including the Board's private cost  $\delta$ ; it plays no role here.

The Board's static optimum is therefore (IA.1) at a discounted marginal cost,

$$w_B(A, \lambda_\tau) = \left( \frac{\alpha A}{(1 - \lambda_\tau) \kappa} \right)^{\frac{1}{1-\alpha}} = (1 - \lambda_\tau)^{-\frac{1}{1-\alpha}} w_S(A) > w_S(A) \quad \text{for } \lambda_\tau > 0. \quad (\text{IA.2})$$

Because input and output increase in pay, a captured Board pays more and its CEO produces more. Yet shareholders are worse off through the operating-income shortfall  $\pi(A, w_S) - \pi(A, w_B)$ . At the estimated curvature and entry wedge, the static gap is  $(1 - \lambda_0)^{-1/(1-\alpha)} = 1.54$ , or 0.43 log points of pay. The equilibrium gap is smaller because the Board prices the failure probability in (14); Section 5.2 quantifies the difference.

Dividing the Board's flow by  $1 + \lambda_\tau$  gives the equivalent reading used in Section 5:

$$(1 - v_\tau)\pi_t + v_\tau \kappa w_t, \quad v_\tau \equiv \frac{\lambda_\tau}{1 + \lambda_\tau},$$

a weighted average of shareholder operating income and the CEO's pay.

**The capture profile.** Capture is linear in CEO tenure with a level shift across observed governance cells,

$$\lambda_{\tau k} = \min\left\{\max\left\{\lambda_0 + \lambda_1 \tau + \gamma(\bar{Z}_k - \bar{Z}), 0\right\}, 0.8\right\}, \quad \bar{Z} = \sum_k \varsigma_k \bar{Z}_k, \quad (\text{IA.3})$$

where  $\bar{Z}_k$  is the mean board co-option index in cell  $k$  and  $\varsigma_k$  is its population share. The bounds keep  $\lambda_{\tau k} \in [0, 1)$  over the observed tenure support. The three pieces are distinct.

$\lambda_0$  is the wedge at hire, before the CEO has influenced who sits on the board. It is a level, read as the CEO's share of match surplus rather than a governance failure.

$\lambda_1$  is the drift from endogenous director selection: the board that hires a CEO is not the board that pays that CEO ten years later. Director turnover and the CEO's influence over nominations raise co-option with tenure and lower the effective cost of high pay (Hermalin and Weisbach, 1998; Coles et al., 2014; Bebchuk and Fried, 2003). Linearity parsimoniously parameterizes this monotone tenure dependence.

$\gamma$  loads on the firm-level co-option index  $Z_f$  of Coles et al. (2014), the share of directors appointed after the CEO took office, purged of its CEO-tenure profile and averaged within firm. The estimation splits firms at its median, solves the model once per half, and mixes the simulated panels. Because the shifts are centered at the share-weighted mean, the shift  $\gamma(\bar{Z}_k - \bar{Z})$  is mean-zero across cells and reallocates capture without changing its population average.

At the estimate (Table 3),  $\lambda_0 = 0.191$ ,  $\lambda_1 = 0.0024$  per year of tenure and  $\gamma = 0.232$ . The tenure drift over twenty years is  $20\lambda_1 \approx 0.047$ , so at the population mean of the co-option index the profile runs from 0.191 at hire to 0.238 at twenty years. The co-option gap is  $\gamma(\bar{Z}_2 - \bar{Z}_1) = 0.033$ , and the more co-opted half sits above that profile by  $\gamma(\bar{Z}_2 - \bar{Z})$  at every

tenure; the bounds never bind. The failure rate across tenure bins, the late-tenure bin above all, is informative about  $\lambda_1$ , while the co-option failure gap is informative about  $\gamma$ ; Section 4.1, Figure IA.2 and Appendix IA2.3 give the mapping from moments to parameters.

**What the wedge is not.** The wedge is not the CEO’s outside option. The model has no competing bidder or participation constraint, but CEO market power yields the same first-order condition. Thus,  $\lambda_0$  is not separately identified from market power. I read  $\lambda_0$  as the CEO’s share of match surplus, consistent with bargaining estimates of roughly half in Taylor (2013) and about 0.26 in Barry et al. (2025). The counterfactuals that vary  $\lambda_0$  are therefore comparative statics of the level of the pay-setting wedge; under the market-power reading, a pay reduction could meet a participation constraint that the model does not have. Removing  $\lambda_0$  is a labor-market experiment, not a governance experiment;  $\lambda_1$  and  $\gamma$  carry the governance content.

The wedge is also not learning-by-doing. Beliefs  $\mu_{at}$  already carry skill accumulation by responding to output and both signals, and pay follows beliefs. For that reason, the pay-tenure profile is not the source of information about  $\lambda_1$ . An unmodeled productivity gain raises the Shareholder’s benchmark  $w_t^S$  and the Board’s wage together, so its first-order effect on dissent is small; it need not be zero, because the benchmark takes the meeting signal at face value while the Board’s posterior shrinks toward the prior more as tenure rises. Capture raises pay relative to the benchmark and makes failure more likely with tenure, the variation that is informative about  $\lambda_1$ . The exposure remains, as the main text flags: if unmodeled skill accumulation also affects dissent,  $\lambda_1$  overstates the governance component of tenure-based pay growth.

### IA1.1.3. Units and measurement

**The scale parameters  $\kappa$  and  $\eta$ .** Write  $w = \kappa^{-1/(1-\alpha)}u$ . Operating income becomes  $\pi = \kappa^{-\alpha/(1-\alpha)}(Au^\alpha - u)$ , the Board’s wage-relevant payoff becomes  $\kappa^{-\alpha/(1-\alpha)}[\mathbb{E}_{b|t|z_b}[A_t]u^\alpha - (1 - \lambda_\tau)u]$  less the expected cost of failure, and the benchmark is  $w_t^S = \kappa^{-1/(1-\alpha)}\tilde{u}$  with  $\tilde{u} = (\alpha e^{Z_{syt}})^{1/(1-\alpha)}$ . The failure event  $w_t^S \leq s_t w_t$  is therefore  $\tilde{u} \leq s_t u$ , and the against-share of Appendix IA1.2 depends on  $\log(s_t w_t / w_t^S) = \log(s_t u / \tilde{u})$ ; neither contains  $\kappa$ . Measure the dissent costs in units of

operating income,  $\chi_s = \kappa^{-\alpha/(1-\alpha)} c_s$  and  $\chi_s + \delta = \kappa^{-\alpha/(1-\alpha)} c$ . Both parties' flow payoffs are then  $\kappa^{-\alpha/(1-\alpha)}$  times a payoff in  $(u, s)$  that does not contain  $\kappa$ , and the continuation values scale by the same factor because neither the separation hazard nor the belief transitions involve pay. The equilibrium  $(u^*, s^*)$  is independent of  $\kappa$ , and the equilibrium wage and output in logs are

$$\log w_t = -\frac{1}{1-\alpha} \log \kappa + \log u^*(\mu_{at}, \tau_t, z_{bt}), \quad (\text{IA.4})$$

$$\log y_t = \log \eta - \frac{\alpha}{1-\alpha} \log \kappa + \log A_t + \alpha \log u^*(\mu_{at}, \tau_t, z_{bt}). \quad (\text{IA.5})$$

Two conclusions follow. First,  $\kappa$  is a unit of account for pay. Every failure rate, every dissent moment, every slope of failure or dissent on log pay, the within-spell variances and covariances of log pay, the pay changes after a vote, and every percentage effect in the counterfactuals are functions of  $(u^*, s^*)$  alone, and the dissent costs are identified in units of operating income, which is how Panel B of Table 3 reports them. Second,  $\kappa$  and  $\eta$  enter only the levels of log pay and log output, each additively. I set  $\kappa$  at 0.366, at which the model reproduces average log pay to within one standard error given the remaining parameters (Table 2, moment 18), and estimate  $\eta$  from average log output. The voting moments identify  $\lambda_0$ , and given  $\lambda_0$  average log pay identifies  $\kappa$ . No other moment is informative about  $\kappa$ , and none depends on it. The same holds under the binding designs of Section 6.1, where the inherited wage scales with every other wage.

**Total compensation as the pay measure.** Total compensation (ExecuComp TDC1) values equity grants at the grant-date price. If boards granted a fixed number of shares, reported pay would carry a price component correlated with output, contaminating the within-spell output elasticity in Table 2. Table IA.1 tests this possibility on the estimation panel. Panel A regresses the change in log pay on the change in log market value within a CEO spell. A 10% change in market value moves total pay by 0.98% and cash pay by 0.27%. About 13% of equity pay behaves as a fixed share count, so grants are predominantly dollar-targeted. Panel B purges the price component from pay or controls for the change in market value; either correction changes the elasticity by 1.6% of its value, inside one standard error. Replacing total pay with cash pay lowers it by 68% and is the wrong correction: the Board moves pay with its beliefs

**Table IA.1.** Grant-date equity valuation and the output elasticity

This table reports grant-date equity valuation and the within-spell output elasticity of pay. All entries are in percent. Panel A reports within CEO-spell regressions of the change in log pay on the one-year-lagged change in log market value, with a tenure control, using the estimation panel of 10,079 firm-years in 1,832 spells. Standard errors clustered by spell are in parentheses. The third row is the pass-through of total pay divided by the equity share of pay, defined as the share of equity pay that behaves as a fixed number of shares. Panel B reports the change in the within-spell output elasticity of pay relative to the targeted specification of moment 2 in Table 2. The three rows purge the price component from pay, add the change in market value as a control, and replace total pay with cash pay, respectively.

|                                                                                  |               |
|----------------------------------------------------------------------------------|---------------|
| <i>Panel A. Pass-through of market value into reported pay</i>                   |               |
| change in reported pay per 10% change in market value, within spell              |               |
| Total pay (TDC1)                                                                 | +0.98% (0.15) |
| Cash pay                                                                         | +0.27% (0.17) |
| Share of equity pay behaving as a fixed share count                              | ≈ 13%         |
| <i>Panel B. Output elasticity of pay, relative to the targeted specification</i> |               |
| Total pay purged of the price component                                          | +1.6%         |
| Total pay, controlling for the change in market value                            | +1.6%         |
| Cash pay in place of total pay                                                   | −68%          |

through the equity component. I therefore estimate the technology from the pay the Board chooses, not from the price at which the market later values it.

### IA1.2. Aggregating shareholders' information into the against-share

Section 3.3 states the Say-on-Pay vote as a screen: the Shareholder (S) chooses  $s_t$ , the vote fails when the meeting-signal benchmark  $w_t^S$  of (IA.12) falls below  $s_t w_t$ , and the Board prices the resulting failure probability given its own signal,  $F_{t|z_{bt}}^S[s_t \times w_t]$ . This appendix shows how the same event generates the continuous against-share observed in the data. S determines the location of the vote distribution through its benchmark and screen, and idiosyncratic noise across votes converts that location into a continuous share. The ballot is cast by a continuum of votes  $i \in [0, 1]$  whose assessments of the benchmark carry idiosyncratic noise around S's,

$$\log w_{it}^S = \log w_t^S + Z_{30} \sigma_\varepsilon + \varepsilon_{it}, \quad \varepsilon_{it} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2), \quad (\text{IA.6})$$

where the  $\varepsilon_{it}$  are independent of everything else in the model. Vote  $i$  is cast against the proposal whenever  $w_{it}^S \leq s_t w_t$ . The continuum is only an aggregation device. The constant  $Z_{30} \equiv -\Phi^{-1}(0.30) = 0.5244$  is a location normalization: it places the institutional failure line of 30% against (Section 2.1) exactly where S's screen binds.

**Proposition IA.1** (The against-share and the failure threshold). *Fix the state, the screen  $s_t$  and the wage  $w_t$ , and let  $\sigma_\varepsilon > 0$ . Under (IA.6), the realized against-share is, almost surely,*

$$share_t = \Phi\left(\frac{\log(s_t w_t) - Z_{30}\sigma_\varepsilon - \log w_t^S}{\sigma_\varepsilon}\right),$$

and for any failure line  $t^* \in (0, 1)$ ,

$$\{share_t > t^*\} = \{w_t^S < s_t w_t e^{-\Delta(t^*)}\}, \quad \Delta(t^*) \equiv \sigma_\varepsilon [Z_{30} + \Phi^{-1}(t^*)].$$

Since  $\Delta(0.30) = 0$ , at the paper's failure line

$$\{SOP fail_t\} = \{w_t^S < s_t w_t\} \quad \text{and} \quad Pr(SOP fail_t) = F_t^S(s_t \times w_t),$$

as in (12).

*Proof.* Given  $w_t^S$ , vote  $i$  is cast against if and only if  $\varepsilon_{it} \leq \log(s_t w_t) - Z_{30}\sigma_\varepsilon - \log w_t^S$ , and a continuum law of large numbers equates the realized share with that probability. The share exceeds  $t^*$  if and only if the argument of  $\Phi$  exceeds  $\Phi^{-1}(t^*)$ , which rearranges to the stated event, and  $\Phi^{-1}(0.30) = -Z_{30}$  gives  $\Delta(0.30) = 0$ . Because  $w_t^S$  is lognormal with positive variance under (IA.12), the boundary  $w_t^S = s_t w_t$  has probability zero, so the failure probability is  $F_t^S(s_t \times w_t)$ . ■

For  $s_t > 0$ ,  $share_t$  is strictly decreasing in  $w_t^S$ , so observing it also recovers  $w_t^S$ , and hence the combined signal  $Z_{synt}$  of (11).

**The noise  $\sigma_\varepsilon$  is a measurement layer, not a mechanism.** At the 30% line,  $\sigma_\varepsilon$  drops out of the failure event, and therefore out of every object the two players optimize: the failure probability the Board prices in (14), the Shareholder's problem (13), and the equilibrium wage and screen policies. Its role is to generate the continuous dissent share observed in the data. It is identified by the moments built from that share, the dissent distribution in particular and the related covariances and regressions, and is estimated jointly with the other parameters at 1.3715 in log-wage units (Table 3).

**The 50% failure line.** Changing the failure line to  $t^*$  shifts the threshold on the benchmark by  $\Delta(t^*)$ . At the statutory majority line,  $\Delta(0.50) = Z_{30}\sigma_\varepsilon > 0$ , so holding the screen fixed the vote fails only when  $w_t^S < s_t w_t e^{-\Delta(0.50)}$ , and S re-optimizes  $s_t$  against the shifted line. Re-estimating at  $t^* = 0.50$  is exactly this reparameterization of the line, not a change in the voting technology.

### IA1.3. Costly dissent without commitment, and the microfoundation of the failure costs

Say-on-Pay is advisory: a failed vote implements no change in pay. The natural objection is that a Shareholder who anticipates paying  $\chi_s$  and changing nothing would never dissent. Two questions are bundled in that objection. Within a period, the screen is a voting policy chosen at the start of the period and applied at the meeting (Assumption 1). The model takes that adherence as given, as the published pay-voting policies of institutions and proxy advisors are applied to each proposal in practice, and the observed failure rate is itself evidence of it: a Shareholder who reconsidered at the meeting would never fail an advisory vote, whereas 6.1% of votes fail (Table 1). Across meetings, no commitment is needed, as shown below. This appendix then shows that the vote is worth its expected cost at the estimate and gives the sources of the two failure costs.

**Timing within the period.** **Assumption 1.** *The Shareholder chooses the screen  $s_t$  at the start-of-period belief state, before any of the period's signals are drawn, and applies it at the meeting.*

Under Assumption 1, the Board observes  $z_{bt}$  and proposes  $w_t(z_{bt}, s_t)$  knowing the screen; the vote fails exactly when  $w_t^S \leq s_t \times w_t$ ; and the proposed wage stands whatever the outcome (Section 3.2 and Figure IA.1). The screen is a standing rule that the proposal responds to, not a reaction to the proposal, and its institutional counterpart is the published voting policy. ISS and Glass Lewis publish their benchmark policies, including quantitative pay-for-performance screens that compare pay with peers and with performance, before each proxy season, and the large institutions publish pay-voting guidelines on the same cycle; realized

votes are then disclosed on Form N-PX. Boards set pay knowing these screens: compensation consultants model the proxy-advisor tests, and firms adjust pay design to avoid a negative recommendation (Ertimur et al., 2013; Larcker et al., 2015), which moves support by about 25 percentage points (Malenko and Shen, 2016). Because the screen is chosen anew at every state, the model's commitment horizon is one policy cycle, which matches the annual revision of these policies. What sustains adherence at the meeting, a policy applied across many firms where a departure at any one of them is visible, lies outside a single-firm model; the observed failure rate is the evidence of it, since a Shareholder who reconsidered at the meeting would never fail an advisory vote. And  $s_t = 0$ , never dissent, is in her choice set at every state, so the vote is exercised only where exercising it is optimal at that state. The second restriction of Section 3.3, that the screen conditions on the proposal only through its gap from the meeting benchmark, is stated in Appendix IA1.4.

**The screen requires no commitment across meetings.** Under the advisory vote the continuation value in (13) does not depend on the current screen: the implemented wage is  $w_t$  in both branches of the vote, the failure costs are flow costs at  $t$ , the Board's flow payoff carries no wage adjustment cost ( $c_w = 0$ ), and the state transition runs through the signals and the exogenous separation hazard, not through pay. Proposition IA.4(iii) shows that both problems therefore reduce to state-by-state problems, so the screen chosen at a state is the one the Shareholder would choose under commitment to a state-contingent path of screens: there is no future self to bind, and no relational-contract or reputational device is needed across meetings. This is a property of the consequence the design attaches to failure, not of voting in general. Proposition 1(c) states it for every state-free design. Under binding reversion it fails: the Board's proposal at  $t$  becomes the fallback the Shareholder screens against at  $t + 1$ , and the Markov screen is no longer the screen she would commit to (Appendix B).

**The vote is worth its expected cost at the estimate.** Let  $\Delta V$  denote the gain in the present value of operating income from having the advisory vote rather than no vote, and let  $B \equiv \chi_s \mathbb{E} \left[ \sum_{j \geq 0} \beta^j \mathbf{1}\{\text{SOP fail}_{t+j}\} \right]$  denote the discounted expected dissent bill. Because  $s_t = 0$  is available at every state, the solution satisfies  $\Delta V \geq B$ ; this is a property of the solution, not

an assumption. At the estimate the inequality is strict: at a new match the advisory vote raises the present value of operating income by 0.97% of no-vote Shareholder welfare and carries a capitalized dissent bill of 0.68%, leaving +0.29% net.

**Where the two costs come from.** The Shareholder bears  $\chi_s$  on a failed vote and the Board bears  $\chi_s + \delta$ . The sum is a parameterization rather than a third primitive, and  $\delta$  is the object the estimation identifies as the Board’s incremental private cost.

Three institutional facts stand behind  $\chi_s$ . First, dissent consumes the holder’s own attention: surveyed institutional investors report reluctance to vote against Say-on-Pay because a negative vote triggers repeated follow-up requests from the firm, “imposing a significant time cost” (Edmans et al., 2023, Online Appendix A). Second, votes and voting policies are observable: Form N-PX makes realized votes public and the large institutions publish pay-voting guidelines, so a vote against is seen by the firm, by proxy advisors, and by the institution’s own clients. Third, the consequences of a low-support outcome are set outside the firm: ISS treats support below 70% as a low-support outcome and may recommend against the compensation committee at the following meeting (ISS, 2022), and institutional engagement and compensation-committee turnover both shift sharply at that cutoff (Dey et al., 2024). A failed vote therefore puts the firm and its holders into at least one cycle of engagement and disclosure that neither side chose.

The Board’s incremental cost  $\delta$  is a career cost falling on directors rather than on the firm’s cash flows, which is why it enters the Board’s objective and not the shareholders’. Table A.2 documents this cost: three years after a failed vote, a compensation-committee director is 4.4 percentage points more likely to have left the board, against a control rate of 22%, and a majority defeat roughly doubles the effect. The estimated  $\delta$  is of the size these consequences suggest, 48% of a year’s captured rent per failed vote (Table 3).

#### **IA1.4. Evolution of Board and Shareholder beliefs**

Two propositions give the law of motion of beliefs across periods; the sequence of beliefs within a period follows.

### IA1.4.1. Evolution of beliefs period to period

Prop. IA.2 gives the update from  $t$  to  $t + 1$ . Prop. IA.3 gives the distribution of next period's beliefs given today's, which enters the continuation values. Figure IA.1 shows when beliefs update and when each strategy is played.

**Proposition IA.2.** *From period  $t$  to  $t + 1$ , the variance of beliefs for both the Board and the Shareholder declines deterministically according to*

$$\sigma_a^2(\tau + 1) = \left[ \sigma_a^{-2}(\tau) + \sigma_{z_b}^{-2} + \sigma_{z_s}^{-2} + \sigma_y^{-2} \right]^{-1} \quad (\text{IA.7})$$

where  $\tau$  is the tenure of the CEO at year  $t$ . Equivalently, as a function of tenure and parameters alone,

$$\sigma_a^2(\tau) = \sigma_0^2 \left[ 1 + \tau (\kappa_{z_b}^{-1} + \kappa_{z_s}^{-1} + \kappa_y^{-1}) \right]^{-1} \quad (\text{IA.8})$$

where  $\kappa_{z_b} = \sigma_{z_b}^2 / \sigma_0^2$ ,  $\kappa_{z_s} = \sigma_{z_s}^2 / \sigma_0^2$  and  $\kappa_y = \sigma_y^2 / \sigma_0^2$ . The mean of beliefs evolves according to

$$\mu_{at+1} = \sigma_a^2(\tau + 1) \left[ \frac{\mu_{at}}{\sigma_a^2(\tau)} + \frac{z_{bt}}{\sigma_{z_b}^2} + \frac{z_{st}}{\sigma_{z_s}^2} + \frac{z_{yt}}{\sigma_y^2} \right] \quad (\text{IA.9})$$

*Proof.* The formulas are standard results in Bayesian learning (e.g., [Pastor and Veronesi, 2009](#); [Taylor, 2010](#)). The Board's and the Shareholder's signals are disclosed each period (Section 3.2), so the two parties hold the same beliefs at the start of each period. ■

**Proposition IA.3.** *The mean and variance of the mean of  $t + 1$  CEO beliefs at  $t$  are*

$$\begin{aligned} \mathbb{E}_t[\mu_{at+1}] &= \mu_{at} \\ \text{Var}_t[\mu_{at+1}] &= \sigma_a^2(\tau) - \sigma_a^2(\tau + 1) \end{aligned}$$

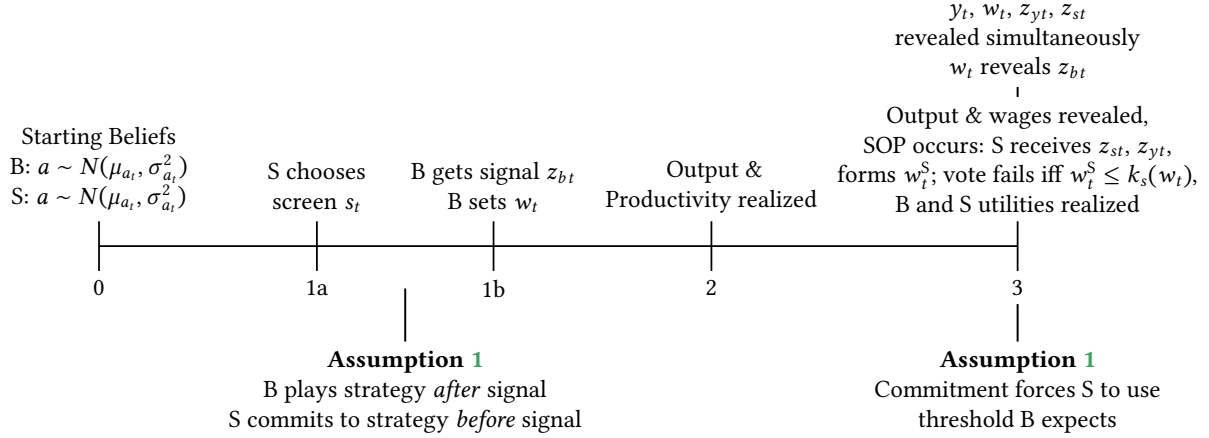
That is,

$$\mu_{at+1} \mid \mu_{at}, \tau \sim N(\mu_{at}, \sigma_a^2(\tau) - \sigma_a^2(\tau + 1))$$

*Proof.* This is the standard identity  $\text{Var}(\mathbb{E}[a \mid I'] \mid I) = \text{Var}(a \mid I) - \text{Var}(a \mid I')$  for a normal  $a$

**Figure IA.1.** Model timeline with exact time that strategies are played

A detailed version of the model timeline in Figure 1, showing when beliefs update and when the Board and the Shareholder play their strategies. The timing of the Shareholder's choice is Assumption 1 in Appendix IA1.3.



and information sets  $I \subset I'$  generated by normal signals about  $a$ . ■

#### IA1.4.2. Board and Shareholder beliefs within the period

Because the wage and the vote reveal  $z_{bt}$  and  $z_{st}$ , B and S share the beliefs  $(\mu_{at}, \sigma_a^2(\tau_t))$  at the start of  $t$ , with  $\sigma_a^2(\tau_t)$  from (IA.8). Within  $t$ :

##### 1. Board beliefs after the compensation committee meeting

The Board receives  $z_{bt}$  and updates to

$$\mu_{bt|z_b} = \sigma_{bt|z_b}^2 \left( \frac{\mu_{at}}{\sigma_a^2(\tau_t)} + \frac{z_{bt}}{\sigma_{z_b}^2} \right) \quad (\text{IA.10})$$

$$\sigma_{bt|z_b}^2 = \frac{\sigma_a^2(\tau_t)\sigma_{z_b}^2}{\sigma_a^2(\tau_t) + \sigma_{z_b}^2} = \sigma_0^2 \left[ 1 + (\tau_t + 1)\kappa_{z_b}^{-1} + \tau_t(\kappa_{z_s}^{-1} + \kappa_y^{-1}) \right]^{-1} \quad (\text{IA.11})$$

It sets pay at these beliefs.

##### 2. Shareholder beliefs when the screen $k_{st}$ is chosen

S chooses the screen at the start of the period, before  $z_{bt}$  is drawn, at the prior  $(\mu_{at}, \sigma_a^2(\tau_t))$ .

S knows that after any  $z_{bt}$  the Board holds  $(\mu_{bt|z_b}, \sigma_{bt|z_b})$  and responds to the screen at that posterior, which S integrates out in (13).

##### 3. The meeting signal and the benchmark wage

The meeting signal is  $Z_{syt} = a + p\varepsilon_{st} + (1-p)\varepsilon_{yt}$ , where  $p = \sigma_{z_s}^{-2} / (\sigma_{z_s}^{-2} + \sigma_y^{-2})$  is the precision

weight and  $\varepsilon_{sy t} \equiv p\varepsilon_{st} + (1 - p)\varepsilon_{yt}$  has variance  $\sigma_{sy}^2 = \sigma_{zs}^2 \sigma_y^2 / (\sigma_{zs}^2 + \sigma_y^2)$ . The benchmark wage is formed from the meeting signal alone,  $w_t^S = \left(\frac{\alpha}{\kappa} \exp(Z_{sy t})\right)^{1/(1-\alpha)}$ . Shareholders observe the proposed wage before voting, but I treat it as the policy being evaluated, not as a signal from which they invert  $z_{bt}$ : the wage enters the vote only through its gap from the benchmark. The screen is nevertheless chosen ex ante against the Board's equilibrium response to its signal. From the prior,  $Z_{sy t} \sim N(\mu_{at}, \sigma_{at}^2 + \sigma_{sy}^2)$ , so when S chooses the screen,

$$w_t^S \mid (\mu_{at}, \sigma_{at}^2) \sim \log N(m_t^S, v_t^S),$$

$$m_t^S \equiv \frac{\mu_{at} + \log \frac{\alpha}{\kappa}}{1 - \alpha}, \quad v_t^S = \frac{\sigma_{at}^2 + \sigma_{sy}^2}{(1 - \alpha)^2}. \quad (\text{IA.12})$$

Let  $F_t^S$  denote the CDF of this distribution, so that the ex ante probability of failure at a wage  $w_t$  is (12). The Board sets pay after observing  $z_{bt}$ , and the meeting noise  $\varepsilon_{sy t}$  is independent of  $z_{bt}$ , so from the Board's information set  $\log w_t^S \mid z_{bt}$  is normal with mean  $(\mu_{bt|z_b} + \log \frac{\alpha}{\kappa}) / (1 - \alpha)$  and variance  $(\sigma_{bt|z_b}^2 + \sigma_{sy}^2) / (1 - \alpha)^2$ . Its CDF,  $F_{t|z_{bt}}^S$ , evaluated at  $s_t \times w_t$  is the failure probability the Board prices at a proposed wage, and (12) is its average over  $z_{bt}$  at a fixed wage.

#### 4. Board and Shareholder beliefs after the annual shareholder meeting

The 10-K and compensation committee report reveal the wage, hence  $z_{bt}$ ; the vote reveals  $Z_{sy t}$ , hence  $z_{st}$ . B and S update to  $(\mu_{at+1}, \sigma_a^2(\tau_t + 1))$  by Prop. IA.2, so beliefs coincide at the start of each period.

### IA1.5. Full derivation of model solution

This appendix states the two problems of Section 3 in the form the numerical solution uses, and shows why, under the advisory vote, both reduce to a problem solved state by state. Appendix B extends the statement to the two binding designs.

**Proposition IA.4** (Reduced form of the two problems). *Fix a state  $(\mu_{at}, \tau_t)$ , with  $\sigma_{at}^2 = \sigma_a^2(\tau_t)$  from (IA.8), and a screen  $s$ .*

(i) The Board's problem (14) can be written as

$$\begin{aligned}
V(\mu_{at}, \tau_t, z_b) = \max_{w(z_b, s)} & \exp\left(\mu_{bt|z_b} + \frac{1}{2}(\sigma_{bt|z_b}^2 + \sigma_y^2)\right) w(z_b, s)^\alpha - (1 - \lambda_{\tau_t})\kappa w(z_b, s) \\
& - (\chi_s + \delta)F_{t|z_b}^S[s \times w(z_b, s)] \\
& + \beta \left[ (1 - f_{\tau_t})\mathbb{E}_{bt|z_b}[V(\mu_{at+1}, \tau_t + 1)] + f_{\tau_t} V^R \right], \tag{IA.13}
\end{aligned}$$

where  $\mu_{bt|z_b}$  and  $\sigma_{bt|z_b}^2$  are the Board's posterior after  $z_{bt}$ , (IA.10) and (IA.11);  $F_{t|z_b}^S$  is the CDF of the benchmark wage conditional on the Board's signal, (IA.12) with  $(\mu_{bt|z_b}, \sigma_{bt|z_b}^2)$  in place of  $(\mu_{at}, \sigma_{at}^2)$ ;  $V(\mu_{at}, \tau_t) = \mathbb{E}_{at}[V(\mu_{at}, \tau_t, z_b)]$  is the value before the signal;  $f_\tau$  is the separation hazard, with  $f_0 = 0$  and  $f_T = 1$ ;  $V^R = V(\mu_0, 0)$  is the entry value; and, from the Board's information set,

$$\mu_{at+1} \mid (\mu_{bt|z_b}, \tau_t) \sim N(\mu_{bt|z_b}, \sigma_{bt|z_b}^2 - \sigma_a^2(\tau_t + 1)).$$

(ii) The Shareholder's problem (13) can be written as

$$\begin{aligned}
V^S(\mu_{at}, \tau_t) = \max_s & \int_{z_b} f(z_b \mid \mu_{at}, \sigma_{at}^2) \left[ \exp\left(\mu_{bt|z_b} + \frac{1}{2}(\sigma_{bt|z_b}^2 + \sigma_y^2)\right) w(z_b, s)^\alpha \right. \\
& \left. - \kappa w(z_b, s) - \chi_s F_{t|z_b}^S[s \times w(z_b, s)] \right] dz_b \\
& + \beta \left[ (1 - f_{\tau_t})\mathbb{E}_{at}[V^S(\mu_{at+1}, \tau_t + 1)] + f_{\tau_t} V^{S,R} \right], \tag{IA.14}
\end{aligned}$$

where  $f(z_b \mid \mu_{at}, \sigma_{at}^2)$  is the  $N(\mu_{at}, \sigma_{at}^2 + \sigma_{z_b}^2)$  density of the Board's signal,  $w(z_b, s)$  is the Board's best response in (i),  $V^{S,R} = V^S(\mu_0, 0)$ , and, from the Shareholder's information set,  $\mu_{at+1} \mid (\mu_{at}, \tau_t) \sim N(\mu_{at}, \sigma_a^2(\tau_t) - \sigma_a^2(\tau_t + 1))$  by Prop. IA.3.

(iii) Under the advisory vote the continuation terms in (IA.13) and (IA.14) depend on neither  $w(z_b, s)$  nor  $s$ . The equilibrium wage and screen at  $(\mu_{at}, \tau_t)$  therefore solve the static problems obtained by deleting those terms, and the Markov-perfect equilibrium of Section 3.5 is the collection of these state-by-state solutions.

*Proof. Part (i).* After the compensation committee meeting the Board's beliefs are  $a \sim N(\mu_{bt|z_b}, \sigma_{bt|z_b}^2)$ , and  $\ln A_t = a + \varepsilon_{yt}$  with  $\varepsilon_{yt} \sim N(0, \sigma_y^2)$  independent of  $a$ , so  $\mathbb{E}_{bt|z_b}[A_t]$  is the lognormal mean

in (IA.13). The failure term is the probability of the event in (12) conditional on the Board's signal, with the conditional law of the benchmark from Appendix IA1.4; the Board's private cost  $\delta$  enters only here. The separation hazard is exogenous, and on separation beliefs reset to the entry state, so the value after separation is the constant  $V^R$ . On continuation,  $\mu_{at+1}$  is the posterior mean (IA.9) after the period's three signals; from the Board's information set it is the mean of a posterior that refines  $(\mu_{bt|z_b}, \sigma_{bt|z_b}^2)$ , so by the law of total variance its mean is  $\mu_{bt|z_b}$  and its variance is  $\sigma_{bt|z_b}^2 - \sigma_a^2(\tau_t + 1)$ .

*Part (ii).* The Shareholder chooses  $s$  at the start of the period, before any signal is drawn (Assumption 1), so she integrates over the Board's signal, which is  $N(\mu_{at}, \sigma_{at}^2 + \sigma_{z_b}^2)$  given  $a \sim N(\mu_{at}, \sigma_{at}^2)$  and  $z_{bt} = a + \varepsilon_{bt}$ . For each realization she knows the Board's posterior, hence its best response  $w(z_b, s)$ , and evaluates the flow at that wage: expected operating income (3), at the same expected productivity as the Board because the Board's posterior is a sufficient statistic for  $z_b$ , less  $\chi_s$  times the failure probability conditional on that signal, so that the integral over  $z_b$  is the ex ante probability that the vote fails at the Board's policy. She does not bear  $\delta$ . The continuation is integrated with the distribution of  $\mu_{at+1}$  from her information set, Prop. IA.3.

*Part (iii).* The state entering  $t + 1$  is either the entry state or  $(\mu_{at+1}, \tau_t + 1)$ . The tenure step and the separation event are exogenous, and  $\mu_{at+1}$  is the function (IA.9) of  $(\mu_{at}, \tau_t)$  and the three signals, all of which are revealed whatever the wage and whatever the vote outcome (Section 3.2). Under the advisory vote the implemented wage is  $w(z_b, s)$  in both branches of the vote, the failure costs are flow costs at  $t$ , and the flow payoffs carry no term in  $w_{t-1}$ . Hence neither  $w(z_b, s)$  nor  $s$  enters the transition or the continuation values, and in each problem the continuation term is a constant with respect to the current choice. Deleting it leaves, at each state, the Stackelberg problem in which the Shareholder chooses  $s$  first and the Board responds at every  $z_b$ ; the screen is chosen from a compact interval on which the objective is continuous, so a maximizer exists. The Shareholder's problem is dynamic, but under the advisory vote its solution is static at each state (Appendix IA1.3 gives the commitment reading). Given the policies, the three value functions  $V$ ,  $V^S$  and  $V^\Pi$  are computed by backward recursion over tenure, with the entry values  $V^R$ ,  $V^{S,R}$  and  $V^{\Pi,R}$  solved as fixed points (Appendix IA2.1).

The argument in part (iii) uses that the fallback after a failed vote depends only on the current proposal and the vote's information. Binding revision has this property and inherits the state-by-state structure; binding reversion does not, because the fallback at  $t + 1$  is the wage implemented at  $t$ , so the inherited wage joins the state and the continuation values depend on the current proposal. Appendix B states the two problems for those designs, and Proposition 1 in Appendix B.3 gives the consequences for the proposal and the screen. ■

**Operating value.** Operating value is the discounted present value of operating income (3) under the equilibrium wage policy  $\omega_t = \omega(\mu_{at}, \tau_t, z_{bt})$ :

$$\begin{aligned} V^\Pi(\mu_{at}, \tau_t) = & \mathbb{E}_{at} \left[ \mathbb{E}_{bt|zb} [A_t] \omega_t^\alpha - \kappa \omega_t \right] \\ & + \beta \left[ (1 - f_{\tau_t}) \mathbb{E}_{at} [V^\Pi(\mu_{at+1}, \tau_t + 1)] + f_{\tau_t} V^{\Pi,R} \right], \end{aligned} \quad (\text{IA.15})$$

where  $V^{\Pi,R} \equiv V^\Pi(\mu_0, \tau = 0)$ .

### IA1.6. Flow-unit normalizers for the dissent costs

Panel B of Table 3 expresses each dissent cost as a share of the annual flow that the party bearing it puts at stake. The normalizers are evaluated in the estimated advisory equilibrium, over the firm-years the model generates. Annual operating income is  $\Pi \equiv \mathbb{E}[A_t w_t^\alpha - \kappa w_t]$ , the per-period operating income (3) averaged over the equilibrium wage policy and the distribution of states; the annual captured rent is  $R \equiv \mathbb{E}[\lambda_\tau \kappa w_t]$ , the Board's private flow benefit from pay in (4); and  $p$  is the equilibrium probability of a failed vote. The four ratios are  $\chi_s/\Pi$  and  $p \chi_s/\Pi$  for the Shareholder, the cost per failed vote and its expected annual value, and  $\delta/R$  and  $p \delta/R$  for the Board. Standard errors in Panel B divide the Panel A standard error of the cost by the same normalizer, held at its point estimate.

## IA2. Estimation appendix

This appendix describes the numerical solution of the model, the simulation, the auxiliary moments and the identification argument, the estimator, and the optimization routine. The sensitivity figure and the robustness tables it refers to are in Internet Appendix IA3.

### IA2.1. Numerical solution

The model has twelve estimated parameters,

$$\mathbb{P} = \left[ \log \eta \quad \sigma_0 \quad \sigma_y \quad \alpha \quad \sigma_{z_b} \quad \sigma_{z_s} \quad \sigma_\varepsilon \quad \lambda_0 \quad \lambda_1 \quad \gamma \quad \chi_s \quad \delta \right], \quad (\text{IA.16})$$

and three calibrated ones: the cost parameter  $\kappa$ , the shift in the separation hazard after a failed vote, and the discount factor  $\beta = 0.9$ . The separation hazards  $f_t$  are the separation rates by tenure in the estimation sample, with  $f_0 = 0$ , so that one-year spells are excluded, and  $f_T = 1$ . Given  $\mathbb{P}$ , the solution proceeds as follows.

1. Discretize the state into a  $(N_\mu, T + 1) = (21, 26)$  grid of belief means and tenures; tenure determines the belief variance, and the state carries no lagged wage. Discretize the Board's signal  $z_{bt}$  on a quadrature grid around each belief mean and, for each node, a grid of candidate wages from half the shareholder-optimal wage to one and a half times the Board's unconstrained wage.
2. At each state, for a candidate screen  $s$ , compute the Board's best response on the wage grid at every signal node and evaluate the Shareholder's objective (13). The screen is chosen on a log-spaced grid and refined around the best grid point by golden-section search. Under the advisory vote neither continuation value depends on the current choices, so the per-state problems are independent and one pass over the state space delivers both policies.
3. Given the policies, compute the value functions by backward induction over tenure, with the value at separation reset to the entry state and solved as a fixed point. These values enter the welfare comparisons and are the starting point for the binding designs, whose failure consequence re-introduces a dependence of the continuation on the current wage (Appendix B).

## IA2.2. Simulation

Each evaluation of the objective simulates  $N_S = 5$  panels of  $N_f = 1,500$  firms, allocated to the two governance cells in proportion to their population shares. For each firm I draw a CEO of skill  $a \sim N(0, \sigma_0^2)$  and, each period, the shocks  $\varepsilon_{zbt}$ ,  $\varepsilon_{zst}$  and  $\varepsilon_{yt}$ . The Board and the Shareholder play the policies of Appendix IA2.1 at the current state, and beliefs update on the realized signals. At the end of each period a CEO with tenure  $\tau > 0$  separates with probability  $f_\tau$ , raised after a failed vote by the log-odds shift described below. The underlying quasi-random draws are the same at every evaluation, so only the parameters change between iterations. Each panel runs for 35 years; the first 20 are burn-in and the last 15 are the analysis sample.

**Turnover after a failed vote.** Failed votes are followed by higher CEO turnover in the data: in a regression of CEO separation on failure, tenure, firm and year effects, a failed vote raises the probability of separation by 3.3 percentage points. So that turnover around votes in the simulated panels matches the data, I raise the hazard of separation at the end of a year with a failed vote by a log-odds shift of 0.15, about 2.5 percentage points at the mean hazard. The shift is not part of the Board’s transition law. Pricing it would treat the reduced-form failure–turnover relation as a causal transition the Board can manipulate and, in a learning model, would give the Board an incentive to induce failure in order to obtain a fresh match. Were it priced, it would enter the Board’s cost of failure additively and be absorbed by  $\delta$ , which the failure moments identify only as part of the Board’s total cost of a failed vote.

## IA2.3. Auxiliary moments and identification

This appendix defines the twenty-four auxiliary moments, numbered as in Table 2, and gives the derivations behind the identification argument of Section 4.1. Every moment is computed by the same code on the observed and the simulated panels. The local sensitivities of Andrews et al. (2017) are reported at the end as a diagnostic of the joint system.

**The auxiliary moments.** The output block (moments 1 to 4) is the mean of firm- and year-purged log revenue, the within-spell slope  $y_1$  of that revenue on log pay in (16), the variance of its residual, and the variance of its CEO effect, corrected for the sampling variance of the spell

means. The voting block (moments 5 to 17, of which 8 and 17 are contrasts listed below) starts from the failure indicator, an against-share above 30%. Moments 5 to 7 are the failure rate in three tenure bins,  $\tau \leq 7$ ,  $8 \leq \tau \leq 15$  and  $\tau \geq 16$ ; moment 9 is the failure rate among votes with an against-share between 25% and 35%; moment 10 is the density-weighted covariance of the failure residual with excess dissent, defined below; moments 11 to 14 are the mean against-share on passing and on failing votes and the shares of votes with less than 5% and with 20% to 30% against; and moments 15 and 16 are the average marginal effects of log pay and of the output residual in the probit (18), which controls for the CEO output effect and is fit on firm-years with an against-share of at least 5%. The Board block (moments 18 to 24, of which 20 is a contrast listed below) is average log pay (18); the variance of the CEO effect in the wage regression (20), formed from log total pay within spell without a firm purge and corrected for sampling variance (19); the within-spell coefficient  $b_1$  of log pay on the lagged failure indicator, controlling for tenure (21); the coefficient of log pay on the lagged against-share (22); the autocovariance of log pay (23); and the change in log pay after a failed vote minus that after a passed vote (24). The tenure coefficient  $b_2$  is not targeted. The three co-option contrasts (moments 8, 17 and 20) are differences between firm-years above the sample median of the co-option index and those at or below it: in the failure rate (8); in the within-spell slope of the against-share on log pay, estimated by cell (17); and in the mean residual of a pooled regression of log pay on log sales within three-digit SIC industry, a size-adjusted pay level (20).

**The screen and the two failure costs.** Conditional on the start-of-period state and the Board's wage, the failure probability is (17): the screen  $s_{it}$  shifts its location and  $v_{it}^S$  its slope. The Shareholder chooses the screen to maximize (13), in which  $\chi_s$  prices every failure; the Board chooses the wage to maximize (14), in which a failure costs  $\chi_s + \delta$ . So  $\chi_s$  enters both problems and  $\delta$  only the Board's. The pooled voting moments describe the screen: the level of failures at each tenure and the response of failure to pay are outcomes of the Shareholder's choice, and a higher  $\chi_s$  lowers the first and flattens the second. The wage moments describe the Board's response to the screen: from (19), the discipline term is proportional to  $(\chi_s + \delta) \partial \text{Pr}(\text{SOP fail}) / \partial w$ , so how far the Board holds pay below what its beliefs and the wedge

would justify measures the total cost of failure. The difference between the two readings is  $\delta$ .

The co-option contrasts are the moments in which the wedge and the failure costs enter differently. Governance varies across cells only through  $\gamma$  in (15). Raising  $\gamma$  moves pay above the benchmark in the high co-option cell and, through (17), raises failures there: the pay gap (moment 20) and the failure gap (moment 8) rise together, in a proportion set by the slope of the failure rule. Raising  $\delta$  acts on the Board's response: it compresses pay toward the screen where the discipline term is largest, which is the high co-option cell, so a given pay gap comes with a smaller failure gap. It also raises the value of the threat to the Shareholder, who screens harder where the threat is worth the most, so the response of dissent to pay steepens in that cell and the dissent-pay slope gap (moment 17) rises. The three contrasts are informative about  $\delta$  and  $\gamma$  jointly, with the pooled voting moments holding  $\chi_s$  in place.

**The governance wedge.** The wedge is  $\lambda_{\tau k} = \lambda_0 + \lambda_1 \tau + \gamma(\bar{Z}_k - \bar{Z})$ . In (19) the level of pay given the technology sits  $-\log(1 - \lambda_{\tau k})/(1 - \alpha)$  above the shareholder-optimal wage, less the discipline term. Because  $\kappa$  is set to reproduce average log pay (Appendix IA1.1), moment 18 identifies  $\kappa$ , and  $\lambda_0$  is disciplined by the voting moments, which depend on pay relative to the benchmark, conditional on  $\alpha$  and on the total cost of failure. The cell difference in  $-\log(1 - \lambda_{\tau k})$  is  $\gamma(\bar{Z}_k - \bar{Z}_{k'})/(1 - \lambda_{\tau})$  to first order, which is how the pay gap (moment 20) enters, jointly with the failure gap and the failure costs. The tenure slope  $\lambda_1$  is informative mainly through the failure profile (moments 5 to 7, the late-tenure rate above all) and the dissent distribution (11 and 13):  $v_{it}^S$  in (17) falls with tenure as beliefs tighten, so a failure rate that rises across tenure requires the Board's proposal to drift away from the benchmark as the CEO ages, and  $\lambda_1$  is that drift.

**Private information: The Shareholder.** Appendix IA1.2 gives the against-share as

$$V_{it}^{\text{against}} = \Phi\left(\frac{\log(s_{it} w_{it}) - Z_{30} \sigma_{\varepsilon} - \log w_t^S}{\sigma_{\varepsilon}}\right), \quad \log w_t^S = \frac{\log \frac{\alpha}{\kappa} + p z_{sit} + (1 - p) z_{yit}}{1 - \alpha},$$

with  $p = \sigma_{z_s}^{-2} / (\sigma_{z_s}^{-2} + \sigma_y^{-2}) = \sigma_y^2 / (\sigma_y^2 + \sigma_{z_s}^2)$ . Let  $N_{it}$  be the index of the probit (18), so that  $\Phi(N_{it})$  is the ex ante failure probability given pay, the output residual and the CEO effect; the index is linear in these regressors, a reduced-form approximation to the model's failure probability.

What the index does not absorb is the private signal, which enters the share through

$$\tilde{z}_{sit} \equiv -\frac{p \varepsilon_{sit}}{(1-\alpha)\sigma_\varepsilon}, \quad V_{it}^{\text{against}} = \Phi(A_{it} + \tilde{z}_{sit}),$$

where  $A_{it}$  is the share's own index with the private noise set to zero.  $A_{it}$  differs from  $N_{it}$  in offset and scale: the share is scaled by  $\sigma_\varepsilon$  and the failure event by  $\sqrt{v_{it}^S}$ , and the probit absorbs only part of what  $A_{it}$  contains. Excess dissent is  $\tilde{V}_{it}^{\text{against}} \equiv V_{it}^{\text{against}} - \Phi(N_{it})$ , and moment 10 is

$$m_{10} = \mathbb{E} \left[ \frac{(\mathbb{1}\{\text{SOP fail}_{it}\} - \Phi(N_{it})) \tilde{V}_{it}^{\text{against}}}{\phi(N_{it})^2} \right],$$

the covariance of the failure residual and the excess share, weighted by the inverse squared density of the index, with both residuals demeaned; the against-share is winsorized at 2.5 percent in each tail before this step, in the observed and the simulated data alike. Expanding around  $\tilde{z}_{sit} = 0$ ,  $\tilde{V}_{it}^{\text{against}} = \Phi(A_{it}) - \Phi(N_{it}) + \phi(A_{it})\tilde{z}_{sit} + \mathcal{O}(\tilde{z}_{sit}^2)$ , so to first order

$$m_{10} = \mathbb{E} \left[ \frac{(\mathbb{1}\{\text{SOP fail}_{it}\} - \Phi(N_{it}))(\Phi(A_{it}) - \Phi(N_{it}))}{\phi(N_{it})^2} \right] + \mathbb{E} \left[ \frac{\phi(A_{it})}{\phi(N_{it})} \frac{(\mathbb{1}\{\text{SOP fail}_{it}\} - \Phi(N_{it}))\tilde{z}_{sit}}{\phi(N_{it})} \right].$$

The second term is the covariance of failure with the private signal, weighted by the density ratio  $\phi(A_{it})/\phi(N_{it})$ . Failure is the event that the same private signal, scaled by  $\sqrt{v_{it}^S}$  rather than by  $\sigma_\varepsilon$ , crosses the threshold, so this term rises with the standard deviation of  $\tilde{z}_{sit}$ ,  $p \sigma_{z_s}/((1-\alpha)\sigma_\varepsilon)$ : the moment rises with the noise in the private signal and falls with the vote scatter. Three features of the mapping matter. The first term does not vanish as  $\sigma_{z_s} \rightarrow 0$ , because the two indices differ in offset and scale; it vanishes when they coincide. The response to  $\sigma_{z_s}$  flattens, because  $p \sigma_{z_s} = \sigma_y^2 \sigma_{z_s}/(\sigma_y^2 + \sigma_{z_s}^2)$  is increasing in  $\sigma_{z_s}$  only while  $\sigma_{z_s} < \sigma_y$ , which holds at the estimates. And  $\sigma_\varepsilon$  scales the moment, so its level is read jointly with  $\sigma_\varepsilon$ , about which the dissent distribution (moments 11 and 13 above all) is most informative.

The failure slopes carry the rest. From (17), the response of failure to log pay is  $\phi(\cdot)/\sqrt{v_{it}^S}$  with  $v_{it}^S = (\sigma_{bt|z_b}^2 + \sigma_{sy}^2)/(1-\alpha)^2$  and  $\sigma_{sy}^2 = \sigma_{z_s}^2 \sigma_y^2/(\sigma_{z_s}^2 + \sigma_y^2)$ : a noisier private signal makes the benchmark a looser standard and flattens  $s_1$  (moment 15). The response to the output residual,  $s_2$  (moment 16), works through the weight  $(1-p)/(1-\alpha)$  the benchmark puts on

the public signal, which rises as the private signal gets noisier. Both slopes also depend on  $\sigma_{z_b}$ , because the wage the Shareholder screens is the Board's response to its own signal, and the two noise parameters enter them with opposite signs. In the joint system the slopes, moment 16 in particular, carry much of the information about  $\sigma_{z_s}$ ; moment 10 is the direct mapping.

**Private information: The Board.** With three signals per period, the shared posterior variance after  $\tau$  periods, (IA.8), is

$$\sigma_a^2(\tau) = (\sigma_0^{-2} + \tau K)^{-1}, \quad K \equiv \sigma_y^{-2} + \sigma_{z_s}^{-2} + \sigma_{z_b}^{-2},$$

and the cross-sectional variance of the shared posterior mean is  $\sigma_0^2 - \sigma_a^2(\tau)$ . The Board sets pay after its own signal, so its posterior mean  $\mu_{bt|z_b}$  in (IA.10) has cross-sectional variance

$$\text{Var}(\mu_{bt|z_b}) = \sigma_0^2 - (\sigma_a^2(\tau) + \sigma_{z_b}^{-2})^{-1},$$

which exceeds the dispersion of the shared belief by the information in  $z_{bt}$  and rises as  $\sigma_{z_b}$  falls. From (19) the CEO effect in log pay is the spell average of  $\log \mathbb{E}_{bt}[A_{it}]/(1 - \alpha)$ , net of the wedge and the discipline term, neither of which is common within a cell: the wedge varies with tenure, and the discipline term with beliefs and the screen. The wage fixed-effect variance (moment 19) is therefore the dispersion of Board beliefs scaled by  $(1 - \alpha)^{-2}$  only to first order. Given  $\sigma_0$  and  $K$ , more precise Board information generates more dispersion in beliefs, and therefore in pay, at every tenure. The post-failure response  $b_1$  (moment 21) is the other side of the same learning. Under disclosure the vote reveals  $z_{st}$ , and a failure is a low realization of it; from (IA.9) the Board's update puts weight  $\sigma_a^2(\tau + 1)/\sigma_{z_s}^2$  on  $z_{st}$ , and pay moves by  $1/(1 - \alpha)$  times the change in the log belief. The revision is larger the more precise the Shareholder's signal is relative to what the Board already knew, which makes  $b_1$  informative about  $\sigma_{z_s}$  relative to  $\sigma_{z_b}$ . It is a learning moment, not a cost moment: the cost of failure shapes the wage proposed before the vote, and the lagged indicator is predetermined at  $t$ .

**Local sensitivity.** Figure IA.2 reports the standardized sensitivities of Andrews et al. (2017) at the estimate: the number of standard errors each parameter moves when one moment is perturbed by one standard error, with all twenty-four moments in the objective.  $\gamma$  responds most to the failure gap (moment 8) and  $\lambda_1$  to the late-tenure failure rate (7), as the argument above implies. The two failure costs load on different moments:  $\chi_s$  on the moments that describe the screen, the dissent distribution (11 and 13) and the failure profile (6 and 7);  $\delta$  on the output fixed-effect variance (4) and on the wage side, the co-option pay gap (20), the responses of pay to a failed vote and to lagged dissent (21 and 22), the pay autocovariance (23) and  $y_1$  (2). The dissent-pay slope gap (17) contributes less and loads about equally on  $\delta$  and  $\gamma$ .  $\sigma_{z_s}$  and  $\sigma_{z_b}$  load on the two failure slopes (15 and 16) with opposite signs, moment 16 the larger for  $\sigma_{z_s}$ , and with opposite signs on the wage moments, above all  $b_1$  (21) and the pay-dynamics moments (22 to 24); the wage fixed-effect variance (19) loads on  $\sigma_{z_s}$  and  $\sigma_\varepsilon$ .  $\sigma_\varepsilon$  responds most to the mean against-share on passing votes (11) and the share of votes below 5% against (13), and also to  $b_1$  and the pay-dynamics moments.  $\lambda_0$  responds to the moments that shift the pay level the technology implies, the dissent distribution, the middle failure bin (6),  $y_1$  and the output fixed-effect variance, rather than to average pay itself, which is measured to two hundredths.  $\log \eta$  is read off mean log output and  $\sigma_y$  off the output residual variance;  $\sigma_0$  loads on the output fixed-effect variance and also on  $s_1$ ,  $b_1$  and the post-failure pay change (24);  $\alpha$  loads on  $y_1$ , the output fixed-effect variance and the dissent distribution, since  $(1 - \alpha)^{-1}$  scales both the wedge and the discipline term. The precision weight mutes moment 10, so the slopes carry more of the local information about  $\sigma_{z_s}$  than the failure–dissent covariance does.

## IA2.4. Estimation

I estimate the twelve parameters in (IA.16); the calibrated inputs are listed in Table A.1. The search is confined to a box, listed in the order of Table 3:  $\sigma_0 \in [0.30, 1.20]$ ,  $\sigma_y \in [0.40, 1.20]$ ,  $\alpha \in [0.30, 0.65]$ ,  $\log \eta \in [6.5, 8.0]$ ,  $\sigma_{z_s} \in [0.30, 2.00]$ ,  $\chi_s \in [0.05, 1.50]$ ,  $\sigma_\varepsilon \in [0.30, 6.0]$ ,  $\sigma_{z_b} \in [0.30, 1.50]$ ,  $\delta$  through the Board’s total failure cost,  $\chi_s + \delta \in [0.20, 1.00]$ ,  $\lambda_0 \in [0.05, 0.35]$ ,  $\lambda_1 \in [0, 0.02]$  and  $\gamma \in [0, 2.0]$ . No estimate lies on a bound. The estimate minimizes the

weighted distance between the model moments  $m(\mathbb{P})$  and the data moments  $m(X)$ ,

$$\min_{\mathbb{P}} \left[ m(\mathbb{P}) - m(X) \right]' W \left[ m(\mathbb{P}) - m(X) \right]. \quad (\text{IA.17})$$

**Weight matrix.** The weight matrix is  $W = V^{-1}$ , the inverse of the covariance matrix of the data moments and the efficient choice for SMM. I estimate  $V$  from influence functions, clustered by firm-CEO spell because the model's learning and shock processes are confined within a spell. For each moment  $m_k$ , let  $IF_{ik}$  be observation  $i$ 's influence on the estimated  $m_k$ . Then

$$\sqrt{N}(\hat{m} - m_0) = \frac{1}{\sqrt{N}} \sum_{i=1}^N IF_i + o_p(1)$$

and

$$\hat{V} = \frac{1}{N} \sum_{g=1}^G \left( \sum_{i \in g} IF_i \right) \left( \sum_{i \in g} IF_i \right)',$$

where  $g$  indexes firm-CEO spells.

**Moment Jacobian.** The Jacobian  $G = \partial m(\mathbb{P}) / \partial \mathbb{P}'$  is computed by central differences on the estimation's own simulated panels, with a two-sided step of one percent of each parameter's search range. To reduce simulation noise, I repeat the calculation across 99 independent seeds and average the per-seed Jacobians; the standard errors and the sensitivities in Figure IA.2 use this averaged  $G$ . The covariance of the estimates is  $(1 + 1/N_S)(G'WG)^{-1}$  with  $N_S = 5$  simulated panels, so the standard errors combine sampling noise in the data moments with simulation noise in the model moments.

## IA2.5. Optimization algorithm

The routine below searches for the global minimum of (IA.17) and returns the best minimum found. It is a version of the TikTak algorithm of Arnoud et al. (2019), modified to run its local searches in parallel.<sup>1</sup>

1. **Parallel local minimization.** Draw a Sobol sequence of  $N$  points in the box above, with  $N = 50,000$  or  $100,000$ . Evaluate the objective at each point and keep the  $N_p$  best, with  $N_p =$

<sup>1</sup>I modified code from <https://github.com/tpapp/MultistartOptimization.jl>, which is based on the original TikTak code, <https://github.com/serdarozkan/TikTak>. See also Liu (2022) for a recent application.

100 or 200. From each of these, run a local minimization (Subplex, NLOpt's LN\_SBPLX) in parallel; the  $N_p$  local minima are the candidates for the global minimum.

2. **Parallel global minimization.** Sort the candidates in ascending order of the objective; the incumbent  $p_i^*$  is the best, with  $i = 1$ . Form  $N_p - i$  convex combinations of the incumbent with each lower-ranked candidate  $p_j$ ,  $p_{ji}^{cand} = \theta_{ji}p_i^* + (1 - \theta_{ji})p_j$ , where  $\theta_{ji} \in [0, 1]$  approaches 1 as  $j$  increases, as in TikTak, and run the local minimizer from each in parallel. If no restart improves on  $p_i^*$ , stop. Otherwise the best improving restart becomes the incumbent,  $i$  advances to its rank, and the step is repeated. The routine also stops when three successive replacements each lower the objective by less than 0.1%.
3. **Polish.** The reported estimate is an iterated Levenberg–Marquardt polish of the moment residuals from the best minimum found.

### IA3. Additional figures and tables

This appendix collects the local-sensitivity figure discussed in Appendix IA2.3, the list of targeted moments and their pairwise correlations, and three re-estimations of the model: at the majority-defeat line, without Board capture, and by passive-ownership half.

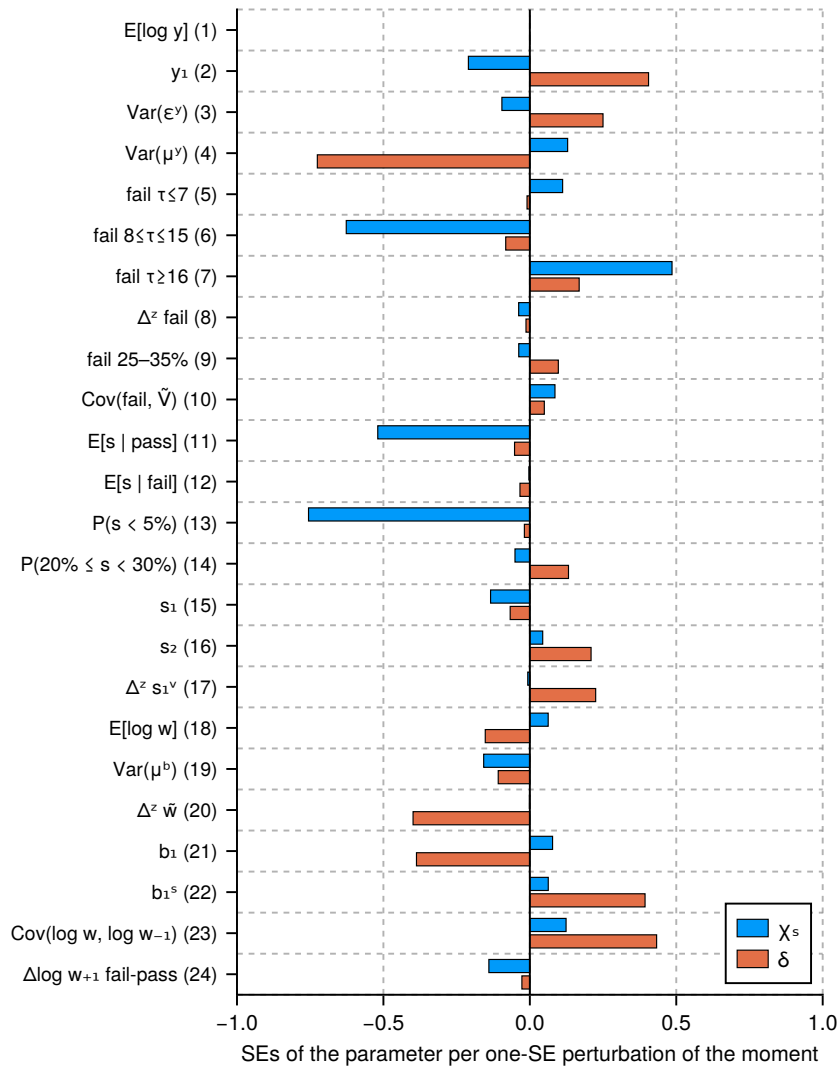
**Figure IA.2.** Sensitivity of the parameter estimates to the targeted moments

This figure reports the standardized sensitivity measure of Andrews et al. (2017),

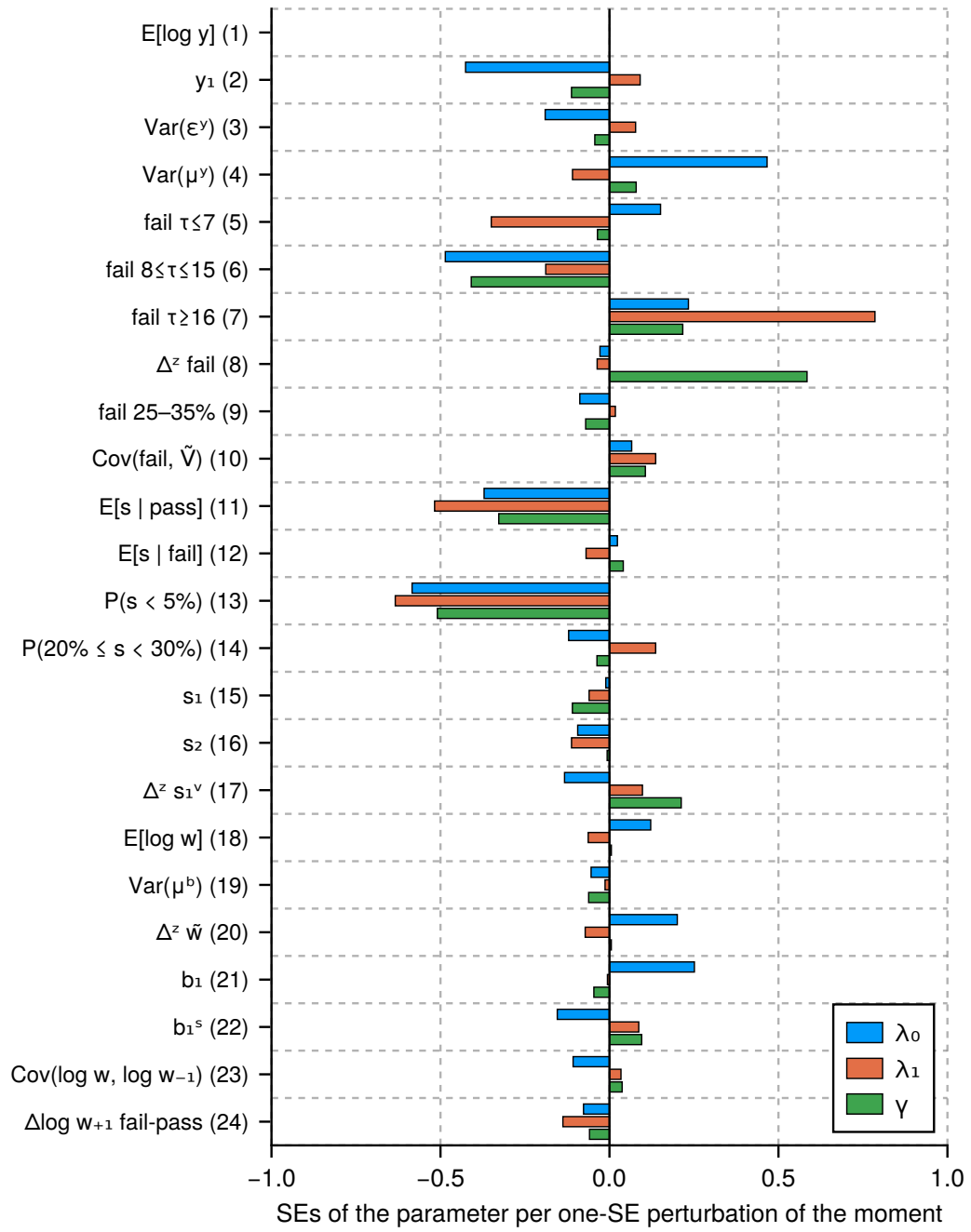
$$\tilde{\Lambda}_{kj} = [(G'WG)^{-1}G'W]_{kj} \hat{\sigma}(m_j)/\hat{\sigma}(\hat{\theta}_k),$$

evaluated at  $\hat{\theta}$ . Each bar gives the change in parameter  $k$ , in standard-error units, from a one-standard-error increase in moment  $j$ . The twenty-four moments are numbered as in Table 2. Panel A reports  $\chi_s$  and  $\delta$ ; Panel B,  $\lambda_0$ ,  $\lambda_1$ , and  $\gamma$ ; Panel C,  $\sigma_{z_b}$  and  $\sigma_{z_s}$ ; Panel D,  $\log \eta$ ,  $\sigma_0$ ,  $\sigma_y$ , and  $\alpha$ ; and Panel E,  $\sigma_\varepsilon$ . Entries are clipped at one standard error in absolute value.

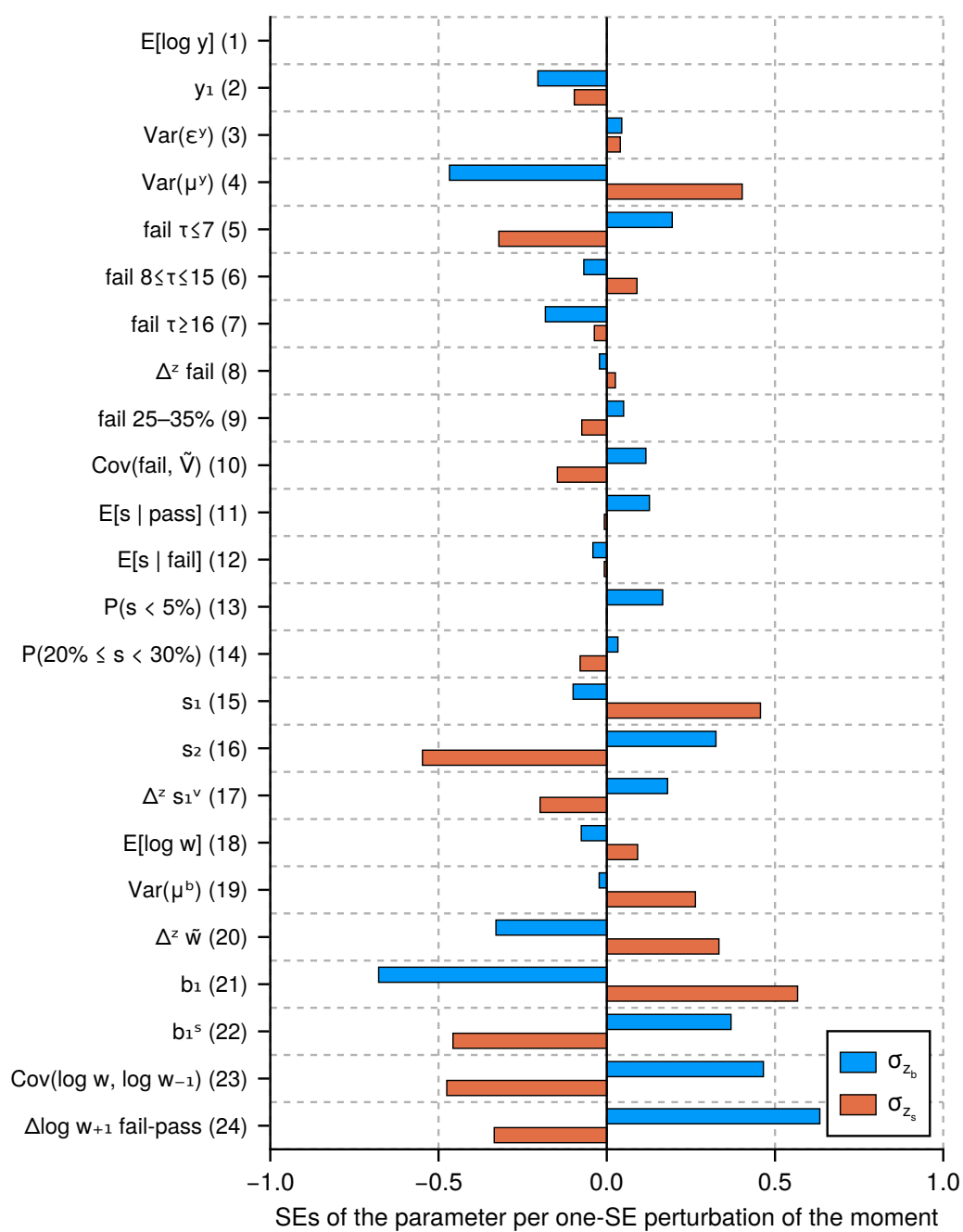
**Panel A. Dissent costs**



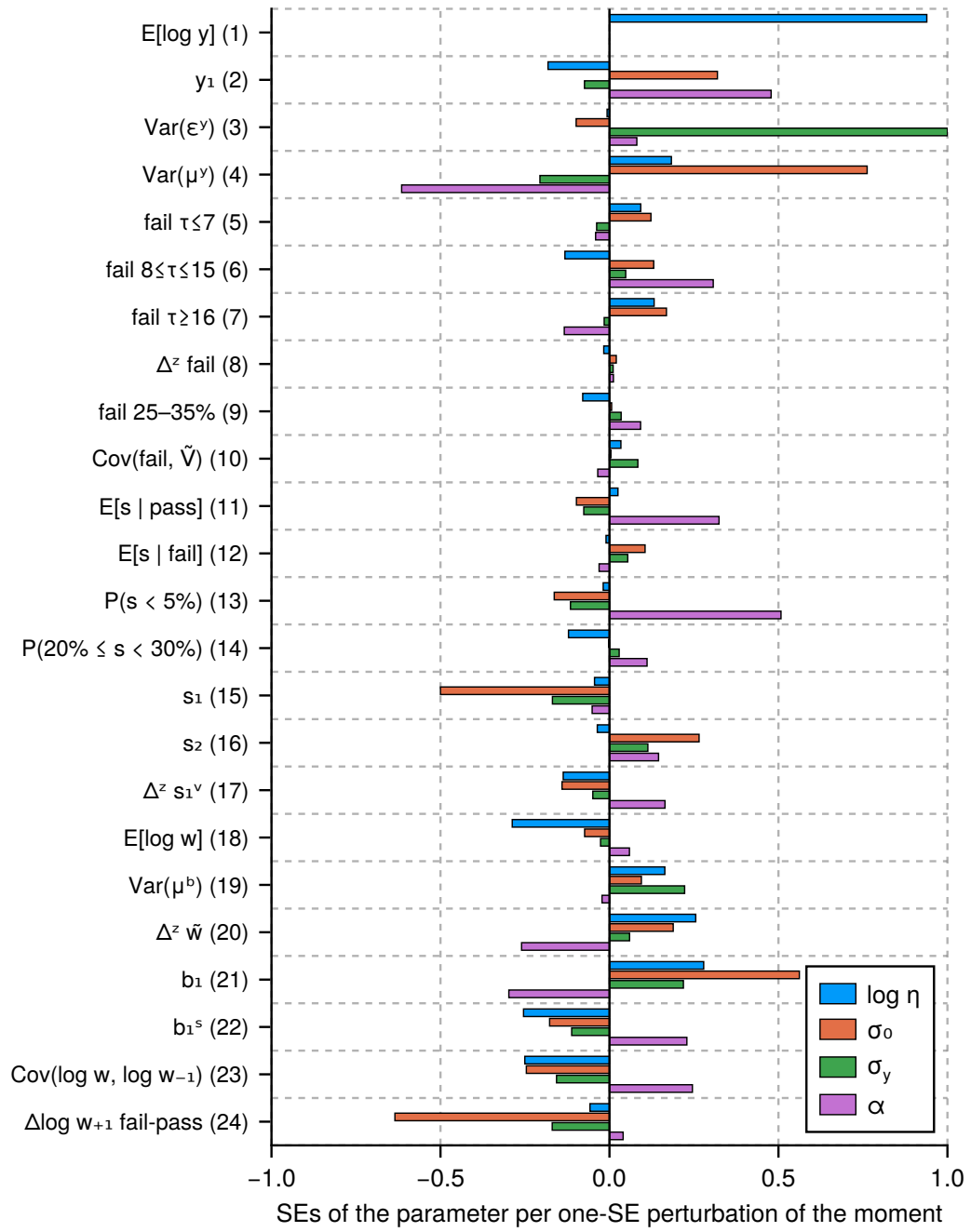
**Panel B. Capture**



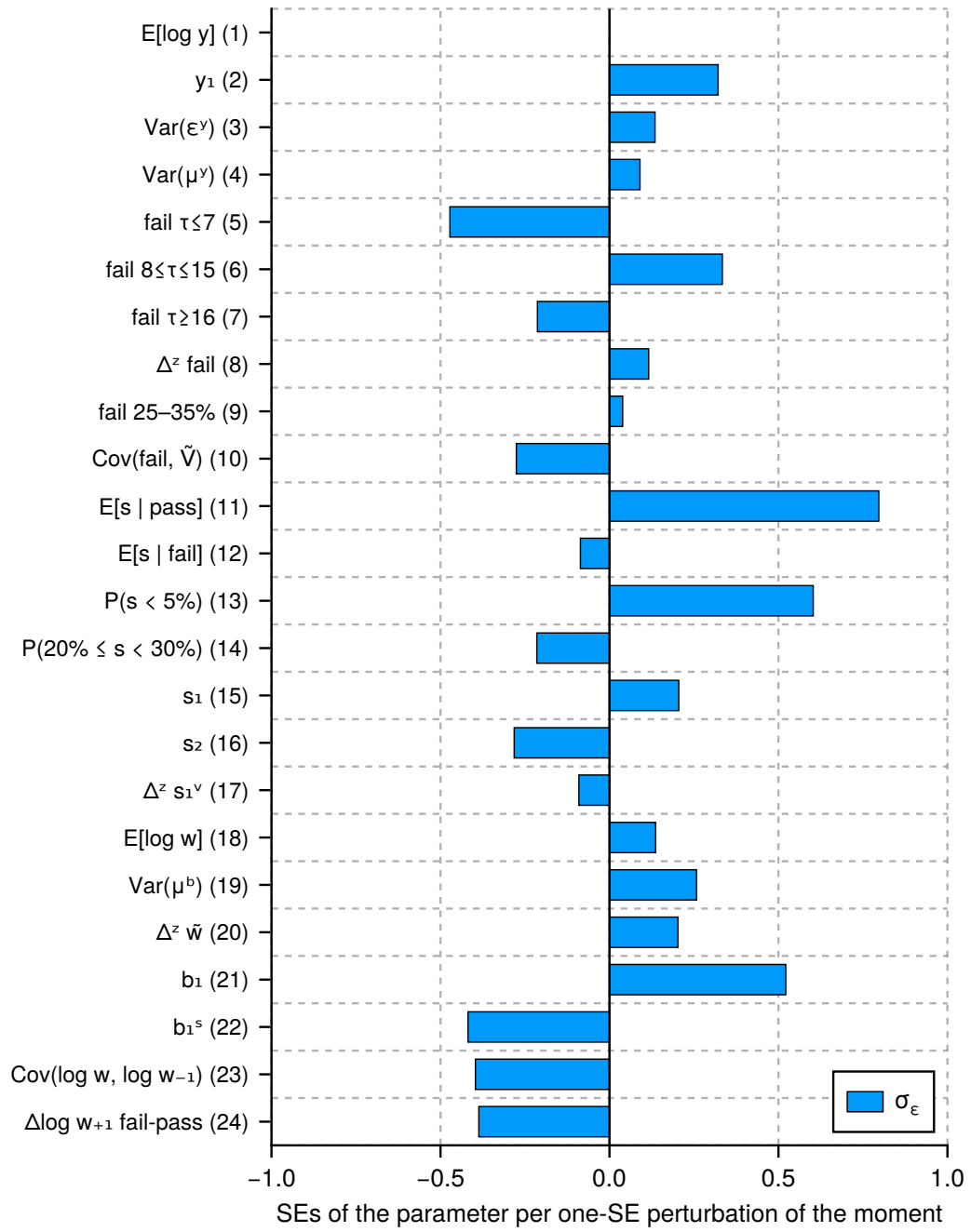
**Panel C. Information**



Panel D. Output



**Panel E.** Dispersion of shareholder thresholds



**Table IA.2.** Targeted moments and their correlations

Panel A lists the twenty-four targeted moments in the order and numbering of Table 2, with their notation. Every parameter is estimated jointly from all twenty-four moments. Section 4.1 gives the identification argument, and Figure IA.2 reports the estimated sensitivities.  $\Delta^Z$  denotes a co-option gap, the moment computed on firm-years above the sample median of the co-option index of Coles et al. (2014) minus the same moment at or below it. Panel B reports pairwise correlations between the empirical moments, numbered as in Panel A. The correlations are constructed from the influence-function variance-covariance matrix described in Appendix IA2.4 and used as the weighting matrix in the estimation. Cells are shaded from negative correlation in red to positive correlation in blue.

**Panel A.** Targeted moments

| Description                   |                                    | Notation                                                                                              |
|-------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Output &amp; CEO skill</b> |                                    |                                                                                                       |
| (1)                           | Mean log output                    | $\mathbb{E}[\log y]$                                                                                  |
| (2)                           | Output–wage elasticity             | $y_1$                                                                                                 |
| (3)                           | Output residual variance           | $\text{Var}(\epsilon^y)$                                                                              |
| (4)                           | Output FE variance                 | $\text{Var}(\mu^y)$                                                                                   |
| <b>Shareholder</b>            |                                    |                                                                                                       |
| (5)                           | Failure rate, tenure $\leq 7$      | $\mathbb{E}[\text{SOP fail} \mid \tau \leq 7]$                                                        |
| (6)                           | Failure rate, tenure 8–15          | $\mathbb{E}[\text{SOP fail} \mid 8 \leq \tau \leq 15]$                                                |
| (7)                           | Failure rate, tenure $\geq 16$     | $\mathbb{E}[\text{SOP fail} \mid \tau \geq 16]$                                                       |
| (8)                           | Failure rate, co-option gap        | $\Delta^Z \mathbb{E}[\text{SOP fail}]$                                                                |
| (9)                           | Failure rate, 25–35% dissent       | $\Pr(\text{SOP fail} \mid 0.25 \leq \text{share} < 0.35)$                                             |
| (10)                          | Failure–dissent covariance         | $\mathbb{E}[e \tilde{V}^{\text{against}} / \phi^2]$                                                   |
| (11)                          | Mean dissent, pass                 | $\mathbb{E}[\text{share} \mid \text{SOP pass}]$                                                       |
| (12)                          | Mean dissent, fail                 | $\mathbb{E}[\text{share} \mid \text{SOP fail}]$                                                       |
| (13)                          | Votes below 5% dissent             | $\Pr(\text{share} < 0.05)$                                                                            |
| (14)                          | Votes with 20–30% dissent          | $\Pr(0.20 \leq \text{share} < 0.30)$                                                                  |
| (15)                          | Failure–wage coef                  | $s_1$                                                                                                 |
| (16)                          | Failure–output shock coef          | $s_2$                                                                                                 |
| (17)                          | Dissent–pay slope, co-option       | $\Delta^Z s_1^V$                                                                                      |
| <b>Board</b>                  |                                    |                                                                                                       |
| (18)                          | Mean log wage                      | $\mathbb{E}[\log w]$                                                                                  |
| (19)                          | Wage FE variance                   | $\text{Var}(\mu^b)$                                                                                   |
| (20)                          | Wage residual, co-option           | $\Delta^Z \mathbb{E}[\tilde{w}]$                                                                      |
| (21)                          | Log wage–lagged failure coef       | $b_1$                                                                                                 |
| (22)                          | Log wage–lagged dissent coef       | $b_1^s$                                                                                               |
| (23)                          | Log wage autocovariance            | $\text{Cov}(\log w_t, \log w_{t-1})$                                                                  |
| (24)                          | $\Delta$ Log wage, fail minus pass | $\mathbb{E}[\Delta \log w_{t+1} \mid \text{fail}] - \mathbb{E}[\Delta \log w_{t+1} \mid \text{pass}]$ |

**Panel B.** Pairwise correlations between the empirical moments

|      | (1)   | (2)   | (3)   | (4)   | (5)   | (6)   | (7)   | (8)   | (9)   | (10)  | (11)  | (12)  | (13)  | (14)  | (15)  | (16)  | (17)  | (18)  | (19) | (20)  | (21) | (22) | (23) | (24) |
|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|------|------|------|------|
| (1)  | 1.00  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (2)  | -0.09 | 1.00  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (3)  | 0.06  | 0.02  | 1.00  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (4)  | 0.19  | 0.03  | 0.72  | 1.00  |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (5)  | 0.01  | -0.13 | 0.09  | 0.01  | 1.00  |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (6)  | 0.01  | -0.04 | 0.13  | 0.07  | 0.11  | 1.00  |       |       |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (7)  | 0.01  | -0.02 | 0.09  | 0.05  | 0.01  | 0.12  | 1.00  |       |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (8)  | -0.01 | -0.02 | 0.04  | 0.02  | 0.22  | 0.01  | 0.28  | 1.00  |       |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (9)  | 0.01  | -0.01 | -0.01 | -0.01 | 0.24  | 0.19  | 0.07  | 0.15  | 1.00  |       |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (10) | -0.16 | 0.13  | -0.12 | -0.13 | -0.52 | -0.46 | -0.33 | -0.19 | -0.04 | 1.00  |       |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (11) | 0.20  | -0.04 | 0.11  | 0.10  | 0.19  | 0.20  | 0.13  | 0.05  | -0.21 | -0.71 | 1.00  |       |       |       |       |       |       |       |      |       |      |      |      |      |
| (12) | 0.00  | -0.08 | 0.02  | -0.02 | 0.03  | 0.04  | 0.14  | -0.05 | -0.33 | -0.08 | 0.01  | 1.00  |       |       |       |       |       |       |      |       |      |      |      |      |
| (13) | -0.30 | 0.06  | -0.10 | -0.11 | -0.29 | -0.27 | -0.16 | -0.08 | -0.00 | 0.69  | -0.81 | -0.04 | 1.00  |       |       |       |       |       |      |       |      |      |      |      |
| (14) | 0.01  | -0.03 | 0.08  | 0.07  | 0.11  | 0.13  | 0.09  | 0.04  | -0.36 | -0.47 | 0.67  | -0.03 | -0.31 | 1.00  |       |       |       |       |      |       |      |      |      |      |
| (15) | -0.05 | 0.06  | 0.02  | -0.09 | 0.28  | 0.28  | 0.23  | 0.20  | 0.11  | -0.27 | 0.04  | 0.11  | -0.09 | 0.04  | 1.00  |       |       |       |      |       |      |      |      |      |
| (16) | -0.05 | 0.10  | 0.01  | -0.09 | -0.05 | 0.04  | 0.09  | 0.06  | -0.02 | 0.01  | -0.01 | -0.04 | 0.03  | 0.01  | 0.50  | 1.00  |       |       |      |       |      |      |      |      |
| (17) | 0.03  | 0.04  | 0.02  | 0.04  | 0.05  | 0.01  | 0.07  | 0.31  | 0.03  | -0.06 | 0.03  | 0.03  | -0.07 | -0.01 | 0.23  | 0.15  | 1.00  |       |      |       |      |      |      |      |
| (18) | 0.76  | 0.01  | 0.07  | 0.10  | 0.14  | 0.11  | 0.09  | 0.01  | 0.00  | -0.13 | 0.36  | 0.04  | -0.42 | 0.11  | 0.13  | 0.01  | 0.04  | 1.00  |      |       |      |      |      |      |
| (19) | -0.06 | -0.03 | 0.02  | 0.24  | 0.03  | -0.02 | -0.04 | 0.03  | -0.01 | -0.12 | 0.01  | -0.00 | -0.03 | -0.00 | -0.09 | -0.03 | 0.00  | -0.28 | 1.00 |       |      |      |      |      |
| (20) | -0.09 | -0.07 | 0.02  | -0.02 | 0.05  | -0.02 | 0.05  | 0.25  | 0.02  | -0.06 | 0.01  | -0.05 | -0.02 | 0.01  | 0.03  | 0.03  | 0.19  | -0.11 | 0.17 | 1.00  |      |      |      |      |
| (21) | -0.01 | 0.17  | -0.03 | -0.07 | 0.00  | 0.08  | 0.09  | 0.06  | 0.03  | -0.05 | 0.02  | -0.03 | -0.03 | 0.03  | 0.24  | 0.23  | 0.11  | 0.01  | 0.01 | -0.00 | 1.00 |      |      |      |
| (22) | 0.00  | 0.18  | -0.03 | -0.07 | -0.01 | 0.03  | 0.04  | 0.01  | 0.01  | -0.02 | 0.02  | 0.01  | -0.03 | 0.03  | 0.19  | 0.17  | 0.09  | 0.03  | 0.01 | -0.02 | 0.87 | 1.00 |      |      |
| (23) | -0.01 | 0.04  | 0.06  | 0.23  | 0.01  | 0.03  | -0.00 | 0.04  | -0.02 | -0.13 | 0.05  | 0.02  | -0.06 | 0.03  | -0.09 | -0.04 | 0.02  | -0.23 | 0.77 | 0.21  | 0.04 | 0.04 | 1.00 |      |
| (24) | 0.02  | 0.29  | -0.04 | -0.03 | -0.01 | 0.12  | 0.07  | -0.02 | 0.05  | -0.06 | 0.04  | -0.09 | -0.06 | 0.04  | 0.01  | 0.16  | -0.07 | 0.03  | 0.03 | -0.03 | 0.62 | 0.54 | 0.05 | 1.00 |

**Table IA.3.** Estimation at the majority-defeat threshold

This table re-estimates the model with SOP failure defined as a vote-against share of at least 50 percent, a majority defeat, in place of 30 percent. Every moment defined by failure is rebuilt at the 50 percent threshold, including the failure band, now 45 to 55 percent dissent, and the near-threshold share bin, now 40 to 50 percent. All twelve parameters are re-estimated with the same estimator, weighting matrix, and calibrated parameters as the main estimate. Panel A reports model fit using the columns of Table 2. Panel B reports the parameter estimates beside the baseline estimates of Table 3. The SMM objective value is 293.9.

**Panel A. Model fit**

| <b>Description</b>                      | <b>Observed</b> | <b>Simulated</b> | <b>Difference</b> | <b>t-stat</b> |
|-----------------------------------------|-----------------|------------------|-------------------|---------------|
| <b>Output &amp; CEO skill</b>           |                 |                  |                   |               |
| (1) Mean log output                     | 7.859           | 7.928            | −0.069            | 1.823         |
| (2) Output–wage elasticity              | 0.284           | 0.251            | 0.033             | −1.958        |
| (3) Output residual variance            | 0.471           | 0.528            | −0.058            | 5.874         |
| (4) Output FE variance                  | 0.450           | 0.636            | −0.186            | 9.659         |
| <b>Shareholder</b>                      |                 |                  |                   |               |
| (5) Failure rate, tenure $\leq 7$       | 0.015           | 0.015            | 0.001             | −0.596        |
| (6) Failure rate, tenure 8–15           | 0.019           | 0.016            | 0.003             | −1.081        |
| (7) Failure rate, tenure $\geq 16$      | 0.038           | 0.023            | 0.015             | −1.906        |
| (8) Failure rate, co-option gap         | 0.008           | 0.008            | 0.000             | −0.079        |
| (9) Failure rate, 45–55% dissent        | 0.442           | 0.414            | 0.028             | −0.744        |
| (10) Failure–dissent covariance         | 0.040           | 0.028            | 0.012             | −2.131        |
| (11) Mean dissent, pass                 | 0.075           | 0.079            | −0.004            | 3.743         |
| (12) Mean dissent, fail                 | 0.622           | 0.613            | 0.009             | −1.496        |
| (13) Votes below 5% dissent             | 0.534           | 0.551            | −0.016            | 2.244         |
| (14) Votes with 40–50% dissent          | 0.016           | 0.018            | −0.002            | 1.205         |
| (15) Failure–wage coef                  | 0.025           | 0.020            | 0.005             | −1.519        |
| (16) Failure–output shock coef          | −0.022          | −0.027           | 0.005             | −1.625        |
| (17) Dissent–pay slope, co-option       | 0.018           | 0.023            | −0.005            | 0.751         |
| <b>Board</b>                            |                 |                  |                   |               |
| (18) Mean log wage                      | 1.659           | 1.696            | −0.037            | 1.992         |
| (19) Wage FE variance                   | 0.574           | 0.523            | 0.051             | −3.178        |
| (20) Wage residual, co-option           | 0.098           | 0.033            | 0.064             | −3.485        |
| (21) Log wage–lagged failure coef       | −0.190          | −0.099           | −0.090            | 1.588         |
| (22) Log wage–lagged dissent coef       | −0.321          | −0.316           | −0.004            | 0.069         |
| (23) Log wage autocovariance            | 0.594           | 0.610            | −0.016            | 0.786         |
| (24) $\Delta$ Log wage, fail minus pass | −0.525          | −0.538           | 0.013             | −0.191        |

**Panel B.** Parameter estimates

| <b>Description</b>                          | <b>Notation</b>      | <b>Baseline<br/>(30%)</b> | <b>Majority<br/>(50%)</b> |
|---------------------------------------------|----------------------|---------------------------|---------------------------|
| <b>Output &amp; CEO skill</b>               |                      |                           |                           |
| Prior std dev of CEO ability                | $\sigma_0$           | 0.7635                    | 0.7369                    |
| Std dev of productivity shock               | $\sigma_y$           | 0.7953                    | 0.8014                    |
| Output–CEO wage elasticity                  | $\alpha$             | 0.5000                    | 0.5008                    |
| Scaling factor (output)                     | $\log \eta$          | 7.0665                    | 7.0703                    |
| <b>Shareholder</b>                          |                      |                           |                           |
| Std dev of Shareholder signal               | $\sigma_{z_s}$       | 0.5826                    | 0.5894                    |
| Shareholder SOP failure cost                | $\chi_s$             | 0.4720                    | 0.4640                    |
| Std dev of dispersed shareholder thresholds | $\sigma_\varepsilon$ | 1.3715                    | 1.3675                    |
| <b>Board</b>                                |                      |                           |                           |
| Std dev of Board signal                     | $\sigma_{z_b}$       | 0.8759                    | 0.9012                    |
| Board incremental private cost              | $\delta$             | 0.2743                    | 0.2922                    |
| CEO board capture (constant)                | $\lambda_0$          | 0.1906                    | 0.1908                    |
| CEO board capture (growth)                  | $\lambda_1$          | 0.0024                    | 0.0023                    |
| CEO board capture (co-option)               | $\gamma$             | 0.2318                    | 0.2123                    |

**Table IA.4.** Estimation without Board capture

This table re-estimates the model imposing  $\lambda_0 = \lambda_1 = \gamma = 0$ . All remaining parameters are re-optimized using the same moments, estimator, weighting matrix, and calibrated parameters as the main estimate. Panel A reports model fit using the columns of Table 2. Panel B reports the parameter estimates beside the baseline estimates of Table 3. The SMM objective value is 1576.1.

**Panel A. Model fit**

| <b>Description</b>                      | <b>Observed</b> | <b>Simulated</b> | <b>Difference</b> | <b><i>t</i>-stat</b> |
|-----------------------------------------|-----------------|------------------|-------------------|----------------------|
| <b>Output &amp; CEO skill</b>           |                 |                  |                   |                      |
| (1) Mean log output                     | 7.859           | 7.723            | 0.136             | −3.560               |
| (2) Output–wage elasticity              | 0.284           | 0.436            | −0.151            | 8.857                |
| (3) Output residual variance            | 0.471           | 0.582            | −0.111            | 11.336               |
| (4) Output FE variance                  | 0.450           | 0.666            | −0.216            | 11.215               |
| <b>Shareholder</b>                      |                 |                  |                   |                      |
| (5) Failure rate, tenure $\leq 7$       | 0.054           | 0.078            | −0.024            | 8.155                |
| (6) Failure rate, tenure 8–15           | 0.066           | 0.078            | −0.012            | 2.083                |
| (7) Failure rate, tenure $\geq 16$      | 0.099           | 0.078            | 0.020             | −1.612               |
| (8) Failure rate, co-option gap         | 0.033           | 0.003            | 0.030             | −5.250               |
| (9) Failure rate, 25–35% dissent        | 0.452           | 0.427            | 0.025             | −1.055               |
| (10) Failure–dissent covariance         | 0.119           | 0.143            | −0.024            | 3.722                |
| (11) Mean dissent, pass                 | 0.061           | 0.048            | 0.013             | −16.470              |
| (12) Mean dissent, fail                 | 0.455           | 0.480            | −0.025            | 4.879                |
| (13) Votes below 5% dissent             | 0.534           | 0.642            | −0.107            | 14.724               |
| (14) Votes with 20–30% dissent          | 0.044           | 0.054            | −0.011            | 5.245                |
| (15) Failure–wage coef                  | 0.058           | 0.052            | 0.006             | −0.966               |
| (16) Failure–output shock coef          | −0.045          | −0.033           | −0.012            | 1.896                |
| (17) Dissent–pay slope, co-option       | 0.018           | −0.001           | 0.019             | −2.671               |
| <b>Board</b>                            |                 |                  |                   |                      |
| (18) Mean log wage                      | 1.659           | 1.509            | 0.150             | −8.017               |
| (19) Wage FE variance                   | 0.571           | 0.511            | 0.059             | −3.737               |
| (20) Wage residual, co-option           | 0.098           | 0.003            | 0.094             | −5.092               |
| (21) Log wage–lagged failure coef       | −0.104          | −0.018           | −0.086            | 3.039                |
| (22) Log wage–lagged dissent coef       | −0.321          | −0.064           | −0.257            | 3.949                |
| (23) Log wage autocovariance            | 0.594           | 0.626            | −0.032            | 1.544                |
| (24) $\Delta$ Log wage, fail minus pass | −0.351          | −0.186           | −0.165            | 4.820                |

**Panel B.** Parameter estimates

| <b>Description</b>                          | <b>Notation</b>      | <b>Baseline</b> | <b>No capture</b> |
|---------------------------------------------|----------------------|-----------------|-------------------|
| <b>Output &amp; CEO skill</b>               |                      |                 |                   |
| Prior std dev of CEO ability                | $\sigma_0$           | 0.7635          | 0.9471            |
| Std dev of productivity shock               | $\sigma_y$           | 0.7953          | 0.8367            |
| Output–CEO wage elasticity                  | $\alpha$             | 0.5000          | 0.5085            |
| Scaling factor (output)                     | $\log \eta$          | 7.0665          | 6.9340            |
| <b>Shareholder</b>                          |                      |                 |                   |
| Std dev of Shareholder signal               | $\sigma_{z_s}$       | 0.5826          | 0.3415            |
| Shareholder SOP failure cost                | $\chi_s$             | 0.4720          | 0.0000            |
| Std dev of dispersed shareholder thresholds | $\sigma_\varepsilon$ | 1.3715          | 1.3715            |
| <b>Board</b>                                |                      |                 |                   |
| Std dev of Board signal                     | $\sigma_{z_b}$       | 0.8759          | 6.9949            |
| Board incremental private cost              | $\delta$             | 0.2743          | 0.1884            |
| CEO board capture (constant)                | $\lambda_0$          | 0.1906          | 0                 |
| CEO board capture (growth)                  | $\lambda_1$          | 0.0024          | 0                 |
| CEO board capture (co-option)               | $\gamma$             | 0.2318          | 0                 |

**Table IA.5.** Parameter estimates by passive ownership

This table re-estimates the model on two halves of the sample split by passive ownership. The measure is the combined stake of Vanguard, BlackRock, and State Street from 13F filings, demeaned by fiscal year and firm-size decile, averaged within a CEO spell, and split at the median. The high half contains the passive-heavy firms. Each half is re-estimated on the same twenty-four moments with the same estimator, weighting matrix, and calibrated parameters. Panel A reports the point estimates in each half and their difference. The full-sample estimates are in Table 3. Panel B reports the targeted moments in each half, observed and simulated at the half's estimate. Moments are numbered as in Table 2. The SMM objective value is 372.2 in the high half and 617.6 in the low half.

**Panel A.** Parameter estimates by half

| Description                    | Notation       | Big Three stake |        |            |
|--------------------------------|----------------|-----------------|--------|------------|
|                                |                | High            | Low    | Difference |
| <b>Output &amp; CEO skill</b>  |                |                 |        |            |
| Prior std dev of CEO ability   | $\sigma_0$     | 0.6666          | 0.7927 | −0.1261    |
| Std dev of productivity shock  | $\sigma_y$     | 0.7711          | 0.8145 | −0.0434    |
| Output–CEO wage elasticity     | $\alpha$       | 0.5039          | 0.5142 | −0.0103    |
| Scaling factor (output)        | $\log \eta$    | 6.9600          | 7.1770 | −0.2169    |
| <b>Shareholder</b>             |                |                 |        |            |
| Std dev of Shareholder signal  | $\sigma_{z_s}$ | 0.5406          | 0.5387 | 0.0019     |
| Shareholder SOP failure cost   | $\chi_s$       | 0.5289          | 0.3476 | 0.1814     |
| <b>Board</b>                   |                |                 |        |            |
| Std dev of Board signal        | $\sigma_{z_b}$ | 0.9503          | 1.0753 | −0.1250    |
| Board incremental private cost | $\delta$       | 0.3008          | 0.4137 | −0.1129    |
| CEO board capture (constant)   | $\lambda_0$    | 0.1840          | 0.1491 | 0.0349     |
| CEO board capture (growth)     | $\lambda_1$    | 0.0016          | 0.0034 | −0.0017    |
| CEO board capture (co-option)  | $\gamma$       | 0.1603          | 0.2225 | −0.0623    |
| Firm-years                     |                | 6753            | 6356   |            |
| CEO spells                     |                | 1145            | 1146   |            |

**Panel B.** Targeted moments by half

| Description                             | High Big Three |           | Low Big Three |           |
|-----------------------------------------|----------------|-----------|---------------|-----------|
|                                         | Observed       | Simulated | Observed      | Simulated |
| <b>Output &amp; CEO skill</b>           |                |           |               |           |
| (1) Mean log output                     | 7.705          | 7.797     | 8.022         | 8.089     |
| (2) Output–wage elasticity              | 0.295          | 0.215     | 0.276         | 0.287     |
| (3) Output residual variance            | 0.448          | 0.486     | 0.497         | 0.543     |
| (4) Output FE variance                  | 0.409          | 0.557     | 0.482         | 0.683     |
| <b>Shareholder</b>                      |                |           |               |           |
| (5) Failure rate, tenure $\leq 7$       | 0.045          | 0.050     | 0.064         | 0.061     |
| (6) Failure rate, tenure 8–15           | 0.065          | 0.067     | 0.067         | 0.091     |
| (7) Failure rate, tenure $\geq 16$      | 0.100          | 0.075     | 0.097         | 0.129     |
| (8) Failure rate, co-option gap         | 0.025          | 0.018     | 0.034         | 0.030     |
| (9) Failure rate, 25–35% dissent        | 0.474          | 0.416     | 0.432         | 0.425     |
| (10) Failure–dissent covariance         | 0.105          | 0.085     | 0.134         | 0.123     |
| (11) Mean dissent, pass                 | 0.059          | 0.063     | 0.064         | 0.065     |
| (12) Mean dissent, fail                 | 0.456          | 0.421     | 0.455         | 0.433     |
| (13) Votes below 5% dissent             | 0.556          | 0.553     | 0.511         | 0.533     |
| (14) Votes with 20–30% dissent          | 0.041          | 0.064     | 0.047         | 0.071     |
| (15) Failure–wage coef                  | 0.060          | 0.054     | 0.056         | 0.050     |
| (16) Failure–output shock coef          | −0.046         | −0.079    | −0.041        | −0.089    |
| (17) Dissent–pay slope, co-option       | 0.000          | 0.019     | 0.022         | 0.023     |
| <b>Board</b>                            |                |           |               |           |
| (18) Mean log wage                      | 1.597          | 1.639     | 1.725         | 1.748     |
| (19) Wage FE variance                   | 0.452          | 0.397     | 0.693         | 0.591     |
| (20) Wage residual, co-option           | 0.064          | 0.023     | 0.128         | 0.026     |
| (21) Log wage–lagged failure coef       | −0.036         | −0.085    | −0.171        | −0.081    |
| (22) Log wage–lagged dissent coef       | −0.218         | −0.336    | −0.426        | −0.303    |
| (23) Log wage autocovariance            | 0.493          | 0.462     | 0.702         | 0.686     |
| (24) $\Delta$ Log wage, fail minus pass | −0.279         | −0.294    | −0.412        | −0.267    |

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